

From Operational Response to Phase Space: A Commutativity No-Go Theorem for Exact Quantum Soldering

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Abstract We investigate the quantum-theoretic consequences of the operational generator–displacement response introduced in the motion–staticity reconstruction of Lorentzian spacetime geometry. Classically, a faithful response

$$\Theta_x(P, \xi) = P(e_x \xi) \quad (1)$$

determines a nondegenerate solder map $e_x : T_x M \rightarrow V_x$ and hence the Lorentzian metric $g = e^* \eta$. We first show that the same response also reconstructs the canonical cotangent geometry: after identifying the physical spacetime covector as $p = e^* P$, the tautological one-form on $T^* M$ satisfies

$$\lambda_{(x, e^* P)}(\Xi) = \Theta_x(P, \pi_* \Xi), \quad (2)$$

so that locally

$$\lambda = P_a e^a, \quad \Omega = d\lambda. \quad (3)$$

Thus the Lorentzian metric and the canonical symplectic structure arise as two distinct geometric consequences of the same operational pairing.

We then ask whether an exact unitary quantization of this response can by itself produce intrinsically noncommutative quantum geometry. Let $W(P, \xi)$ be a strongly continuous unitary response satisfying exact composition in both generator and displacement arguments,

$$W(P + Q, \xi) = W(P, \xi)W(Q, \xi), \quad (4)$$

$$W(P, \xi + \zeta) = W(P, \xi)W(P, \zeta). \quad (5)$$

Consistency of the two composition laws implies

$$[W(P, \xi), W(Q, \zeta)] = 0 \quad (6)$$

for all admissible generators and displacements. Consequently, the C^* -algebra generated by the exact responses is commutative. By the joint spectral theorem, the response admits a representation of the form

$$W(P, \xi) = \int_{\mathcal{E}} \exp\left[-\frac{i}{\hbar} P(e\xi)\right] dE(e), \quad (7)$$

where $\mathcal{E}$ is a spectral space of classical solder maps. Every state restricted to this algebra is therefore representable by an ordinary probability measure on $\mathcal{E}$. In this precise sense, fluctuations observable through the exact response algebra alone are indistinguishable from classical stochastic fluctuations of soldered geometry.

We further show that unitary parameter orbits generated solely by this commuting response have vanishing Berry curvature, while their quantum information metric is a positive semidefinite covariance form. These structures therefore cannot by themselves supply either a noncommutative geometric algebra or a Lorentz-signature momentum-space metric. The result is a no-go theorem rather than a model of quantum gravity: an intrinsically quantum extension of the operationally reconstructed geometry must modify at least one ingredient of the exact response structure, for example through projective composition, noncommuting transport, finite calibration geometry, or an additional conjugate response sector.

Keywords: operational soldering; quantum geometry; commutativity no-go theorem; unitary response; Liouville form; symplectic geometry; spectral theorem; classical stochastic geometry; Berry curvature; quantum geometric tensor; motion–staticity geometry

Contents

1	Introduction	7
2	Operational Response and Canonical Cotangent Geometry	12
2.1	Generator–displacement response and soldering	12
2.2	Physical cotangent generators	14
2.3	The response as the Liouville one-form	14
2.4	Canonical symplectic structure	15
2.5	Local Lorentz covariance	16
2.6	Mass-shell restriction and presymplectic geometry	17
2.7	Metric–symplectic bifurcation of the operational response	17
2.8	From classical response to unitary response	18
3	Exact Unitary Operational Response	19
3.1	From the massive response cone to the additive generator space	19
3.2	Definition of an exact unitary response	20
3.3	Immediate consequences of the separate composition laws	21
3.4	Unitary group completion of the positive generator response	21
3.5	Infinitesimal generators for fixed response arguments	23
3.6	Classical correspondence and the role of $\hbar$	23
3.7	The exact response algebra	24
3.8	Exactness versus projective composition	25
3.9	Scope of the exact-response assumptions	25
3.10	Algebraic question	26
4	Commutativity of the Exact Response Algebra	26
4.1	Compatibility of the two exact composition laws	26
4.2	Exact-response commutativity theorem	27
4.3	Abelianness of the response algebra	28
4.4	A rectangular composition identity	29
4.5	Vanishing mixed group commutators	29
4.6	No nontrivial central commutator from exact composition	29
4.7	Infinitesimal consequence	30

4.8	No tetrad commutator is assumed	30
4.9	Faithfulness and commutativity	31
4.10	Fiberwise and spacetime-local character of the theorem	31
4.11	A no-go statement at the algebraic level	32
4.12	Comparison with canonical projective quantization	32
4.13	Transition to the spectral representation	33
5	Joint Spectral Representation and Classical Stochastic Geometry	33
5.1	The Gelfand spectrum of the response algebra	33
5.2	Continuity of the scalar spectral responses	34
5.3	Classification of continuous scalar bicharacters	35
5.4	Linear response maps associated with spectral points	36
5.5	Spectral response theorem	37
5.6	Sharp classical geometry as a special spectral sector	37
5.7	Spectral nondegeneracy	38
5.8	Spectral soldering operators	38
5.9	States and probability measures on response space	39
5.10	Classical stochastic representation theorem	40
5.11	Operational equivalence to classical random geometry	40
5.12	Moments of the spectral soldering distribution	41
5.13	Mean coframe versus mean metric	41
5.14	No intrinsic contextuality within the response algebra	42
5.15	Local character of the stochastic representation	42
5.16	The exact-response quantum-geometry no-go theorem	43
5.17	What the no-go theorem does not imply	43
5.18	Transition to quantum information geometry	44
6	Quantum Geometric Tensor of the Exact Response Orbit	44
6.1	Response-generated state orbits	45
6.2	Quantum geometric tensor	45
6.3	Quantum geometric tensor of the generator orbit	46
6.4	Vanishing Berry curvature	46
6.5	The surviving information metric	47
6.6	Spectral interpretation of the information metric	47
6.7	Quantum Fisher information	48
6.8	No identification with the Lorentzian generator metric	48
6.9	Fixed-generator displacement orbits	49
6.10	The full product-parameter orbit	49
6.11	No Berry reconstruction of the canonical symplectic form	50
6.12	No momentum-space curvature from the exact response alone	51
6.13	Absence of an intrinsic momentum scale	51
6.14	Observer-dependent positive information geometry	52
6.15	Information geometry does not detect intrinsic quantumness	52
6.16	Quantum-geometric corollary of exact additivity	52
6.17	What would be required for a nonzero Berry curvature	53
6.18	Transition to projective response composition	53

7	Projective Response Composition and the Missing Quantum Structure	54
7.1	The combined response label space	54
7.2	Relation to the canonical cotangent symplectic form	56
7.3	The Weyl projective composition law	56
7.4	A soldered Weyl system	57
7.5	Why the exact response is not a Weyl system	58
7.6	Two marginal translation sectors	58
7.7	The response phase as a possible Weyl commutator	59
7.8	A structural incompatibility theorem	59
7.9	Infinitesimal Heisenberg relation	60
7.10	Why the solder map cannot yet be promoted to the central operator	60
7.11	Central spectral soldering does not evade the no-go theorem	61
7.12	Projective deformation of only one argument	61
7.13	The minimal algebraic change required for noncommutativity	62
7.14	A quantum-response design principle	62
7.15	Relation between the classical symplectic form and a quantum cocycle	62
7.16	Local projective quantization versus inter-event transport	63
7.17	No derivation of quantum gravity	63
7.18	Exact and projective response structures	64
7.19	Transition to the discussion of admissible quantum extensions	64
8	Admissible Quantum Extensions Beyond Exact Response	65
8.1	The assumptions that must be crossed	65
8.2	Projective composition	66
8.3	Noncommuting inter-event transport	66
8.4	Quantum transport curvature	67
8.5	Path dependence as additional response data	68
8.6	The classical closed-response phase and torsion	68
8.7	Additional conjugate response sectors	69
8.8	Configuration subalgebras can be classical	69
8.9	Finite calibration transformations	70
8.10	A linearity obstruction for exact response automorphisms	70
8.11	Curved momentum space requires additional comparison structure	71
8.12	Associativity and nonlinear composition	72
8.13	Nonassociative response composition	72
8.14	Classical stochasticity is not sufficient	73
8.15	Superposition language does not evade commutativity	73
8.16	Necessary condition for coherent geometry interference	74
8.17	Response classicality and superselection	74
8.18	A hierarchy of possible quantum extensions	74
9	The role of $\hbar$ after the no-go theorem	75
9.1	No emergence of Newton's constant from exact response	76
9.2	Three logically distinct structures	76
9.3	What counts as a genuine evasion of the theorem	76
9.4	A minimal escape theorem	77

9.5	Scope of the no-go theorem	77
9.6	Implications for the MS reconstruction programme	78
9.7	A possible research hierarchy	78
9.8	Transition to the final discussion	79
10	Discussion and Conclusions	79
10.1	Summary of the exact-response results	80
10.2	Classical stochasticity versus intrinsic quantumness	81
10.3	The role of the quantum geometric tensor	81
10.4	Why the appearance of $\hbar$ is not sufficient	82
10.5	Exact response and projective quantization	83
10.6	A change in the role of the operational phase	83
10.7	What the no-go theorem excludes	84
10.8	What the no-go theorem does not exclude	84
10.9	The distinction between local response and transport	84
10.10	Implications for quantum extensions of MS geometry	85
10.11	Possible locations of the missing quantum input	85
10.12	No derivation of a fundamental ultraviolet scale	86
10.13	Relation to canonical quantization	86
10.14	Novelty and relation to standard mathematics	87
10.15	Limitations	87
10.16	A hierarchy of reconstructed and additional structures	88
10.17	Final conclusion	88
A	Positive Generator Cone and Additive Group Completion	89
A.1	Additive closure of the future cone	90
A.2	The cone generates the full vector space	90
A.3	Uniqueness of the unitary group completion	91
B	An Interchange-Law Proof of Response Commutativity	91
B.1	Why unitarity is used	92
B.2	Semigroup-valued responses	92
C	Classification of Continuous Bicharacters on Real Vector Spaces	92
C.1	Characters of a real vector group	93
C.2	Linearity in the first argument	93
C.3	Application to operational soldering	94
D	Joint Spectral Construction of the Soldering Operators	94
D.1	Bilinear Stone generators	94
D.2	Reconstruction for arbitrary labels	95
D.3	Joint projection-valued measure	95
E	Strong Commutativity and Domains of Unbounded Generators	96
E.1	Strong commutativity	96
E.2	Ordinary commutators	96
E.3	Joint functional calculus	96

F	Faithfulness Does Not Imply Spectral Nondegeneracy	97
F.1	A finite-dimensional counterexample	97
F.2	Spectral nondegeneracy as an additional condition	98
G	Probability Measures, Moments, and Response Tomography	98
G.1	First moments	98
G.2	Second moments	98
G.3	Cumulant functional	99
G.4	Mean metric and fluctuation correction	99
H	Detailed Derivation of the Quantum Geometric Tensor	99
H.1	Parameter derivatives	99
H.2	Symmetric and antisymmetric parts	100
H.3	Positivity	100
H.4	Application to the full (P, ξ) orbit	100
I	Weyl Cocycles, Coboundaries, and Central Extensions	101
I.1	Projective composition	101
I.2	Change of phase convention	101
I.3	Symplectic Weyl cocycle	101
I.4	Soldered cocycle	102
J	Lorentz Covariance of the Soldered Symplectic Pairing	102
K	Covariant Decomposition of the Liouville Two-Form	103
K.1	Cancellation of connection terms	103
K.2	Torsion-free branch	103
K.3	Zero section	103
L	Counterexamples When Exact Double Additivity Is Relaxed	104
L.1	Exact generator composition alone	104
L.2	Exact displacement composition alone	104
L.3	Projective composition	105
L.4	Noncommuting inter-event transport	105
M	Local Algebras and Inter-Event Comparison	105
M.1	A bundle of response algebras	105
M.2	Transport automorphisms	106
M.3	Closed transport	106
N	A Hierarchy of Classicality and Quantumness	106
N.1	Level I: sharp classical soldering	106
N.2	Level II: classical stochastic soldering	107
N.3	Level III: quantum matter on a spectral geometry	107
N.4	Level IV: projective local response	107
N.5	Level V: noncommutative geometric response	107
N.6	Level VI: quantum inter-event comparison	107

O	Assumption Map for the Exact-Response No-Go Theorem	107
O.1	Classical reconstruction assumptions	108
O.2	Cotangent reconstruction assumptions	108
O.3	Quantum commutativity assumptions	108
O.4	Spectral representation assumptions	108
O.5	Classical stochastic representation assumptions	108
O.6	Quantum geometric tensor assumptions	109
P	Summary of the Appendix Results	109

1 Introduction

The passage from classical spacetime geometry to quantum geometry is often formulated as a quantization problem for geometric variables that are already given. In metric approaches one promotes $g_{\mu\nu}$, or functions thereof, to quantum degrees of freedom; in first-order formulations the corresponding variables may be coframes, Lorentz connections, fluxes, or holonomies. The conceptual order is then

$$\text{classical geometric variable} \longrightarrow \text{quantization rule} \longrightarrow \text{quantum geometric algebra.} \quad (8)$$

The purpose of the present work is to examine a different question. Suppose that the spacetime geometry itself has first been reconstructed from an operational response law rather than postulated as a primitive field. How much quantum structure is then obtained by quantizing that response law while preserving its classical composition properties exactly?

This question arises naturally in the motion–staticity (MS) reconstruction programme developed in the preceding works. The starting point of that programme is a compact state geometry for the classical massive sector,

$$(\mathbf{D}, S), \quad |\mathbf{D}|^2 + S^2 = 1, \quad S > 0, \quad (9)$$

whose operational realization identifies

$$\mathbf{D} = \frac{\mathbf{v}}{c}, \quad S = \frac{d\tau}{dt}. \quad (10)$$

The projective map

$$\Phi(\mathbf{D}, S) = \frac{1}{S}(1, \mathbf{D}) \quad (11)$$

sends the open compact hemisphere to the future unit Lorentz hyperboloid. In the flat free-particle sector, consistency of the same primitive scalar with the Hamiltonian Legendre remainder leads, under regularity and nondegeneracy assumptions, to the relativistic Hamiltonian

$$H(\mathbf{p}) = c \sqrt{\mathbf{p}^2 + Q^2}, \quad (12)$$

while universality and composition covariance of inertial observer changes recover the proper orthochronous Lorentz action on energy–momentum. The resulting construction is empirically equivalent to the classical free massive sector of special relativity; its purpose is reconstructive rather than phenomenological.

The curved-spacetime extension requires an additional ingredient. An abstract internal Lorentz space does not by itself determine how internal translation generators couple to spacetime displacements. This missing incidence relation was encoded in the scalar operational response

$$\Theta_x(P, \xi), \quad P \in \Sigma_{Q,x}^+, \quad \xi \in T_x M, \quad (13)$$

where $\Sigma_{Q,x}^+$ is the positive massive generator shell. Additivity and homogeneity in the generator, additivity and continuity in the infinitesimal displacement, smooth dependence on x , and a faithfulness condition imply the existence of a unique linear solder map

$$e_x : T_x M \longrightarrow V_x \quad (14)$$

such that

$$\boxed{\Theta_x(P, \xi) = P(e_x \xi)} \quad (1)$$

If the response is faithful and the tangent and internal spaces have equal dimension, e_x is nondegenerate. The internal Lorentz metric η therefore induces

$$\boxed{g = e^* \eta} \quad (2)$$

In this reconstruction, a Lorentzian spacetime metric is not introduced independently: it records how internal generators respond to infinitesimal physical displacements.

The same response data were subsequently extended to a restricted classical gravitational completion and, for asymptotically isolated configurations, to differentiable translation generators and their Lorentzian radial invariant. Those constructions remain explicitly conditional at the dynamical level: the numerical gravitational coupling, cosmological constant, more general torsional sectors, higher-derivative completions, and quantum dynamics are not determined by the compact MS state relation. The present paper therefore does not attempt to continue the classical reconstruction by postulating a particular quantum-gravitational action. Instead, it asks a logically prior question:

$$\boxed{\text{What follows if the operational response itself is quantized exactly?}} \quad (15)$$

There are two reasons to ask this question before introducing additional quantum-geometric structure.

First, the classical response contains more geometry than is apparent from Eq. (1). Once

$$p = e^* P \quad (16)$$

is interpreted as the physical spacetime covector, the response coincides with the canonical evaluation underlying the tautological one-form on the cotangent bundle. If $\pi : T^* M \rightarrow M$ is the cotangent projection and $\Xi \in T_{(x,p)} T^* M$, the Liouville one-form is

$$\lambda_{(x,p)}(\Xi) = p(\pi_* \Xi). \quad (17)$$

At the response-defined point $p = e^* P$,

$$\lambda_{(x,e^* P)}(\Xi) = e^* P(\pi_* \Xi) \quad (18)$$

$$= P(e_x \pi_* \Xi) \quad (19)$$

$$= \Theta_x(P, \pi_* \Xi). \quad (20)$$

Thus

$$\boxed{\lambda = P_a e^a} \quad (3)$$

and consequently

$$\boxed{\Omega = d\lambda} \quad (4)$$

The standard symplectic geometry of T^*M is therefore not an unrelated structure appended to the soldered Lorentzian geometry. Both arise from the same generator–displacement pairing:

$$\begin{array}{c}
\nearrow \\
\Theta \\
\searrow \\
\lambda = P_a e^a, \quad \Omega = d\lambda.
\end{array}$$

$g = e^* \eta$

(5)

No novelty is claimed for the tautological one-form or for the canonical symplectic form themselves. The point is the reconstruction order: the operational response that determines soldering also determines the canonical phase-space potential once the physical cotangent generator is identified.

Second, the appearance of the phase

$$\exp\left[-\frac{i}{\hbar}\Theta(P, \xi)\right] \quad (6)$$

might suggest that quantizing the response automatically turns the reconstructed geometry into a quantum geometry. This implication is false. A central result of this paper is that preserving the classical response composition laws too faithfully has the opposite consequence.

We introduce a unitary operational response $W(P, \xi)$ and impose the exact quantum analogues of generator and displacement composition:

$$\boxed{W(P + Q, \xi) = W(P, \xi)W(Q, \xi)} \quad (7)$$

$$\boxed{W(P, \xi + \zeta) = W(P, \xi)W(P, \zeta)} \quad (8)$$

We assume the corresponding identity conditions and strong continuity but do not assume a tetrad operator or a pre-existing quantum spacetime algebra. Equations (7) and (8) are simply the exact preservation of the two classical additive structures of Θ .

The consistency condition is unexpectedly restrictive. Evaluating $W(P + Q, \xi + \zeta)$ first by generator composition and then by displacement composition gives

$$W(P, \xi)W(P, \zeta)W(Q, \xi)W(Q, \zeta), \quad (21)$$

whereas reversing the order of the two decompositions gives

$$W(P, \xi)W(Q, \xi)W(P, \zeta)W(Q, \zeta). \quad (22)$$

Unitarity allows the common factors to be cancelled, yielding

$$\boxed{W(P, \zeta)W(Q, \xi) = W(Q, \xi)W(P, \zeta)} \quad (9)$$

Since the arguments are arbitrary,

$$\boxed{[W(P, \xi), W(Q, \zeta)] = 0} \quad (10)$$

throughout the exact response family.

Equation (10) is the first principal result of this work. It shows that exact additivity in both operational arguments forces the quantum response algebra to be Abelian. The result does not depend on a perturbative expansion in $\hbar$, on a semiclassical approximation, or on a particular Hilbert-space representation. It follows directly from the compatibility of the two exact composition laws.

The consequence is stronger than the commutativity of a chosen set of tetrad operators. Because the response algebra is Abelian, standard spectral theory applies. The unitary family can be decomposed into scalar characters. Since the scalar characters are continuous and additive in both P and ξ , they are continuous bicharacters and hence take the bilinear form

$$\chi_e(P, \xi) = \exp\left[-\frac{i}{\hbar}P(e\xi)\right], \quad (11)$$

where e is an ordinary real linear map of the same algebraic type as a classical solder form. The resulting joint spectral representation is

$$\boxed{W(P, \xi) = \int_{\mathcal{E}} \exp\left[-\frac{i}{\hbar}P(e\xi)\right] d\mathbf{E}(e)} \quad (12)$$

Here $\mathbf{E}$ is a projection-valued spectral measure and $\mathcal{E}$ denotes the corresponding space of classical response maps.

Equivalently, when the relevant generators are represented infinitesimally one may write

$$W(P, \xi) = \exp\left[-\frac{i}{\hbar}P_a \xi^\mu \hat{e}_\mu^a\right], \quad (13)$$

but Eq. (13) should be read as a conclusion of the spectral representation rather than as a prior tetrad quantization. The operators $\hat{e}_\mu^a$ obtained in this way strongly commute.

For any normal state ω , Eq. (12) therefore gives

$$\boxed{\omega(W(P, \xi)) = \int_{\mathcal{E}} \exp\left[-\frac{i}{\hbar}P(e\xi)\right] d\mu_\omega(e)} \quad (14)$$

with $d\mu_\omega$ an ordinary probability measure on the classical spectral space. Hence all statistical predictions accessible solely through the exact response algebra admit a classical stochastic representation. This leads to the main no-go statement of the paper:

$$\boxed{\text{Exact unitary quantization of the additive operational response does not by itself produce intrinsically noncommuting observables}} \quad (15)$$

The phrase “does not produce” is important. Equation (15) does not assert that the underlying Hilbert space must be classical, nor does it exclude quantum matter, noncommuting transport observables, projective representations, or additional quantum-geometric sectors. It states only that the algebra generated by the exact response observables $W(P, \xi)$ is commutative and therefore admits a

classical probability representation. A superposition or mixed state of different solder maps, when probed only by $\mathcal{A}_W$, is operationally indistinguishable from a classical random geometry.

This distinction also clarifies the role of $\hbar$. The occurrence of $\hbar$ in Eq. (6) establishes an action-to-phase conversion, but a phase alone does not imply a noncommutative quantum algebra:

$$\boxed{\hbar \text{ in an operational character } \not\Rightarrow \text{intrinsic quantum geometry}} \quad (16)$$

Noncommutativity requires additional algebraic information.

A related conclusion follows from the quantum geometric tensor. Fix a displacement ξ and define

$$A_a = \xi^\mu \hat{e}_{a\mu}. \quad (23)$$

The response orbit of a reference state is

$$|\psi(P)\rangle = \exp\left[-\frac{i}{\hbar} P^a A_a\right] |\psi_0\rangle. \quad (24)$$

Since $[A_a, A_b] = 0$, its quantum geometric tensor reduces to the covariance matrix

$$Q_{ab} = \frac{1}{\hbar^2} (\langle A_a A_b \rangle - \langle A_a \rangle \langle A_b \rangle). \quad (17)$$

The antisymmetric imaginary part vanishes:

$$\boxed{\Omega_{ab}^{\text{Berry}} = 0} \quad (18)$$

whereas the real part is positive semidefinite:

$$\boxed{g_{ab}^{\text{FS}} = \frac{1}{\hbar^2} \text{Cov}(A_a, A_b) \geq 0} \quad (19)$$

Accordingly, the quantum information geometry generated by the exact response cannot simply be identified with the Lorentzian generator metric or with a curved Lorentz-signature momentum space. This removes a potential but incorrect route from response fluctuations to momentum-space curvature.

The contrast with ordinary canonical quantization is instructive. A Weyl system does not preserve ordinary additive composition exactly. Instead its unitary translations form a projective representation,

$$W(z)W(z') = \exp\left[\frac{i}{2\hbar}\Omega(z, z')\right] W(z + z'), \quad (20)$$

and the nontrivial two-cocycle encodes the symplectic structure responsible for the Heisenberg commutation relations. The exact operational response considered here contains no such cocycle. Classical response additivity and quantum unitarity therefore do not, by themselves, reproduce the canonical mechanism of quantumness.

The resulting lesson is constructive rather than purely negative. Any genuinely noncommutative quantum extension of the operationally reconstructed spacetime geometry must change at least one of the assumptions leading to Eq. (10). Possible directions include projective rather than exact response composition, noncommuting inter-event transport, finite nonlinear generator calibration, or the introduction of an additional response sector canonically conjugate to the exact soldering data. The present

paper does not select among these alternatives. Its purpose is to establish the boundary that any such extension must cross.

The logical structure may be summarized as

$$\begin{array}{l}
\Theta \implies e, g, \\
\Theta \implies \lambda, \Omega, \\
\Theta \xrightarrow{\text{exact unitary response}} W, \\
\text{double exact additivity} \implies \mathcal{A}_W \text{ commutative,} \\
\mathcal{A}_W \text{ commutative} \implies \text{spectral classical-random representation,} \\
\implies \text{no intrinsic quantum geometry from } W \text{ alone.}
\end{array} \tag{21}$$

The remainder of the paper is organized as follows. Section II reviews only the operational response structures required for the present argument and derives the Liouville and symplectic reconstruction. Section III formulates the exact unitary response axioms. Section IV proves the commutativity theorem. Section V develops the joint spectral representation and the induced classical stochastic interpretation of the response algebra. Section VI studies the associated quantum geometric tensor and derives the vanishing-Berry-curvature and positive-information-metric corollaries. Section VII compares exact response composition with projective Weyl composition and identifies the structural ingredient missing from a genuinely noncommutative quantum extension. Section VIII discusses the scope of the no-go theorem and its implications for future quantum extensions of motion–staticity geometry.

2 Operational Response and Canonical Cotangent Geometry

The purpose of this section is to isolate the classical geometric content of the operational generator–displacement response that will be needed for the quantum analysis below. The soldering and Lorentzian metric reconstruction have been established in the preceding MS construction. Here we show that the same response also determines the canonical Liouville structure on the corresponding cotangent phase space.

The result does not introduce a new symplectic structure. Rather, it identifies the standard tautological one-form on T^*M as the natural phase-space realization of the same operational pairing that reconstructs the spacetime coframe. This provides the classical geometric interface between operational soldering and the unitary response considered in Sec. 3.

2.1 Generator–displacement response and soldering

Let M be a smooth four-dimensional spacetime manifold, and let

$$V \longrightarrow M$$

be a rank-four real vector bundle equipped fiberwise with a Lorentzian bilinear form

$$\eta_x : V_x \times V_x \longrightarrow \mathbb{R}$$

of signature

$$(-, +, +, +).$$

Its dual bundle is denoted by V^* .

At each point $x \in M$, let

$$\Sigma_{Q,x}^+ = \{P \in V_x^* : \eta_x^*(P, P) = -Q^2, P \text{ future}\}, \quad Q > 0, \quad (25)$$

be the positive massive generator shell. The operational response is a real-valued map

$$\Theta_x : \Sigma_{Q,x}^+ \times T_x M \longrightarrow \mathbb{R}, \quad (P, \xi) \longmapsto \Theta_x(P, \xi). \quad (26)$$

We assume the response satisfies the following properties.

1. For fixed ξ , the response is compatible with additive and positively homogeneous composition of physically realizable generators.
2. For fixed P , the response is additive and continuous in infinitesimal displacements.
3. The response varies smoothly with x .
4. The response is locally faithful:

$$\Theta_x(P, \xi) = 0 \quad \text{for all } P \in \Sigma_{Q,x}^+ \implies \xi = 0. \quad (27)$$

Since the positive massive shell spans V_x^* , these assumptions extend the response uniquely to a bilinear pairing between V_x^* and $T_x M$. Consequently there exists a unique linear map

$$e_x : T_x M \longrightarrow V_x \quad (28)$$

such that

$$\Theta_x(P, \xi) = P(e_x \xi) \quad (29)$$

for all $P \in V_x^*$ and $\xi \in T_x M$.

Faithfulness implies

$$\ker e_x = \{0\}.$$

Since

$$\dim T_x M = \dim V_x = 4,$$

the map e_x is an isomorphism. The collection of the e_x therefore defines a nondegenerate solder form

$$e \in \Omega^1(M, V).$$

In a local internal frame $\{E_a\}$ with dual frame $\{E^a\}$, write

$$e = e^a E_a, \quad e^a = e^a_\mu dx^\mu. \quad (30)$$

If

$$P = P_a E^a, \quad (31)$$

then Eq. (29) becomes

$$\Theta_x(P, \xi) = P_a e^a_\mu \xi^\mu. \quad (32)$$

The reconstructed spacetime metric is

$$g = e^* \eta, \quad (33)$$

or in components,

$$g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu. \quad (34)$$

Thus the same response pairing that associates translation generators with infinitesimal displacements also determines the Lorentzian tangent geometry.

2.2 Physical cotangent generators

The solder map induces a dual isomorphism

$$e_x^* : V_x^* \longrightarrow T_x^* M. \quad (35)$$

For an internal generator $P \in V_x^*$, define the corresponding physical spacetime covector by

$$p = e_x^* P. \quad (36)$$

By definition,

$$p(\xi) = P(e_x \xi) = \Theta_x(P, \xi). \quad (37)$$

Thus the operational response itself is the evaluation of the physical cotangent generator on the displacement.

In local coordinates,

$$p_\mu = P_a e^a{}_\mu. \quad (38)$$

The internal massive shell is thereby transferred to the standard spacetime cotangent mass shell. Indeed,

$$g^{-1}(p, p) = \eta^*(P, P). \quad (39)$$

Hence

$$\eta^*(P, P) = -Q^2 \quad \Longleftrightarrow \quad g^{-1}(p, p) = -Q^2. \quad (40)$$

The radial invariant of the internal generator geometry is therefore unchanged by soldering. What changes is its realization: internally it is a Lorentzian norm on V_x^* , whereas physically it becomes the cotangent mass shell determined by the reconstructed spacetime metric.

2.3 The response as the Liouville one-form

Let

$$\pi : T^* M \longrightarrow M$$

be the canonical cotangent projection. Recall that the tautological, or Liouville, one-form

$$\lambda \in \Omega^1(T^* M)$$

is defined at a point $(x, p) \in T^* M$ by

$$\lambda_{(x,p)}(\Xi) = p(\pi_* \Xi), \quad \Xi \in T_{(x,p)} T^* M. \quad (41)$$

We now evaluate this canonical form on the response-defined cotangent point

$$p = e_x^* P.$$

Proposition 2.1 (Operational realization of the Liouville form). *Let the operational response satisfy Eq. (29), and let*

$$p = e_x^* P.$$

Then, for every

$$\Xi \in T_{(x,p)} T^* M,$$

the tautological one-form satisfies

$$\lambda_{(x,e_x^* P)}(\Xi) = \Theta_x(P, \pi_* \Xi). \quad (42)$$

Proof. By the definition of the tautological one-form,

$$\begin{aligned}\lambda_{(x, e_x^* P)}(\Xi) &= (e_x^* P)(\pi_* \Xi) \\ &= P(e_x \pi_* \Xi).\end{aligned}\tag{43}$$

Using the defining response relation

$$\Theta_x(P, \xi) = P(e_x \xi)$$

with

$$\xi = \pi_* \Xi$$

gives Eq. (42). $\square$

In a local Lorentz frame, Eq. (42) takes the familiar form

$$\lambda = p_\mu dx^\mu = P_a e^a.\tag{44}$$

Equation (44) is standard as a local expression for the cotangent Liouville form once a coframe has been specified. Its significance here is logical rather than formal: the coframe appearing in it is not introduced independently but is the unique solder map reconstructed from the operational response.

Accordingly, the response determines two different bilinear structures through two different constructions:

$$g = e^* \eta,\tag{45}$$

$$\lambda = P_a e^a.\tag{46}$$

The first uses the internal Lorentz quadratic form, while the second uses the canonical pairing between generators and displacements.

2.4 Canonical symplectic structure

The canonical symplectic form on T^*M is obtained by exterior differentiation:

$$\Omega = d\lambda.\tag{47}$$

With the convention

$$\lambda = p_\mu dx^\mu,$$

this gives

$$\Omega = dp_\mu \wedge dx^\mu.\tag{48}$$

Using Eq. (44),

$$\Omega = d(P_a e^a).\tag{49}$$

Introduce a Lorentz connection

$$\omega^a_b$$

and define

$$DP_a = dP_a - \omega^b_a P_b,\tag{50}$$

$$T^a = De^a = de^a + \omega^a_b \wedge e^b.\tag{51}$$

A direct calculation yields

$$\Omega = DP_a \wedge e^a + P_a T^a. \quad (52)$$

To verify this identity, note that

$$\begin{aligned} DP_a \wedge e^a + P_a T^a &= (dP_a - \omega^b_a P_b) \wedge e^a \\ &\quad + P_a (de^a + \omega^a_b \wedge e^b). \end{aligned} \quad (53)$$

The connection terms cancel after relabeling the internal indices, leaving

$$dP_a \wedge e^a + P_a de^a = d(P_a e^a).$$

Equation (52) separates two different contributions to the canonical two-form. The first is the covariant generator–solder variation,

$$DP_a \wedge e^a,$$

whereas the second measures the coupling of the generator to torsion,

$$P_a T^a.$$

In the torsion-free branch,

$$T^a = 0, \quad (54)$$

and therefore

$$\Omega = DP_a \wedge e^a. \quad (55)$$

The canonical symplectic structure does not require the torsion-free condition; Eq. (52) is valid for an arbitrary Lorentz connection. The torsion-free expression is merely its specialization to the minimal metric branch considered in the classical gravitational reconstruction.

2.5 Local Lorentz covariance

The construction is invariant under local Lorentz changes of internal frame. Let

$$\Lambda(x) \in SO^+(1, 3).$$

The solder form transforms as

$$e \mapsto e' = \Lambda e, \quad (56)$$

while an internal covector transforms dually,

$$P \mapsto P' = \Lambda^{-T} P. \quad (57)$$

Therefore

$$P'_a e'^a = P_a e^a. \quad (58)$$

Consequently,

$$\lambda' = \lambda, \quad (59)$$

$$\Omega' = \Omega. \quad (60)$$

Likewise,

$$g = e^* \eta$$

is invariant because Λ preserves the internal Lorentz form.

Hence the metric and symplectic branches reconstructed from the response are simultaneously independent of the internal Lorentz representative:

$$(e, P) \sim (\Lambda e, \Lambda^{-T} P) \implies (g, \lambda, \Omega) \text{ unchanged.} \quad (61)$$

This distinction will be important below. The internal coframe and generator components are representation-dependent quantities, whereas the physical cotangent covector, the Liouville form, and the canonical symplectic structure are gauge-invariant geometric objects.

2.6 Mass-shell restriction and presymplectic geometry

The full cotangent bundle T^*M is symplectic, but a relativistic massive particle is constrained to the hypersurface

$$C_Q = \{(x, p) \in T^*M : g^{-1}(p, p) + Q^2 = 0, \text{ } p \text{ future}\}. \quad (62)$$

Let

$$\iota_Q : C_Q \hookrightarrow T^*M$$

denote the inclusion. The pullback

$$\Omega_Q = \iota_Q^* \Omega \quad (63)$$

is in general presymplectic rather than symplectic. Its characteristic direction is generated by the Hamiltonian flow of the mass-shell constraint

$$\mathcal{H}(x, p) = \frac{1}{2} (g^{-1}(p, p) + Q^2). \quad (64)$$

Accordingly, the physical particle phase space is obtained only after quotienting the constrained hypersurface by the reparametrization flow, at least locally where the quotient is regular:

$$\Gamma_Q \simeq C_Q / \text{flow}(X_{\mathcal{H}}). \quad (65)$$

This standard constrained-Hamiltonian structure is included here to avoid a possible conflation. The operational response reconstructs the canonical cotangent symplectic geometry before imposition of the relativistic mass-shell constraint. The on-shell particle phase space is a reduced symplectic space obtained from that canonical structure.

2.7 Metric–symplectic bifurcation of the operational response

The preceding results may be summarized as follows.

Theorem 2.2 (Metric–symplectic reconstruction from operational response). *Assume that the generator–displacement response is smooth, additive, homogeneous, and faithful in the sense required for the nondegenerate soldering reconstruction. Then:*

1. *there exists a unique solder isomorphism*

$$e : TM \longrightarrow V$$

satisfying

$$\Theta(P, \xi) = P(e\xi);$$

2. *the internal Lorentz form induces the spacetime metric*

$$g = e^* \eta;$$

3. *the dual solder map identifies the internal generator with the physical cotangent covector*

$$p = e^* P;$$

4. *the tautological one-form satisfies*

$$\lambda_{(x, e^* P)}(\Xi) = \Theta_x(P, \pi_* \Xi);$$

5. *the canonical cotangent symplectic form is*

$$\Omega = d\lambda = d(P_a e^a).$$

The theorem exhibits a bifurcation of the classical response geometry:

$$\begin{array}{l} \Theta \longrightarrow e \longrightarrow g = e^* \eta, \\ \Theta \longrightarrow p = e^* P \longrightarrow \lambda = P_a e^a \longrightarrow \Omega = d\lambda. \end{array} \quad (66)$$

The two branches should not be identified. The Lorentzian metric is obtained by pulling back the internal quadratic form η , while the symplectic structure is obtained by differentiating the canonical generator–displacement pairing. Their common origin is the soldering response, not an identification of metric and symplectic tensors.

This distinction will also be essential in the quantum discussion. A Berry curvature, a quantum information metric, the spacetime Lorentzian metric, and the canonical symplectic form are mathematically different objects and need not encode the same geometry.

2.8 From classical response to unitary response

The classical response

$$\Theta(P, \xi) = P(e\xi)$$

has the dimensions of action when P is interpreted as a translation generator and ξ as a physical displacement. It therefore defines a natural phase

$$\chi(P, \xi) = \exp \left[-\frac{i}{\hbar} \Theta(P, \xi) \right]. \quad (67)$$

For a fixed classical solder map,

$$\chi(P, \xi) = \exp \left[-\frac{i}{\hbar} P(e\xi) \right]. \quad (68)$$

Because Θ is bilinear, this phase satisfies

$$\chi(P + Q, \xi) = \chi(P, \xi)\chi(Q, \xi), \quad (69)$$

$$\chi(P, \xi + \zeta) = \chi(P, \xi)\chi(P, \zeta). \quad (70)$$

Thus the classical response phase is a continuous bicharacter of the generator and displacement groups.

The central question of the remainder of this paper is whether these two composition laws may be promoted simultaneously to exact operator identities in a Hilbert-space representation while still allowing an intrinsically noncommutative quantum geometry. We therefore replace the scalar character χ by a strongly continuous unitary family

$$W(P, \xi)$$

and examine the consequences of retaining Eqs. (69) and (70) exactly at the operator level.

As shown in the next section, this apparently conservative quantization assumption is already strong enough to determine the algebraic character of the resulting response theory.

3 Exact Unitary Operational Response

We now pass from the classical generator–displacement response to a Hilbert-space realization. The purpose of this section is not to assume a particular quantization of the solder form, nor to introduce a noncommutative spacetime algebra. Instead, we ask for the most direct operator analogue of the classical response composition laws derived in Sec. 2.

The central assumption is deliberately conservative: the additivity of the classical response is preserved exactly after exponentiation. We shall see in the next section that this assumption is already sufficiently strong to determine the algebraic character of the resulting unitary response family.

Throughout this section the spacetime point x is fixed unless otherwise indicated, and the subscript x is suppressed. All statements are therefore fiberwise. Smooth dependence on x may be restored after the local algebraic structure has been determined.

3.1 From the massive response cone to the additive generator space

The classical response is physically introduced on positive massive generators. For a fixed mass scale $Q > 0$, let

$$\Sigma_Q^+ \subset V^*$$

denote the future massive shell,

$$\Sigma_Q^+ = \{P \in V^* : \eta^*(P, P) = -Q^2, P \text{ future}\}. \quad (71)$$

The fixed shell itself is not closed under addition. The natural additive domain associated with the physically positive massive sector is therefore its positive cone,

$$C^+ = \text{cone}(\Sigma_Q^+). \quad (72)$$

For four-dimensional Lorentzian generator space this is the open future timelike cone together with the origin, with the appropriate closure under positive scaling understood when needed.

The cone satisfies

$$P, Q \in C^+ \implies P + Q \in C^+. \quad (73)$$

Moreover, it generates the full real vector space V^* : every $R \in V^*$ can be written as

$$R = P - Q \quad (74)$$

for some

$$P, Q \in C^+.$$

This distinction between the physical cone and its additive group completion will be important below. The operational response is initially required only on physically positive generators, but exact unitary additivity permits a unique extension to the vector group V^* .

3.2 Definition of an exact unitary response

Let $\mathcal{H}$ be a complex Hilbert space. An exact unitary operational response is a family

$$W(P, \xi) \in \mathcal{U}(\mathcal{H}), \quad P \in C^+, \quad \xi \in T_x M, \quad (75)$$

satisfying the following conditions.

Definition 3.1 (Exact unitary operational response). *A family $W(P, \xi)$ is called an exact unitary operational response if:*

1. unitarity:

$$W(P, \xi)^\dagger = W(P, \xi)^{-1}; \quad (76)$$

2. identity normalization:

$$W(0, \xi) = I, \quad (77)$$

$$W(P, 0) = I; \quad (78)$$

3. exact generator composition:

$$W(P + Q, \xi) = W(P, \xi)W(Q, \xi), \quad P, Q \in C^+; \quad (79)$$

4. exact displacement composition:

$$W(P, \xi + \zeta) = W(P, \xi)W(P, \zeta); \quad (80)$$

5. strong continuity: for every $\psi \in \mathcal{H}$, the map

$$(P, \xi) \mapsto W(P, \xi)\psi \quad (81)$$

is continuous on its domain.

The two composition laws are the direct operator analogues of the classical relations

$$\Theta(P + Q, \xi) = \Theta(P, \xi) + \Theta(Q, \xi), \quad (82)$$

$$\Theta(P, \xi + \zeta) = \Theta(P, \xi) + \Theta(P, \zeta). \quad (83)$$

No commutation relation between responses with different generator and displacement labels is assumed in Definition 3.1. In particular, we do not assume

$$[W(P, \xi), W(Q, \zeta)] = 0.$$

Whether such commutativity follows from the composition laws is the question addressed in Sec. 4.

3.3 Immediate consequences of the separate composition laws

Before combining the two exact composition laws, it is useful to record the consequences of each one separately.

For fixed P , Eq. (80) gives

$$W(P, \xi + \zeta) = W(P, \xi)W(P, \zeta). \quad (84)$$

Since

$$\xi + \zeta = \zeta + \xi,$$

one also has

$$W(P, \xi)W(P, \zeta) = W(P, \zeta)W(P, \xi). \quad (85)$$

Thus for each fixed generator P , the displacement response forms a strongly continuous unitary representation of the additive tangent group.

Furthermore,

$$W(P, -\xi) = W(P, \xi)^{-1} = W(P, \xi)^\dagger. \quad (86)$$

Similarly, for fixed ξ , Eq. (79) implies

$$W(P + Q, \xi) = W(P, \xi)W(Q, \xi). \quad (87)$$

Since

$$P + Q = Q + P,$$

it follows that

$$W(P, \xi)W(Q, \xi) = W(Q, \xi)W(P, \xi) \quad (88)$$

for all

$$P, Q \in C^+.$$

Thus each argument separately already generates an Abelian unitary family. What is not yet established is the cross commutativity of responses carrying simultaneously different generator and displacement labels.

3.4 Unitary group completion of the positive generator response

For the spectral analysis developed later, it is convenient to extend the generator variable from the positive cone

$$C^+$$

to the full additive vector group V^* .

Let

$$R \in V^*$$

and choose a decomposition

$$R = P - Q, \quad P, Q \in C^+. \quad (89)$$

Define

$$\widetilde{W}(R, \xi) = W(P, \xi)W(Q, \xi)^{-1}. \quad (90)$$

We first verify that this definition is independent of the chosen decomposition.

Proposition 3.2 (Generator group completion). *Let W satisfy unitarity and exact generator composition on C^+ . Then Eq. (90) defines a unique unitary extension*

$$\widetilde{W} : V^* \times T_x M \longrightarrow \mathcal{U}(\mathcal{H})$$

satisfying

$$\widetilde{W}(R + S, \xi) = \widetilde{W}(R, \xi) \widetilde{W}(S, \xi). \quad (91)$$

Proof. Suppose

$$R = P_1 - Q_1 = P_2 - Q_2, \quad (92)$$

with all $P_i, Q_i \in C^+$. Then

$$P_1 + Q_2 = P_2 + Q_1.$$

Exact generator composition gives

$$W(P_1, \xi) W(Q_2, \xi) = W(P_2, \xi) W(Q_1, \xi). \quad (93)$$

All four factors commute for fixed ξ by Eq. (88). Multiplying by the relevant unitary inverses therefore yields

$$W(P_1, \xi) W(Q_1, \xi)^{-1} = W(P_2, \xi) W(Q_2, \xi)^{-1}. \quad (94)$$

Hence the extension is well defined.

For

$$R = P - Q, \quad S = P' - Q',$$

one has

$$R + S = (P + P') - (Q + Q').$$

Consequently,

$$\begin{aligned} \widetilde{W}(R + S, \xi) &= W(P + P', \xi) W(Q + Q', \xi)^{-1} \\ &= W(P, \xi) W(P', \xi) W(Q', \xi)^{-1} W(Q, \xi)^{-1} \\ &= \widetilde{W}(R, \xi) \widetilde{W}(S, \xi). \end{aligned} \quad (95)$$

The extension is unique because C^+ generates V^* . □

From now on we suppress the tilde and write

$$W(P, \xi)$$

for the group-completed response unless the distinction between the physical positive cone and its additive completion is relevant.

Thus, for each fixed displacement,

$$P \longmapsto W(P, \xi)$$

is a strongly continuous unitary representation of the real additive group V^* .

3.5 Infinitesimal generators for fixed response arguments

Strong continuity permits the use of Stone's theorem separately in each argument. For fixed P , consider the one-parameter displacement

$$\xi = tX, \quad X \in T_x M.$$

Then

$$U_{P,X}(t) := W(P, tX) \tag{96}$$

is a strongly continuous one-parameter unitary group. Stone's theorem therefore gives a self-adjoint generator

$$G(P, X)$$

such that

$$W(P, tX) = \exp \left[-\frac{i}{\hbar} t G(P, X) \right]. \tag{97}$$

Equivalently, after choosing a tangent basis

$$\{e_\mu\},$$

one may write

$$W(P, \xi) = \exp \left[-\frac{i}{\hbar} \xi^\mu G_\mu(P) \right] \tag{98}$$

within the joint spectral representation of the commuting displacement generators associated with fixed P .

At this stage the operators

$$G_\mu(P)$$

need not yet be assumed linear in P , and no operator-valued coframe is introduced. The dependence of $G_\mu(P)$ on the generator label will instead be constrained by exact generator composition.

Similarly, for fixed ξ and a generator direction $A \in V^*$,

$$t \mapsto W(tA, \xi) \tag{99}$$

is a strongly continuous one-parameter unitary group after generator group completion. It therefore possesses a self-adjoint infinitesimal generator.

The existence of these generators should not be confused with the existence of a noncommutative spacetime or tetrad algebra. Stone's theorem determines the infinitesimal generators of individual one-parameter unitary representations; the commutation relations among generators associated with different response directions remain to be determined from the full two-argument composition law.

3.6 Classical correspondence and the role of $\hbar$

For a sharp classical solder map e , the unitary response reduces to the scalar character

$$W_e(P, \xi) = \exp \left[-\frac{i}{\hbar} P(e\xi) \right]. \tag{100}$$

More generally, one may require that a family of states

$$\omega_\hbar$$

have a semiclassical response satisfying

$$\omega_{\hbar}(W(P, \xi)) \longrightarrow \exp \left[-\frac{i}{\hbar} \Theta(P, \xi) \right] \quad (101)$$

in an appropriate semiclassical regime. The precise form of such a limit is not needed for the commutativity theorem below.

It is important to separate two logically distinct facts. The factor

$$\hbar^{-1}$$

converts an action-valued response into a dimensionless phase. By itself, however, it imposes no noncommutative operator algebra. Whether the resulting response is intrinsically quantum depends on the composition law of the unitary operators, not merely on the occurrence of a complex phase.

Thus the present construction deliberately refrains from identifying

$$\exp[-i\Theta/\hbar]$$

with a sufficient criterion for quantum geometry.

3.7 The exact response algebra

Let

$$\mathcal{A}_W$$

denote the unital C^* -algebra generated by the complete exact response family:

$$\mathcal{A}_W = C^* (\{W(P, \xi) : P \in V^*, \xi \in T_x M\}). \quad (102)$$

At this point no assumption is made that

$$\mathcal{A}_W$$

is commutative.

The distinction between the full Hilbert-space observable algebra

$$\mathcal{B}(\mathcal{H})$$

and the response algebra

$$\mathcal{A}_W$$

will be essential. Even if the underlying theory contains noncommuting observables outside $\mathcal{A}_W$, the question posed here concerns only the geometry detectable by the exact generator–displacement responses themselves.

Accordingly, any no-go result obtained for $\mathcal{A}_W$ will have a restricted but precise meaning: it constrains the quantum geometry carried by the operational response sector, not every possible observable in an enlarged quantum theory.

3.8 Exactness versus projective composition

Definition 3.1 should also be distinguished from a projective unitary representation.

A projective composition law in the generator argument would have the form

$$W(P + Q, \xi) = e^{i\alpha(P, Q; \xi)} W(P, \xi) W(Q, \xi), \quad (103)$$

while a projective displacement law would take the form

$$W(P, \xi + \zeta) = e^{i\beta(P; \xi, \zeta)} W(P, \xi) W(P, \zeta). \quad (104)$$

The functions

$$\alpha \quad \text{and} \quad \beta$$

would be constrained by associativity through cocycle conditions. Such phases can carry nontrivial algebraic information even when the underlying label groups are Abelian.

The present section sets

$$\alpha = 0, \quad \beta = 0. \quad (105)$$

This is the precise meaning of *exact* response composition.

The distinction is conceptually important. In the canonical Weyl system, noncommutativity is encoded precisely by a nontrivial projective phase associated with the symplectic form. We do not introduce such a phase here because the purpose of the present analysis is to determine what follows from preserving the classical operational composition laws without modification.

3.9 Scope of the exact-response assumptions

The assumptions introduced above are intentionally stronger than what a general quantum theory need satisfy. Their role is to define a sharply testable quantization problem.

In particular, the analysis does not assume:

1. a fundamental operator-valued metric

$$\hat{g}_{\mu\nu};$$

2. a fundamental operator-valued tetrad

$$\hat{e}^a_{\mu};$$

3. noncommuting spacetime coordinates;
4. a curved momentum manifold;
5. a canonical Heisenberg algebra for the response variables;
6. a path-integral measure over geometries;
7. a quantum gravitational field equation.

The only quantum input at this stage is a Hilbert space, unitarity, strong continuity, and exact preservation of the two classical composition laws.

This minimality makes the result of the next section particularly restrictive. If the exact response algebra turns out to be commutative, the conclusion cannot be attributed to a special choice of tetrad representation or to a particular gravitational Hamiltonian. It will follow from the exact composition structure itself.

3.10 Algebraic question

We are now in a position to state the central algebraic problem.

Given

$$W(P + Q, \xi) = W(P, \xi)W(Q, \xi), \quad (106)$$

$$W(P, \xi + \zeta) = W(P, \xi)W(P, \zeta), \quad (107)$$

does there exist an exact unitary response family for which

$$[W(P, \xi), W(Q, \zeta)] \neq 0 \quad (108)$$

for some choices of the labels?

The answer is negative.

The reason is not contained in either Eq. (106) or Eq. (107) separately. It arises from the compatibility of the two laws when the same response

$$W(P + Q, \xi + \zeta)$$

is decomposed in the two possible orders.

The next section develops this argument and proves that every pair of exact response operators commutes. The resulting commutativity theorem will then provide the starting point for the joint spectral representation and the classical stochastic interpretation of the response sector.

4 Commutativity of the Exact Response Algebra

We now prove the central algebraic result of the paper. The two exact composition laws introduced in Sec. 3 are individually compatible with ordinary Abelian parameter groups. The nontrivial question is whether their simultaneous implementation can nevertheless produce noncommuting response operators when both the generator and displacement labels are varied.

The answer is negative. Compatibility of exact generator composition with exact displacement composition forces every pair of response operators to commute.

The argument is elementary, but its consequence is structural. It does not depend on a chosen representation of a tetrad operator, on a semiclassical expansion, or on a gravitational field equation. It follows directly from the requirement that the same two-argument response may be composed exactly in either order.

4.1 Compatibility of the two exact composition laws

Recall the exact response laws

$$W(P + Q, \xi) = W(P, \xi)W(Q, \xi), \quad (109)$$

$$W(P, \xi + \zeta) = W(P, \xi)W(P, \zeta). \quad (110)$$

Consider the response

$$W(P + Q, \xi + \zeta). \quad (111)$$

There are two natural ways to reduce this expression using the exact composition laws.

First compose in the generator argument. Equation (109) gives

$$W(P + Q, \xi + \zeta) = W(P, \xi + \zeta)W(Q, \xi + \zeta). \quad (112)$$

Applying displacement composition separately to the two factors yields

$$W(P + Q, \xi + \zeta) = W(P, \xi)W(P, \zeta)W(Q, \xi)W(Q, \zeta). \quad (113)$$

Alternatively, compose first in the displacement argument:

$$W(P + Q, \xi + \zeta) = W(P + Q, \xi)W(P + Q, \zeta). \quad (114)$$

Applying generator composition to each factor gives

$$W(P + Q, \xi + \zeta) = W(P, \xi)W(Q, \xi)W(P, \zeta)W(Q, \zeta). \quad (115)$$

Equating Eqs. (113) and (115), we obtain

$$\begin{aligned} & W(P, \xi)W(P, \zeta)W(Q, \xi)W(Q, \zeta) \\ &= W(P, \xi)W(Q, \xi)W(P, \zeta)W(Q, \zeta). \end{aligned} \quad (116)$$

Since every response operator is unitary, the leftmost factor $W(P, \xi)$ and the rightmost factor $W(Q, \zeta)$ may be cancelled without assuming any additional commutativity. This gives

$$W(P, \zeta)W(Q, \xi) = W(Q, \xi)W(P, \zeta). \quad (117)$$

The labels in Eq. (117) are arbitrary. Renaming them immediately gives the full pairwise commutativity relation.

4.2 Exact-response commutativity theorem

Theorem 4.1 (Exact-response commutativity). *Let*

$$W(P, \xi)$$

be an exact unitary operational response satisfying Definition 3.1. Then, for every

$$P, Q \in C^+$$

and every

$$\xi, \zeta \in T_x M,$$

one has

$$[W(P, \xi), W(Q, \zeta)] = 0. \quad (118)$$

After the generator group completion of Proposition 3.2, the same relation holds for all

$$P, Q \in V^*.$$

Proof. For positive generators, the result follows directly from Eq. (117), since the labels P, Q, ξ, ζ are arbitrary.

Now let

$$R = P_1 - P_2, \quad S = Q_1 - Q_2,$$

with

$$P_1, P_2, Q_1, Q_2 \in C^+.$$

The group-completed response is

$$W(R, \xi) = W(P_1, \xi)W(P_2, \xi)^{-1}, \quad (119)$$

$$W(S, \zeta) = W(Q_1, \zeta)W(Q_2, \zeta)^{-1}. \quad (120)$$

Every factor in the first expression commutes with every factor in the second by the positive-cone result. Their inverses commute as well. Hence

$$W(R, \xi)W(S, \zeta) = W(S, \zeta)W(R, \xi),$$

which establishes Eq. (118) on the full additive generator space. $\square$

4.3 Abelianness of the response algebra

The theorem immediately determines the C^* -algebra generated by the exact responses.

Corollary 4.2 (Commutative response algebra). *The unital C^* -algebra*

$$\mathcal{A}_W = C^* (\{W(P, \xi) : P \in V^*, \xi \in T_x M\})$$

is commutative.

Proof. The generating family is pairwise commuting by Theorem 4.1. The algebra generated by finite linear combinations and products of these operators is therefore commutative. Taking the norm closure preserves commutativity. $\square$

Thus

$$\mathcal{A}_W \subset \mathcal{B}(\mathcal{H}) \quad (121)$$

is an Abelian subalgebra even when the complete Hilbert-space observable algebra

$$\mathcal{B}(\mathcal{H})$$

is highly noncommutative.

This distinction is essential. The theorem does not imply that every observable of an enlarged quantum theory commutes. It implies only that the geometry accessible through the exact operational responses $W(P, \xi)$ belongs to a commutative observable sector.

4.4 A rectangular composition identity

Once pairwise commutativity has been established, the two exact composition laws may be combined without ordering ambiguities. In particular,

$$W(P + Q, \xi + \zeta) = W(P, \xi)W(P, \zeta)W(Q, \xi)W(Q, \zeta), \quad (122)$$

and the four factors on the right-hand side may be reordered arbitrarily.

This relation has a useful interpretation. The value of the response at the corner

$$(P + Q, \xi + \zeta)$$

is determined by the four pairings

$$(P, \xi), \quad (P, \zeta), \quad (Q, \xi), \quad (Q, \zeta),$$

without any ordering-dependent correction.

For a scalar classical bicharacter,

$$\chi(P, \xi) = \exp \left[-\frac{i}{\hbar} P(e\xi) \right],$$

this property is automatic because the phase exponent is bilinear. The exact unitary theorem shows that promoting the scalar response to operators while retaining both exact composition laws introduces no new ordering structure: the operator family remains algebraically equivalent to an Abelian bicharacter-valued response.

4.5 Vanishing mixed group commutators

The commutativity theorem may equivalently be expressed through group commutators. Define

$$\mathcal{K}(P, Q; \xi, \zeta) = W(P, \xi)W(Q, \zeta)W(P, \xi)^{-1}W(Q, \zeta)^{-1}. \quad (123)$$

Then

$$\mathcal{K}(P, Q; \xi, \zeta) = I. \quad (124)$$

No infinitesimal approximation is involved in Eq. (124). The result holds at finite values of the generator and displacement labels wherever the exact response laws are defined.

This observation will later distinguish the present response from a canonical Weyl system. In a projective Weyl representation, the corresponding group commutator is generally a nontrivial central phase controlled by the symplectic form. Here it is identically trivial.

4.6 No nontrivial central commutator from exact composition

One might attempt to weaken the desired noncommutativity by asking only for a central relation of the form

$$W(P, \xi)W(Q, \zeta) = e^{i\Gamma(P, Q; \xi, \zeta)} W(Q, \zeta)W(P, \xi). \quad (125)$$

Theorem 4.1 immediately implies

$$e^{i\Gamma(P, Q; \xi, \zeta)} = 1 \quad (126)$$

for all labels.

Thus exact double additivity excludes not only ordinary noncommutativity but also a nontrivial central commutator within the response sector.

In particular, no Heisenberg-type phase can emerge while Eqs. (109) and (110) remain exact operator identities.

4.7 Infinitesimal consequence

The finite commutativity result also constrains the infinitesimal generators introduced in Sec. 3.5. For fixed generator labels P, Q and tangent directions X, Y , define

$$U_{P,X}(s) = W(P, sX), \quad (127)$$

$$U_{Q,Y}(t) = W(Q, tY). \quad (128)$$

Theorem 4.1 gives

$$U_{P,X}(s)U_{Q,Y}(t) = U_{Q,Y}(t)U_{P,X}(s) \quad (129)$$

for all $s, t \in \mathbb{R}$.

Hence the corresponding one-parameter unitary groups commute strongly. Their spectral measures therefore commute. If

$$G(P, X) \quad \text{and} \quad G(Q, Y)$$

denote their Stone generators, these self-adjoint operators are strongly commuting in the spectral sense.

Where an ordinary operator commutator is defined on a common invariant core, one consequently has

$$[G(P, X), G(Q, Y)] = 0. \quad (130)$$

The spectral formulation is the more precise statement because the infinitesimal generators need not be bounded. The important point is that the finite unitary response already supplies the stronger commutation relation from which the joint spectral structure follows.

4.8 No tetrad commutator is assumed

It is useful to emphasize the logical direction of the argument.

We have not postulated operators

$$\hat{e}^a_{\mu}$$

and then imposed

$$[\hat{e}^a_{\mu}, \hat{e}^b_{\nu}] = 0.$$

Indeed, no operator-valued tetrad was needed to prove Theorem 4.1.

Instead, the chain of reasoning is

exact generator composition + exact displacement composition $\implies$ pairwise commutativity of W $\implies$ commutative response C^* -algebra.	(131)
--	-------

Only after this conclusion has been established will the next section use joint spectral theory to show that the commuting response may be represented in terms of a spectral family of ordinary solder maps.

Thus the eventual appearance of commuting soldering variables is a consequence of the exact operational response algebra, not an assumption about the quantum geometry.

4.9 Faithfulness and commutativity

The classical soldering reconstruction used a faithfulness condition to exclude displacements invisible to all physical generators. An analogous condition may be imposed on the unitary response.

Definition 4.3 (Quantum response faithfulness). *The exact response is called displacement-faithful if*

$$W(P, \xi) = I \quad \text{for all } P \in V^* \quad \implies \quad \xi = 0, \quad (132)$$

and generator-faithful if

$$W(P, \xi) = I \quad \text{for all } \xi \in T_x M \quad \implies \quad P = 0. \quad (133)$$

Faithfulness does not alter the commutativity theorem. It restricts the spectral support of the response representation but cannot turn the Abelian algebra into a non-Abelian one.

This point will be important in the next section. A faithful response may distinguish all physical displacements and generators while still being representable by a commutative spectral measure. Operational completeness of a response family is therefore logically distinct from intrinsic quantum noncommutativity.

4.10 Fiberwise and spacetime-local character of the theorem

The proof above was carried out at a fixed spacetime point x . Hence the theorem is local and fiberwise:

$$[W_x(P, \xi), W_x(Q, \zeta)] = 0. \quad (134)$$

No statement has yet been made about response operators associated with distinct spacetime points,

$$W_x(P, \xi) \quad \text{and} \quad W_y(Q, \zeta), \quad x \neq y.$$

Comparing such operators requires an additional transport structure between the Hilbert spaces or response algebras attached to different events.

Consequently, the present theorem does not imply

$$[W_x, W_y] = 0 \quad (135)$$

without further assumptions.

This distinction identifies one possible location at which genuinely noncommutative quantum structure may enter. Even if every local exact response algebra

$$\mathcal{A}_{W,x}$$

is Abelian, the transport or comparison of these algebras between different events need not be.

The theorem therefore excludes intrinsic noncommutativity of the local exact response sector, not every possible noncommutative structure in a quantum theory of inter-event comparison.

4.11 A no-go statement at the algebraic level

The preceding results may be collected into the following statement.

Theorem 4.4 (Algebraic no-go for noncommutative exact response). *Consider a Hilbert-space realization of the operational generator–displacement response with the following properties:*

1. *the response operators are unitary;*
2. *the response is strongly continuous;*
3. *generator composition is represented exactly;*
4. *displacement composition is represented exactly;*
5. *the quantum geometric observables under consideration are generated solely by the resulting local response family.*

Then the corresponding local response C^ -algebra is commutative. In particular, no intrinsically noncommutative local quantum geometry can be encoded in this algebra alone.*

The final qualification is essential. The theorem does not exclude an enlarged quantum theory containing additional noncommuting observables. It states that such noncommutativity cannot originate solely from an exact unitary implementation of the two classical response addition laws.

Accordingly, any noncommutative extension must relax or enlarge at least one ingredient of the construction. Possibilities include projective composition laws, noncommuting inter-event transport, finite nonlinear calibration transformations, or additional response variables not contained in the exact local soldering sector.

4.12 Comparison with canonical projective quantization

The contrast with canonical quantum mechanics may already be stated at the algebraic level.

For canonical phase-space translations, the Weyl operators obey a projective composition law of the schematic form

$$\mathcal{W}(z)\mathcal{W}(z') = \exp\left[\frac{i}{2\hbar}\Omega(z, z')\right]\mathcal{W}(z + z'), \quad (136)$$

where

$$\Omega(z, z')$$

is the classical symplectic pairing. The central phase obstructs ordinary commutativity and gives the Heisenberg algebra at the infinitesimal level.

By contrast, the exact operational response satisfies composition with no cocycle:

$$W(P + Q, \xi) = W(P, \xi)W(Q, \xi), \quad (137)$$

$$W(P, \xi + \zeta) = W(P, \xi)W(P, \zeta). \quad (138)$$

The resulting group commutator is therefore trivial.

This comparison will be developed more fully after the spectral structure of the response has been established. For the present purpose, it shows that the occurrence of the classical symplectic form

$$\Omega = d\lambda$$

in Sec. 2 does not by itself force the unitary response to realize the corresponding Weyl cocycle. The classical symplectic structure and the quantum projective composition law are logically distinct ingredients.

4.13 Transition to the spectral representation

Theorem 4.1 reduces the next stage of the problem to the spectral analysis of an Abelian unitary family.

Since all

$$W(P, \xi)$$

commute and are normal operators, they admit a joint spectral description. The two exact additivity laws further constrain the scalar functions appearing in that spectral representation: almost every spectral component must itself be a continuous character in each argument.

The next section shows that these scalar components are continuous bicharacters and hence are represented by ordinary bilinear generator–displacement pairings. This leads to a spectral representation of the form

$$W(P, \xi) = \int_{\mathcal{E}} \exp \left[-\frac{i}{\hbar} P(e\xi) \right] dE(e), \quad (139)$$

with $\mathcal{E}$ a spectral space of classical soldering maps.

The resulting interpretation is stronger than mere commutativity: states restricted to the exact response algebra will admit ordinary probability representations over $\mathcal{E}$. Establishing this statement rigorously is the subject of the following section.

5 Joint Spectral Representation and Classical Stochastic Geometry

The commutativity theorem of Sec. 4 determines the algebraic type of the exact response sector. We now develop its spectral consequences.

The central result is that every spectral component of the exact response is a continuous scalar bicharacter of the generator and displacement groups. Since both groups are finite-dimensional real vector spaces, such bicharacters are generated by ordinary real bilinear pairings. Consequently, the full operator-valued response is a spectral superposition of classical generator–displacement response maps.

This statement is stronger than pairwise commutativity. It implies that every state restricted to the exact response algebra admits an ordinary probability representation over a space of classical linear response maps. In this precise operational sense, fluctuations detectable through the exact response sector alone are classical stochastic fluctuations, even when the response acts on a Hilbert space.

5.1 The Gelfand spectrum of the response algebra

By Corollary 4.2, the unital C^* -algebra

$$\mathcal{A}_W$$

generated by the exact response operators is commutative.

Let

$$\mathfrak{X}_W = \text{Spec}(\mathcal{A}_W) \quad (140)$$

denote its Gelfand spectrum, namely the space of nonzero multiplicative linear functionals

$$\chi : \mathcal{A}_W \longrightarrow \mathbb{C}.$$

For every

$$\omega \in \mathfrak{X}_W,$$

define the scalar response

$$w_\omega(P, \xi) := \omega(W(P, \xi)). \quad (141)$$

Since $W(P, \xi)$ is unitary and a character of a unital C^* -algebra sends unitary elements to complex numbers of unit modulus,

$$w_\omega(P, \xi) \in U(1). \quad (142)$$

The exact operator composition laws descend pointwise to the Gelfand spectrum:

$$w_\omega(P + Q, \xi) = w_\omega(P, \xi)w_\omega(Q, \xi), \quad (143)$$

$$w_\omega(P, \xi + \zeta) = w_\omega(P, \xi)w_\omega(P, \zeta). \quad (144)$$

Thus every spectral point determines a scalar function that is a character in each argument separately.

5.2 Continuity of the scalar spectral responses

The operator family $W(P, \xi)$ was assumed strongly continuous. Strong continuity does not, in complete generality, imply norm continuity of a unitary representation. For the classification below, what is required is continuity of the scalar spectral characters.

Accordingly, we isolate the regular spectral sector relevant to the operational reconstruction.

Definition 5.1 (Regular spectral response). *A spectral point*

$$\omega \in \mathfrak{X}_W$$

is called regular if the scalar function

$$(P, \xi) \mapsto w_\omega(P, \xi)$$

is continuous on

$$V^* \times T_x M.$$

The set of regular spectral points will be denoted by

$$\mathfrak{X}_W^{\text{reg}}.$$

In the direct spectral realization of a strongly continuous Abelian unitary representation, the relevant measurable characters are continuous almost everywhere with respect to the associated spectral measure. The following analysis is therefore applied on the support of that regular spectral representation.

The regularity assumption excludes discontinuous algebraic characters that have no continuous classical-response interpretation.

5.3 Classification of continuous scalar bicharacters

We now classify the scalar response

$$w(P, \xi)$$

at a fixed regular spectral point.

Lemma 5.2 (Continuous bicharacter lemma). *Let V^* and T be finite-dimensional real vector spaces, and let*

$$w : V^* \times T \longrightarrow U(1)$$

be continuous and satisfy

$$w(P + Q, \xi) = w(P, \xi)w(Q, \xi), \quad (145)$$

$$w(P, \xi + \zeta) = w(P, \xi)w(P, \zeta), \quad (146)$$

$$w(0, \xi) = w(P, 0) = 1. \quad (147)$$

Then there exists a unique real bilinear form

$$b : V^* \times T \longrightarrow \mathbb{R}$$

such that

$$w(P, \xi) = \exp \left[-\frac{i}{\hbar} b(P, \xi) \right]. \quad (148)$$

Proof. Fix $P \in V^*$. The map

$$\xi \longmapsto w(P, \xi)$$

is a continuous character of the additive vector group T . Every continuous character of a finite-dimensional real vector group has the form

$$w(P, \xi) = \exp \left[-\frac{i}{\hbar} \ell_P(\xi) \right] \quad (149)$$

for a unique

$$\ell_P \in T^*.$$

Generator composition gives

$$\begin{aligned} & \exp \left[-\frac{i}{\hbar} \ell_{P+Q}(\xi) \right] \\ &= \exp \left[-\frac{i}{\hbar} (\ell_P(\xi) + \ell_Q(\xi)) \right]. \end{aligned} \quad (150)$$

The uniqueness of the infinitesimal generator of a continuous one-parameter character implies

$$\ell_{P+Q} = \ell_P + \ell_Q. \quad (151)$$

Similarly, continuity in the real scaling parameter gives

$$\ell_{tP} = t\ell_P, \quad t \in \mathbb{R}. \quad (152)$$

Hence

$$P \longmapsto \ell_P$$

is a real linear map from V^* to T^* .

Define

$$b(P, \xi) := \ell_P(\xi). \quad (153)$$

It is linear in P and linear in ξ , hence bilinear. Equation (148) follows immediately.

Uniqueness follows from the uniqueness of the continuous character generator ℓ_P . $\square$

5.4 Linear response maps associated with spectral points

A real bilinear form

$$b : V^* \times T_x M \longrightarrow \mathbb{R}$$

is canonically equivalent, in finite dimensions, to a linear map

$$e_b : T_x M \longrightarrow V$$

defined by

$$b(P, \xi) = P(e_b \xi). \quad (154)$$

Thus every regular spectral point

$$\omega \in \mathfrak{X}_W^{\text{reg}}$$

determines a unique real linear response map

$$e_\omega : T_x M \longrightarrow V \quad (155)$$

such that

$$w_\omega(P, \xi) = \exp \left[-\frac{i}{\hbar} P(e_\omega \xi) \right]. \quad (156)$$

This motivates the spectral response space

$$\mathcal{E}_x \subset \text{Hom}(T_x M, V), \quad (157)$$

defined as the collection of linear maps arising from the regular spectral support.

It is important not to assume at this stage that every

$$e \in \mathcal{E}_x$$

is nondegenerate. A linear map with

$$\det e = 0$$

still defines a perfectly valid scalar bicharacter through Eq. (156). Such a map does not, however, define a nondegenerate soldering of tangent and internal spaces.

We therefore distinguish

$$\mathcal{E}_x^{\text{nd}} = \{e \in \mathcal{E}_x : \det e \neq 0\} \quad (158)$$

from the possibly degenerate part

$$\mathcal{E}_x^{\text{deg}} = \{e \in \mathcal{E}_x : \det e = 0\}. \quad (159)$$

Only the nondegenerate sector consists of genuine classical solder maps and therefore determines ordinary Lorentzian metrics through

$$g_e = e^* \eta.$$

5.5 Spectral response theorem

We can now formulate the principal representation result.

Theorem 5.3 (Spectral representation of the exact response). *Let $W(P, \xi)$ be an exact unitary operational response satisfying the assumptions of Theorem 4.1. Assume that the associated Abelian unitary representation is supported on regular continuous bicharacters. Then there exists a projection-valued measure*

$$E$$

on a measurable spectral response space

$$\mathcal{E}_x \subset \text{Hom}(T_x M, V)$$

such that

$$W(P, \xi) = \int_{\mathcal{E}_x} \exp \left[-\frac{i}{\hbar} P(e\xi) \right] dE(e). \quad (160)$$

Proof. The exact response algebra is commutative by Corollary 4.2. The joint spectral theorem therefore represents its commuting normal generators as multiplication by scalar spectral functions on a common measure space.

At almost every regular spectral point, the scalar response satisfies the two exact character laws Eqs. (143) and (144). By Lemma 5.2, it is generated by a unique real bilinear form. Finite-dimensional duality then identifies that bilinear form with a unique

$$e \in \text{Hom}(T_x M, V).$$

Pushing the joint spectral measure forward under this identification gives the projection-valued measure E on $\mathcal{E}_x$, yielding Eq. (160). $\square$

The theorem shows that the exact quantum response is spectrally assembled from the same algebraic type of object that appeared in the classical soldering reconstruction: a linear map from displacement space to internal Lorentz space.

The difference is that a general quantum representation may have spectral support on many such maps rather than on a single sharp map.

5.6 Sharp classical geometry as a special spectral sector

A classical response geometry corresponds to a spectral measure concentrated at a single nondegenerate map

$$e_0 : T_x M \longrightarrow V.$$

If

$$dE(e) = I \delta_{e_0}(de), \quad (161)$$

then

$$W(P, \xi) = \exp \left[-\frac{i}{\hbar} P(e_0 \xi) \right] I. \quad (162)$$

This is precisely the scalar response phase associated with the classical operational soldering

$$\Theta(P, \xi) = P(e_0 \xi).$$

A nontrivial spectral measure therefore generalizes the sharp classical geometry by allowing a distribution of linear response maps, while retaining exact response composition.

5.7 Spectral nondegeneracy

The classical operational soldering theorem used faithfulness to infer that the unique response map e was nondegenerate. In the spectral theory the situation is subtler.

Define the spectral projector onto the degenerate sector by

$$\Pi_{\text{deg}} = \mathbf{E}(\mathcal{E}_x^{\text{deg}}). \quad (163)$$

We call the response spectrally nondegenerate if

$$\Pi_{\text{deg}} = 0. \quad (164)$$

Under this condition, the response is supported almost everywhere on invertible solder maps:

$$e : T_x M \xrightarrow{\sim} V. \quad (165)$$

Each spectral point then defines an ordinary Lorentzian metric

$$g_e = e^* \eta. \quad (166)$$

Quantum response faithfulness in the sense of Definition 4.3 does not by itself imply Eq. (164). It may be possible for different degenerate spectral maps collectively to distinguish all displacement directions even though no individual map is invertible.

Thus

$$\text{global response faithfulness} \not\Rightarrow \text{pointwise spectral nondegeneracy}. \quad (167)$$

If a quantum theory is intended to describe a stochastic ensemble of ordinary nondegenerate space-time geometries, spectral nondegeneracy must therefore be imposed or derived as an additional condition.

5.8 Spectral soldering operators

The projection-valued measure in Theorem 5.3 may be used to define coordinate operators associated with the spectral response maps.

Choose bases

$$\{E_a\} \subset V, \quad \{e_\mu\} \subset T_x M,$$

and write

$$e(e_\mu) = e^a_\mu E_a.$$

Whenever the coordinate functions

$$e \mapsto e^a_\mu$$

are integrable on an appropriate dense domain, define

$$\hat{e}^a_\mu = \int_{\mathcal{E}_x} e^a_\mu d\mathbf{E}(e). \quad (168)$$

These operators need not be bounded. Their natural domains are determined by the corresponding spectral integrability conditions.

Because all of them are functions of the same projection-valued measure, they strongly commute:

$$\hat{e}^a_\mu \quad \text{and} \quad \hat{e}^b_\nu \quad \text{strongly commute for all } a, b, \mu, \nu. \quad (169)$$

Where their products and ordinary commutators are defined on a common invariant domain,

$$[\hat{e}^a_\mu, \hat{e}^b_\nu] = 0. \quad (170)$$

The response may then be written formally as

$$W(P, \xi) = \exp \left[-\frac{i}{\hbar} P_a \xi^\mu \hat{e}^a_\mu \right], \quad (171)$$

with the exponential understood through the joint functional calculus.

The logical order is important:

$W \longrightarrow \text{commutative response algebra}$ $\longrightarrow \text{joint spectral measure}$ $\longrightarrow \hat{e}^a_\mu.$
--

(172)

Thus Eq. (171) is not obtained by postulating a quantum tetrad and exponentiating it. The commuting spectral soldering operators arise as a consequence of the exact response laws.

5.9 States and probability measures on response space

Let

$$\omega$$

be a state on the commutative C^* -algebra $\mathcal{A}_W$. By the Riesz representation theorem applied to the Gelfand representation of $\mathcal{A}_W$, there exists a probability measure

$$\mu_\omega$$

on the relevant spectral space such that

$$\omega(A) = \int_{\mathfrak{X}_W} \widehat{A}(\chi) d\mu_\omega(\chi), \quad A \in \mathcal{A}_W. \quad (173)$$

After identifying regular characters with response maps

$$e \in \mathcal{E}_x,$$

the expectation value of an exact response becomes

$$\omega(W(P, \xi)) = \int_{\mathcal{E}_x} \exp \left[-\frac{i}{\hbar} P(e\xi) \right] d\mu_\omega(e). \quad (174)$$

Equation (174) is the characteristic functional of an ordinary classical random linear map

$$e : T_x M \longrightarrow V.$$

If the probability measure is supported on

$$\mathcal{E}_x^{\text{nd}},$$

it may be interpreted as an ordinary probability distribution over nondegenerate soldered Lorentzian geometries.

5.10 Classical stochastic representation theorem

We can now formulate the principal interpretive consequence of the spectral representation.

Theorem 5.4 (Classical stochastic representation of the exact response). *Let $\mathcal{A}_W$ be the local C^* -algebra generated by an exact unitary operational response satisfying the assumptions of Theorem 5.3. Then every state*

$$\omega|_{\mathcal{A}_W}$$

is equivalent, with respect to all observables in $\mathcal{A}_W$, to an ordinary classical probability measure on the spectral space of linear response maps.

In particular, all finite collections of exact response expectation values are reproducible by classical stochastic averages of the characters

$$\exp\left[-\frac{i}{\hbar}P(e\xi)\right].$$

Proof. The algebra $\mathcal{A}_W$ is commutative by Corollary 4.2. Every state on a unital commutative C^* -algebra is represented by a probability measure on its Gelfand spectrum. The regular spectral characters of the exact response are identified with linear maps

$$e : T_x M \rightarrow V$$

by Lemma 5.2 and Eq. (154). Therefore the expectation value of every response observable is an ordinary probability average over classical response maps. Since products and norm limits of the response generators remain functions on the same spectrum, the conclusion holds for all of $\mathcal{A}_W$. $\square$

5.11 Operational equivalence to classical random geometry

Theorem 5.4 should be interpreted carefully.

It does not assert that the complete quantum state of the underlying Hilbert space is classical. A state on

$$\mathcal{B}(\mathcal{H})$$

may contain entanglement, interference with other sectors, or expectation values of observables not belonging to

$$\mathcal{A}_W.$$

The theorem states only that after restricting attention to observables generated by the exact local response,

$$\omega \mapsto \omega|_{\mathcal{A}_W}, \tag{175}$$

all resulting statistics possess a classical probability model.

Accordingly, if two theories have the same probability measure

$$\mu_\omega(e)$$

on the response spectrum, then no measurement belonging solely to the exact response algebra can distinguish between

1. a Hilbert-space state with commuting spectral soldering operators, and
2. a classical stochastic ensemble of ordinary linear response maps e .

This motivates the following terminology.

Definition 5.5 (Response-classical geometry). *A quantum geometric sector is called response-classical with respect to $\mathcal{A}_W$ if all of its response statistics admit an ordinary probability representation on a classical response space.*

The exact operational response sector is response-classical in this sense.

5.12 Moments of the spectral soldering distribution

When the relevant moments exist, derivatives of the characteristic functional

$$\omega(W(P, \xi))$$

at the identity recover moments of the random solder map.

For example, differentiating with respect to the displacement coordinate gives

$$\begin{aligned} i\hbar \frac{\partial}{\partial \xi^\mu} \omega(W(P, \xi)) \Big|_{\xi=0} &= \int_{\mathcal{E}_x} P_a e^a_\mu d\mu_\omega(e) \\ &= P_a \langle e^a_\mu \rangle_\omega. \end{aligned} \quad (176)$$

If this identity holds for a spanning family of P , the mean response map is reconstructed as

$$\bar{e}^a_\mu = \int_{\mathcal{E}_x} e^a_\mu d\mu_\omega(e). \quad (177)$$

Similarly,

$$-\hbar^2 \frac{\partial^2}{\partial \xi^\mu \partial \xi^\nu} \omega(W(P, \xi)) \Big|_{\xi=0} = P_a P_b \langle e^a_\mu e^b_\nu \rangle_\omega. \quad (178)$$

Mixed derivatives with independent generator labels may be used to recover the covariance tensor

$$\text{Cov}_\omega(e^a_\mu, e^b_\nu) = \langle e^a_\mu e^b_\nu \rangle_\omega - \langle e^a_\mu \rangle_\omega \langle e^b_\nu \rangle_\omega. \quad (179)$$

These quantities describe fluctuations of the response map, but their existence does not make the response algebra noncommutative. They are ordinary moments and covariances of the probability measure

$$\mu_\omega.$$

5.13 Mean coframe versus mean metric

A stochastic response distribution also clarifies an important distinction between the mean solder map and the mean metric.

At each nondegenerate spectral point,

$$g_{\mu\nu}(e) = \eta_{ab} e^a_\mu e^b_\nu. \quad (180)$$

The stochastic mean metric is therefore

$$\langle g_{\mu\nu} \rangle_\omega = \eta_{ab} \langle e^a_\mu e^b_\nu \rangle_\omega. \quad (181)$$

By contrast, the metric constructed from the mean solder map is

$$g_{\mu\nu}(\bar{e}) = \eta_{ab} \bar{e}^a_{\mu} \bar{e}^b_{\nu}. \quad (182)$$

Their difference is

$$\langle g_{\mu\nu} \rangle_{\omega} - g_{\mu\nu}(\bar{e}) = \eta_{ab} \text{Cov}_{\omega}(e^a_{\mu}, e^b_{\nu}). \quad (183)$$

Thus stochastic response fluctuations may modify metric expectation values even though the underlying response algebra is commutative.

This provides a useful warning: nontrivial geometric fluctuations are not, by themselves, evidence of intrinsically quantum geometry.

5.14 No intrinsic contextuality within the response algebra

Since

$$\mathcal{A}_W$$

is commutative, all of its observables admit a joint classical spectral description. In particular, any finite collection

$$A_1, \dots, A_n \in \mathcal{A}_W$$

may be represented simultaneously as ordinary functions

$$a_1(e), \dots, a_n(e)$$

on the same probability space.

Therefore the exact response algebra by itself exhibits no obstruction to a global joint assignment of values to its observables.

This statement concerns only the response sector. It should not be read as a claim that a larger quantum theory containing

$$\mathcal{A}_W$$

cannot exhibit contextuality or Bell-type nonclassicality through additional observables.

5.15 Local character of the stochastic representation

The spectral representation constructed above is fiberwise at a fixed spacetime point x . Accordingly,

$$\mu_{\omega,x}$$

is initially a probability measure on

$$\mathcal{E}_x \subset \text{Hom}(T_x M, V_x).$$

To construct a stochastic field

$$x \longmapsto e_x$$

over an extended spacetime region, one requires a consistent family of joint probability distributions at distinct points. Such a structure does not follow from the local response algebra alone.

Likewise, the present theorem does not determine correlations such as

$$\langle e^a_{\mu}(x) e^b_{\nu}(y) \rangle$$

for

$$x \neq y.$$

Their definition requires additional information concerning inter-event transport, field-theoretic dynamics, or a global state on an enlarged algebra.

This is one of the locations at which a genuinely noncommutative extension may enter without contradicting the local no-go theorem.

5.16 The exact-response quantum-geometry no-go theorem

The preceding results may now be assembled into the main no-go statement of the paper.

Theorem 5.6 (Exact-response quantum-geometry no-go). *Consider a local Hilbert-space realization of the operational generator–displacement response satisfying:*

1. *unitarity;*
2. *strong continuity;*
3. *exact additive composition in the generator argument;*
4. *exact additive composition in the displacement argument;*
5. *regularity of the associated spectral characters;*
6. *restriction of the geometric observable sector to the C^* -algebra generated by the exact responses.*

Then:

1. *the exact response algebra is commutative;*
2. *the response admits a joint spectral representation by ordinary real linear maps*

$$e : T_x M \rightarrow V_x;$$

3. *every state on the response algebra is represented by an ordinary probability measure on that spectral response space;*
4. *all response statistics are therefore reproducible by a classical stochastic model of the local soldering data.*

Consequently, intrinsically noncommutative local quantum geometry cannot arise from the exact response algebra alone.

The qualification “from the exact response algebra alone” is part of the theorem. The result neither excludes nor constructs quantum geometry in an enlarged theory.

5.17 What the no-go theorem does not imply

Several conclusions do not follow from Theorem 5.6.

First, the theorem does not imply that the probability distribution

$$\mu_\omega(e)$$

must be sharply concentrated. Large stochastic fluctuations are compatible with exact response composition.

Second, it does not imply that

$$\langle g_{\mu\nu} \rangle$$

must satisfy a classical gravitational field equation. No dynamics has been imposed on the spectral measure.

Third, it does not imply that observables outside

$$\mathcal{A}_W$$

commute.

Fourth, it does not imply that inter-event comparison operators commute with local response observables.

Fifth, it does not exclude a projective deformation of the exact composition laws.

The theorem therefore identifies a boundary rather than a complete quantum theory.

5.18 Transition to quantum information geometry

The spectral theorem establishes that the exact response sector is classically representable at the level of its observable algebra. There is, however, another frequently used diagnostic of quantumness: the geometry of Hilbert-space state orbits.

One may ask whether the family

$$W(P, \xi)|\psi\rangle$$

carries a nontrivial Berry curvature or a quantum information metric that could nevertheless be interpreted as an emergent spacetime or momentum-space geometry.

The commutativity result already places strong restrictions on this possibility. Since the infinitesimal generators of the exact response strongly commute, the antisymmetric part of their quantum geometric tensor vanishes, while the symmetric part reduces to an ordinary covariance matrix.

The next section derives these statements explicitly and shows why the resulting positive semidefinite information geometry cannot be identified with the Lorentz-signature generator geometry reconstructed from the classical MS response.

6 Quantum Geometric Tensor of the Exact Response Orbit

The preceding sections established that the exact local response algebra is commutative and admits a classical stochastic representation over linear solder maps. We now examine a different notion of geometry that can be associated with a Hilbert-space realization of the response: the quantum geometric tensor of a parameterized family of states.

This question is conceptually distinct from the spectral analysis of Sec. 5. Even when an observable algebra is commutative, a family of normalized Hilbert-space states may carry a nontrivial information metric. It is therefore necessary to determine whether the exact response orbit can generate a Berry curvature or a quantum information geometry that could be identified with the Lorentzian geometry of the generator space, with spacetime geometry, or with a curved momentum-space structure.

The answer is again restrictive. Because the infinitesimal generators of the exact response strongly commute, the antisymmetric part of the quantum geometric tensor vanishes. The symmetric part survives, but it is a positive semidefinite covariance form. It therefore has a different mathematical character from the Lorentzian bilinear form that defines the massive generator shell.

6.1 Response-generated state orbits

Fix a spacetime point x and a displacement

$$\xi \in T_x M.$$

Let

$$|\psi_0\rangle \in \mathcal{H}$$

be a normalized reference vector belonging to the common domain required for the expectation values below.

The generator-labeled response orbit is defined by

$$|\psi(P; \xi)\rangle = W(P, \xi)|\psi_0\rangle, \quad P \in V_x^*. \quad (184)$$

Since $W(P, \xi)$ is unitary,

$$\langle \psi(P; \xi) | \psi(P; \xi) \rangle = 1. \quad (185)$$

Choose an internal basis and write

$$P = P^a E_a^*.$$

The exact generator composition law implies that, for fixed ξ , the family

$$P \mapsto W(P, \xi)$$

is a strongly continuous unitary representation of the additive group V_x^* .

Because the generators strongly commute, there exist strongly commuting self-adjoint operators

$$A_a(\xi)$$

such that

$$W(P, \xi) = \exp \left[-\frac{i}{\hbar} P^a A_a(\xi) \right]. \quad (186)$$

In the spectral soldering representation of Sec. 5.8,

$$A_a(\xi) = \xi^\mu \hat{e}_{a\mu}. \quad (187)$$

The operators satisfy

$$A_a(\xi) \quad \text{and} \quad A_b(\xi) \quad \text{strongly commute} \quad (188)$$

for all a, b .

6.2 Quantum geometric tensor

For a smooth normalized family

$$|\psi(\lambda)\rangle$$

depending on real parameters λ^A , the quantum geometric tensor is

$$Q_{AB} = \langle \partial_A \psi | (I - |\psi\rangle\langle\psi|) | \partial_B \psi \rangle. \quad (189)$$

Its real part defines the Fubini–Study metric,

$$g_{AB}^{\text{FS}} = \text{Re } Q_{AB}, \quad (190)$$

while its imaginary part determines the Berry curvature according to the convention

$$\mathcal{F}_{AB}^{\text{B}} = 2 \text{Im } Q_{AB}. \quad (191)$$

Different sign conventions for $\mathcal{F}^{\text{B}}$ occur in the literature. None of the conclusions below depends on that convention because the imaginary part will vanish identically.

6.3 Quantum geometric tensor of the generator orbit

Differentiate Eq. (184) with respect to P^a . Since the self-adjoint generators $A_a(\xi)$ strongly commute,

$$\frac{\partial}{\partial P^a} |\psi(P; \xi)\rangle = -\frac{i}{\hbar} A_a(\xi) |\psi(P; \xi)\rangle. \quad (192)$$

Substituting this expression into Eq. (189) gives

$$Q_{ab}^{(\xi)} = \frac{1}{\hbar^2} [\langle A_a(\xi) A_b(\xi) \rangle_P - \langle A_a(\xi) \rangle_P \langle A_b(\xi) \rangle_P], \quad (193)$$

where

$$\langle O \rangle_P = \langle \psi(P; \xi) | O | \psi(P; \xi) \rangle.$$

Because every $A_a(\xi)$ commutes with

$$W(P, \xi),$$

their expectation values are independent of the orbit parameter:

$$\langle A_a(\xi) \rangle_P = \langle A_a(\xi) \rangle_0, \quad (194)$$

and likewise

$$\langle A_a(\xi) A_b(\xi) \rangle_P = \langle A_a(\xi) A_b(\xi) \rangle_0. \quad (195)$$

Hence the quantum geometric tensor is constant over the generator orbit:

$$Q_{ab}^{(\xi)}(P) = Q_{ab}^{(\xi)}(0). \quad (196)$$

6.4 Vanishing Berry curvature

For self-adjoint generators, the antisymmetric part of the quantum geometric tensor is controlled by their commutator. Explicitly,

$$Q_{ab}^{(\xi)} - Q_{ba}^{(\xi)} = \frac{1}{\hbar^2} \langle [A_a(\xi), A_b(\xi)] \rangle. \quad (197)$$

Since the exact response theorem gives strong commutativity,

$$[A_a(\xi), A_b(\xi)] = 0 \quad (198)$$

on every common invariant domain on which the ordinary commutator is defined.

It follows that

$$\text{Im } Q_{ab}^{(\xi)} = 0. \quad (199)$$

Therefore

$$\mathcal{F}_{ab}^B = 0. \quad (200)$$

We may summarize this as follows.

Proposition 6.1 (Vanishing Berry curvature of the exact response orbit). *Let the generator-labeled state orbit be generated solely by an exact unitary operational response. Wherever the corresponding quantum geometric tensor is well defined, its Berry curvature vanishes identically.*

The result is not a consequence of choosing a special reference state. It follows from the commutative response algebra itself.

6.5 The surviving information metric

Although the Berry curvature vanishes, the symmetric part of the quantum geometric tensor need not vanish.

Equation (193) gives

$$g_{ab}^{\text{FS}}(\xi) = \frac{1}{\hbar^2} \text{Cov}_{\psi_0}(A_a(\xi), A_b(\xi)), \quad (201)$$

where

$$\text{Cov}_{\psi_0}(A_a, A_b) = \frac{1}{2} \langle \{\Delta A_a, \Delta A_b\} \rangle_{\psi_0} \quad (202)$$

and

$$\Delta A_a = A_a - \langle A_a \rangle_{\psi_0}.$$

Since the generators commute, the symmetrization in Eq. (202) is redundant, but it is useful for comparison with the general noncommutative expression.

For every real vector v^a ,

$$\begin{aligned} v^a v^b g_{ab}^{\text{FS}} &= \frac{1}{\hbar^2} \langle (v^a \Delta A_a)^2 \rangle_{\psi_0} \\ &\geq 0. \end{aligned} \quad (203)$$

Thus

$$g_{ab}^{\text{FS}} \geq 0 \quad (204)$$

as a quadratic form.

The information metric may be degenerate. A direction v^a lies in its null space precisely when

$$v^a A_a(\xi) \quad (205)$$

has zero variance in the reference state.

6.6 Spectral interpretation of the information metric

Using the spectral soldering representation,

$$A_a(\xi) = \xi^\mu \hat{e}_{a\mu},$$

Eq. (201) becomes

$$g_{ab}^{\text{FS}}(\xi) = \frac{1}{\hbar^2} \xi^\mu \xi^\nu \text{Cov}_{\psi_0}(\hat{e}_{a\mu}, \hat{e}_{b\nu}). \quad (206)$$

If the reference state restricted to the response algebra is represented by the probability measure

$$\mu_{\psi_0}(e),$$

this is simply

$$g_{ab}^{\text{FS}}(\xi) = \frac{1}{\hbar^2} \xi^\mu \xi^\nu \text{Cov}_{\mu_{\psi_0}}(e_{a\mu}, e_{b\nu}). \quad (207)$$

Hence the nonzero Fubini–Study metric does not contradict the classical stochastic representation theorem. It measures the statistical distinguishability of different parameter values in a family generated by random soldering data.

A positive quantum information metric can therefore coexist with a fully commutative response algebra.

6.7 Quantum Fisher information

For a pure-state family, the quantum Fisher information matrix is related to the Fubini–Study metric by

$$F_{ab}^Q = 4g_{ab}^{\text{FS}} \quad (208)$$

under the standard normalization convention.

Accordingly,

$$F_{ab}^Q(\xi) = \frac{4}{\hbar^2} \text{Cov}_{\psi_0}(A_a(\xi), A_b(\xi)). \quad (209)$$

The Fisher matrix is therefore positive semidefinite:

$$F_{ab}^Q \geq 0. \quad (210)$$

Its interpretation is informational: it controls the local distinguishability of nearby states

$$|\psi(P; \xi)\rangle \quad \text{and} \quad |\psi(P + dP; \xi)\rangle.$$

It does not define the causal norm of the generator label P .

6.8 No identification with the Lorentzian generator metric

The massive generator geometry inherited from the MS construction is Lorentzian:

$$\eta^{ab} P_a P_b = -Q^2. \quad (211)$$

Its quadratic form has signature

$$(-, +, +, +).$$

By contrast,

$$g_{ab}^{\text{FS}}$$

is positive semidefinite.

Therefore no nondegenerate proportionality relation of the form

$$g_{ab}^{\text{FS}} = C \eta_{ab} \quad (212)$$

with real nonzero C can hold on the full four-dimensional generator space.

Proposition 6.2 (Signature obstruction). *A nondegenerate Fubini–Study or quantum Fisher information metric generated by a unitary response orbit cannot be identified with the full Lorentzian generator metric of the massive MS sector.*

Proof. The Fubini–Study metric is positive semidefinite by Eq. (203), whereas the Lorentzian generator metric possesses one negative and three positive directions. The two bilinear forms therefore have different signatures. $\square$

This conclusion does not prevent an information metric from being defined on a positive-definite submanifold, such as an observer hyperboloid or a compact stabilizer orbit. It prevents only the direct identification of the full information metric with the ambient Lorentzian generator metric.

6.9 Fixed-generator displacement orbits

A dual construction may be performed by fixing

$$P \in V_x^*$$

and varying the displacement parameter.

Define

$$|\psi(\xi; P)\rangle = W(P, \xi)|\psi_0\rangle. \quad (213)$$

By strong continuity and exact displacement composition, there exist strongly commuting self-adjoint generators

$$B_\mu(P)$$

such that

$$W(P, \xi) = \exp\left[-\frac{i}{\hbar}\xi^\mu B_\mu(P)\right]. \quad (214)$$

In the spectral soldering representation,

$$B_\mu(P) = P_a \hat{e}^a_\mu. \quad (215)$$

The displacement-space quantum geometric tensor is then

$$Q_{\mu\nu}^{(P)} = \frac{1}{\hbar^2} \text{Cov}_{\psi_0}(B_\mu(P), B_\nu(P)). \quad (216)$$

Again,

$$\text{Im } Q_{\mu\nu}^{(P)} = 0, \quad (217)$$

and

$$g_{\mu\nu}^{\text{FS}}(P) = \frac{1}{\hbar^2} P_a P_b \text{Cov}_{\psi_0}(\hat{e}^a_\mu, \hat{e}^b_\nu) \geq 0. \quad (218)$$

Thus varying displacements rather than generators does not evade the signature obstruction.

6.10 The full product-parameter orbit

One may also consider the family

$$|\psi(P, \xi)\rangle = W(P, \xi)|\psi_0\rangle \quad (219)$$

on the product parameter space

$$V_x^* \times T_x M.$$

Because the response exponent

$$P_a \xi^\mu \hat{e}^a_\mu$$

is bilinear in the labels, the parameter derivatives now contain explicit dependence on the complementary argument:

$$\partial_{P_a} |\psi(P, \xi)\rangle = -\frac{i}{\hbar} \xi^\mu \hat{e}_{a\mu} |\psi(P, \xi)\rangle, \quad (220)$$

$$\partial_{\xi^\mu} |\psi(P, \xi)\rangle = -\frac{i}{\hbar} P_a \hat{e}^a_\mu |\psi(P, \xi)\rangle. \quad (221)$$

The diagonal blocks of the quantum geometric tensor are

$$Q_{ab} = \frac{1}{\hbar^2} \xi^\mu \xi^\nu \text{Cov}(\hat{e}_{a\mu}, \hat{e}_{b\nu}), \quad (222)$$

$$Q_{\mu\nu} = \frac{1}{\hbar^2} P_a P_b \text{Cov}(\hat{e}^a_\mu, \hat{e}^b_\nu). \quad (223)$$

The mixed block is

$$Q_{a\mu} = \frac{1}{\hbar^2} \xi^\nu P_b \text{Cov}(\hat{e}_{a\nu}, \hat{e}^b_\mu). \quad (224)$$

Because all spectral soldering operators strongly commute, the full tensor is real:

$$\text{Im } Q_{AB} = 0 \quad (225)$$

for all product-space parameter indices A, B .

Hence the entire exact local response orbit has vanishing Berry two-form.

6.11 No Berry reconstruction of the canonical symplectic form

Section 2 showed that the classical operational response reconstructs the canonical cotangent symplectic form

$$\Omega_{\text{can}} = d(P_a e^a). \quad (226)$$

The present section gives

$$\mathcal{F}^B = 0 \quad (227)$$

for the exact local response orbit.

Therefore

$$\mathcal{F}^B \neq \Omega_{\text{can}} \quad (228)$$

whenever the canonical symplectic form is nonzero.

This distinction is fundamental. The canonical form

$$\Omega_{\text{can}}$$

is a geometric structure on the classical cotangent phase space. The Berry curvature

$$\mathcal{F}^B$$

is a curvature on a bundle of quantum states over a parameter space. They are mathematically different objects, and the existence of one does not imply the existence of the other.

In the present exact-response theory they provide an especially clear contrast:

$$\boxed{\begin{aligned} \Theta &\longrightarrow \Omega_{\text{can}} = d(P_a e^a), \\ W_{\text{exact}} &\longrightarrow \mathcal{F}^B = 0. \end{aligned}} \quad (229)$$

6.12 No momentum-space curvature from the exact response alone

A nonzero positive information metric does not by itself define a nontrivial momentum-space curvature.

For fixed ξ , the generator-space information metric

$$g_{ab}^{\text{FS}}(\xi)$$

is independent of P by Eq. (196). Therefore, in the affine coordinates P^a ,

$$\partial_c g_{ab}^{\text{FS}} = 0. \quad (230)$$

Whenever the covariance matrix is nondegenerate on the parameter subspace under consideration, its Levi–Civita connection vanishes in these coordinates:

$$\Gamma_{bc}^a = 0, \quad (231)$$

and consequently

$$R_{bcd}^a = 0. \quad (232)$$

Thus the simplest quantum information geometry of the exact response is flat as well as positive semidefinite.

This yields a second obstruction:

$$\text{exact local response fluctuations} \not\Rightarrow \text{curved momentum geometry}. \quad (233)$$

A nontrivial curved momentum-space geometry would require additional parameter dependence, nonlinear composition, transport data, or other structures absent from the exact local response orbit.

6.13 Absence of an intrinsic momentum scale

The covariance form

$$g_{ab}^{\text{FS}}$$

may contain scales inherited from the reference state or from the spectral probability distribution. Such scales are state-dependent response widths.

They should not be confused with a universal momentum-curvature scale

$$M_*.$$

Nothing in exact additivity, commutativity, or the spectral response theorem generates a distinguished universal constant with dimensions of momentum. In particular, the presence of

$$\hbar$$

in the denominator of Eq. (201) merely converts generator variance into the dimensionless normalization appropriate to Hilbert-space distance.

Hence an identification such as

$$g^{\text{FS}} \sim \frac{1}{M_*^2} g^{(p)} \quad (234)$$

would require additional dynamical or constitutive input.

6.14 Observer-dependent positive information geometry

The signature obstruction applies to the identification of a positive information metric with the full ambient Lorentzian metric. It does not exclude observer-dependent positive geometries.

Let

$$U^a$$

be a unit future timelike observer,

$$\eta_{ab}U^aU^b = -1. \quad (235)$$

The associated spatial projector is

$$h_{(U)}^{ab} = \eta^{ab} + U^aU^b. \quad (236)$$

An $SO(3)_U$ -invariant positive bilinear form on the generator labels may have the general structure

$$G_{(U)}^{ab} = A h_{(U)}^{ab} + B U^aU^b, \quad A \geq 0, \quad B \geq 0. \quad (237)$$

Such a metric is compatible with the positivity of quantum Fisher information because the observer supplies the additional timelike direction required to convert the Lorentzian decomposition into a positive response metric.

However,

$$G_{(U)}^{ab} \neq \eta^{ab} \quad (238)$$

as bilinear forms.

Thus observer dressing may explain the tensorial structure of a positive information metric, but it does not convert that metric into the physical Lorentzian generator metric.

6.15 Information geometry does not detect intrinsic quantumness

The results above illustrate an important conceptual point.

A nonzero Fubini–Study metric is often associated with quantum state geometry. In the present exact response sector, however, the same metric is expressible entirely as the covariance matrix of a classical probability distribution on the spectral soldering space.

Consequently,

$$g^{\text{FS}} \neq 0 \quad (239)$$

does not imply

$$\text{intrinsically noncommutative geometry.} \quad (240)$$

The metric measures distinguishability of response distributions, not the failure of a classical joint probability description.

This distinction is especially important when attempting to infer quantum spacetime structure from fidelity susceptibilities or Fisher information alone.

6.16 Quantum-geometric corollary of exact additivity

The conclusions of this section can be summarized in the following theorem.

Theorem 6.3 (Quantum-geometric restriction of exact response). *Let*

$$W(P, \xi)$$

be an exact unitary operational response satisfying the assumptions of Theorem 5.6, and let the relevant reference vector belong to the common domains required for the second moments of the infinitesimal generators.

Then the Hilbert-space state orbits generated solely by the local exact response satisfy:

1. *the Berry curvature vanishes;*
2. *the Fubini–Study metric is a positive semidefinite covariance form;*
3. *for fixed displacement, the generator-space information metric is constant in the affine generator coordinates;*
4. *the information metric cannot be identified with the full Lorentzian generator metric;*
5. *the local exact response alone does not generate a nonzero momentum-space curvature.*

The theorem strengthens the algebraic no-go result. Not only is the response observable algebra classically representable, but the most immediate Hilbert-space geometric structures associated with its unitary orbit also fail to supply the missing noncommutative or Lorentzian quantum geometry.

6.17 What would be required for a nonzero Berry curvature

For a unitary orbit generated by self-adjoint operators T_A , the antisymmetric part of the quantum geometric tensor is schematically controlled by

$$\text{Im } Q_{AB} \propto \frac{1}{2i\hbar^2} \langle [T_A, T_B] \rangle. \quad (241)$$

Therefore a nonzero Berry curvature requires noncommuting generators, or an equivalent projective or parameter-dependent structure.

The exact local response cannot provide this because its two composition laws force those generators to commute.

Hence any attempt to obtain a nontrivial Berry curvature from the MS response must introduce structure beyond the exact local bicharacter. Possible mechanisms include:

1. projective response composition with a nontrivial two-cocycle;
2. noncommuting inter-event transport operators;
3. parameter-dependent generators whose comparison requires an additional connection;
4. an enlarged set of response observables containing canonically conjugate sectors;
5. nonlinear finite calibration transformations not reducible to the exact additive local response.

These possibilities lie outside the no-go theorem because each changes the algebraic assumptions from which commutativity was derived.

6.18 Transition to projective response composition

The preceding analysis identifies the precise obstruction.

The classical operational response contains a nontrivial canonical symplectic form through

$$\Omega_{\text{can}} = d(P_a e^a),$$

but its exact unitary realization contains no phase that couples two distinct response transformations. As a consequence, the finite group commutator is trivial, the infinitesimal response generators commute, and the Berry curvature vanishes.

Canonical quantum mechanics resolves the analogous problem by replacing ordinary representations of phase-space translations with projective ones. The symplectic form then appears not merely as a classical two-form but as a central group cocycle.

This suggests the next question:

Can the canonical symplectic pairing reconstructed from Θ enter a nontrivial projective deformation of the exact response law?	(242)
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The next section compares the exact response with Weyl-type projective composition and determines which algebraic ingredient must change before intrinsic quantum noncommutativity can appear.

7 Projective Response Composition and the Missing Quantum Structure

The preceding sections identify a precise obstruction to obtaining intrinsically noncommutative geometry from the exact operational response. The classical response reconstructs a nontrivial canonical symplectic form,

$$\Omega_{\text{can}} = d\lambda, \quad \lambda = P_a e^a,$$

whereas the exact unitary response

$$W(P, \xi)$$

generates a commutative C^* -algebra and has vanishing Berry curvature.

This section examines the structural difference between these two facts. The central issue is not the absence of a classical symplectic form. Such a form has already been reconstructed. What is absent is a nontrivial projective composition law in which that symplectic structure appears as a central phase.

The comparison with the Weyl form of canonical quantization is therefore useful, but it must be made carefully. The bivariate response $W(P, \xi)$ is not itself a Weyl operator on phase space. A genuine Weyl system acts on a combined additive phase-space label and represents its symplectic pairing through a two-cocycle. Consequently, passing from the exact operational response to a noncommutative quantum response requires an enlargement or deformation of the response structure rather than merely inserting $\hbar$ into the classical phase.

7.1 The combined response label space

At a fixed spacetime point x , let

$$T = T_x M, \quad V^* = V_x^*.$$

The exact response depends on a generator

$$P \in V^*$$

and a displacement

$$\xi \in T.$$

The natural combined real vector space is

$$\mathcal{Z}_x = V^* \oplus T. \quad (243)$$

We write its elements as

$$z = (P, \xi), \quad z' = (Q, \zeta). \quad (244)$$

Once a nondegenerate solder map

$$e : T \longrightarrow V$$

has been specified, the internal generator P corresponds to the physical cotangent covector

$$p = e^* P.$$

The canonical symplectic pairing on infinitesimal cotangent directions then induces an antisymmetric bilinear form on $\mathcal{Z}_x$:

$$\omega_e((P, \xi), (Q, \zeta)) = P(e\zeta) - Q(e\xi). \quad (245)$$

The form ω_e is antisymmetric because

$$\begin{aligned} \omega_e(z', z) &= Q(e\xi) - P(e\zeta) \\ &= -\omega_e(z, z'). \end{aligned} \quad (246)$$

It is bilinear in both arguments. If e is nondegenerate, then ω_e is nondegenerate on

$$V^* \oplus T.$$

Indeed, suppose

$$\omega_e((P, \xi), (Q, \zeta)) = 0$$

for every (Q, ζ) . Taking $Q = 0$ gives

$$P(e\zeta) = 0$$

for every ζ , and nondegeneracy of e implies

$$P = 0.$$

Taking $\zeta = 0$ then gives

$$Q(e\xi) = 0$$

for every Q , hence

$$\xi = 0.$$

Thus

$$(\mathcal{Z}_x, \omega_e) \quad (247)$$

is a finite-dimensional symplectic vector space whenever the soldering is nondegenerate.

7.2 Relation to the canonical cotangent symplectic form

The form

$$\omega_e$$

is not an additional symplectic structure independent of Ω_{can} . It is the fiberwise linearized realization of the canonical cotangent symplectic pairing after the internal generator has been mapped to a physical covector by the solder form.

In local coordinates,

$$p_\mu = P_a e^a{}_\mu.$$

A tangent vector to the local affine cotangent phase space may be written schematically as

$$(\delta p, \delta x).$$

For two such variations,

$$(\delta p, \delta x) \quad \text{and} \quad (\delta p', \delta x'),$$

the canonical symplectic form gives

$$\Omega_{\text{can}}((\delta p, \delta x), (\delta p', \delta x')) = \delta p(\delta x') - \delta p'(\delta x). \quad (248)$$

Under

$$\delta p = e^* P, \quad \delta p' = e^* Q,$$

and

$$\delta x = \xi, \quad \delta x' = \zeta,$$

Eq. (248) becomes precisely Eq. (245).

Accordingly,

$$\omega_e = \text{the soldered fiberwise representation of } \Omega_{\text{can}}. \quad (249)$$

This observation will allow us to compare the exact response composition with the Weyl representation of a symplectic vector space.

7.3 The Weyl projective composition law

Let

$$(\mathcal{Z}, \omega)$$

be a real symplectic vector space. A Weyl system is a strongly continuous family of unitary operators

$$\mathcal{W}(z), \quad z \in \mathcal{Z},$$

satisfying a projective composition law of the form

$$\mathcal{W}(z)\mathcal{W}(z') = \exp\left[-\frac{i}{2\hbar}\omega(z, z')\right]\mathcal{W}(z + z'). \quad (250)$$

The numerical coefficient and overall sign depend on convention. The essential structure is the presence of the antisymmetric bilinear two-cocycle

$$\omega(z, z').$$

Associativity requires

$$\omega(z, z') + \omega(z + z', z'') = \omega(z', z'') + \omega(z, z' + z''). \quad (251)$$

For bilinear ω , this identity follows immediately.

Reversing the order of the two operators gives

$$\mathcal{W}(z')\mathcal{W}(z) = \exp\left[\frac{i}{2\hbar}\omega(z, z')\right]\mathcal{W}(z + z'). \quad (252)$$

Consequently,

$$\mathcal{W}(z)\mathcal{W}(z') = \exp\left[-\frac{i}{\hbar}\omega(z, z')\right]\mathcal{W}(z')\mathcal{W}(z). \quad (253)$$

The corresponding group commutator is

$$\mathcal{W}(z)\mathcal{W}(z')\mathcal{W}(z)^{-1}\mathcal{W}(z')^{-1} = \exp\left[-\frac{i}{\hbar}\omega(z, z')\right]I. \quad (254)$$

Thus the classical symplectic structure enters the quantum algebra as a central phase.

7.4 A soldered Weyl system

Applying the preceding construction to the response label space

$$\mathcal{Z}_x = V_x^* \oplus T_x M$$

and the soldered symplectic form

$$\omega_e$$

gives the formal projective law

$$\begin{aligned} & \mathcal{W}_e(P, \xi)\mathcal{W}_e(Q, \zeta) \\ &= \exp\left[-\frac{i}{2\hbar}(P(e\zeta) - Q(e\xi))\right]\mathcal{W}_e(P + Q, \xi + \zeta). \end{aligned} \quad (255)$$

Its finite group commutator is

$$\begin{aligned} & \mathcal{W}_e(P, \xi)\mathcal{W}_e(Q, \zeta)\mathcal{W}_e(P, \xi)^{-1}\mathcal{W}_e(Q, \zeta)^{-1} \\ &= \exp\left[-\frac{i}{\hbar}(P(e\zeta) - Q(e\xi))\right]I. \end{aligned} \quad (256)$$

Equation (255) is the canonical Weyl construction applied to the symplectic vector space defined by a fixed classical solder map. It is not derived from the exact operational response axioms.

This distinction is central to the present paper.

7.5 Why the exact response is not a Weyl system

The exact response has the form

$$W(P, \xi) \quad (257)$$

and satisfies separately

$$W(P + Q, \xi) = W(P, \xi)W(Q, \xi), \quad (258)$$

$$W(P, \xi + \zeta) = W(P, \xi)W(P, \zeta). \quad (259)$$

These equations do not define a projective representation of the combined group

$$V^* \oplus T.$$

Rather, they define a function that is separately a representation in each argument.

The distinction can be seen already in the sharp classical case:

$$W_e(P, \xi) = \exp \left[-\frac{i}{\hbar} P(e\xi) \right] I. \quad (260)$$

By contrast, a Weyl operator

$$\mathcal{W}_e(P, \xi)$$

is not simply the phase associated with the scalar pairing $P(e\xi)$. It represents a combined translation in a symplectic phase space.

Therefore the passage

$$W(P, \xi) \longrightarrow \mathcal{W}(P, \xi) \quad (261)$$

is not a change of notation. It introduces additional operator structure.

7.6 Two marginal translation sectors

The distinction becomes especially transparent if the Weyl system is decomposed into two marginal translation families.

Define

$$U(P) = \mathcal{W}(P, 0), \quad (262)$$

$$V(\xi) = \mathcal{W}(0, \xi). \quad (263)$$

Since

$$\omega_e((P, 0), (Q, 0)) = 0,$$

the U -sector is an ordinary Abelian representation:

$$U(P)U(Q) = U(P + Q). \quad (264)$$

Likewise,

$$\omega_e((0, \xi), (0, \zeta)) = 0,$$

so

$$V(\xi)V(\zeta) = V(\xi + \zeta). \quad (265)$$

However, the mixed symplectic pairing is

$$\omega_e((P, 0), (0, \xi)) = P(e\xi). \quad (266)$$

Hence

$$U(P)V(\xi) = \exp\left[-\frac{i}{\hbar}P(e\xi)\right]V(\xi)U(P). \quad (267)$$

The classical operational response phase therefore reappears naturally as the central commutator between two distinct translation sectors.

This is structurally different from the exact response

$$W(P, \xi),$$

where the same phase is itself the response amplitude.

7.7 The response phase as a possible Weyl commutator

Equation (267) suggests a useful reinterpretation of the classical scalar response:

$$\chi_e(P, \xi) = \exp\left[-\frac{i}{\hbar}P(e\xi)\right]. \quad (268)$$

In a Weyl-type extension, the same scalar may be written as

$$\chi_e(P, \xi)I = U(P)V(\xi)U(P)^{-1}V(\xi)^{-1}. \quad (269)$$

Thus one possible quantum extension changes the logical role of the classical response phase:

exact theory: $\chi_e(P, \xi) = \text{response itself},$ Weyl-type theory: $\chi_e(P, \xi) = \text{central commutator of two response sectors}.$	(270)
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This is a substantive algebraic enlargement. It cannot be obtained while retaining the exact two-argument response laws unchanged.

7.8 A structural incompatibility theorem

The preceding observation may be formulated as a no-go statement.

Theorem 7.1 (Exact–projective incompatibility). *Let $W(P, \xi)$ be a unitary response satisfying exact additive composition separately in the generator and displacement arguments:*

$$W(P + Q, \xi) = W(P, \xi)W(Q, \xi), \quad (271)$$

$$W(P, \xi + \zeta) = W(P, \xi)W(P, \zeta). \quad (272)$$

Then the finite group commutator of any two responses is trivial.

Consequently, a nontrivial Weyl-type central relation

$$W(z)W(z') = e^{-i\omega(z, z')/\hbar}W(z')W(z) \quad (273)$$

with

$$\omega \neq 0$$

cannot arise within the same exact response structure.

Proof. The triviality of the group commutator follows from Theorem 4.1. A nonzero symplectic phase in Eq. (273) would give a nontrivial group commutator and therefore contradict exact commutativity. $\square$

Hence a genuinely projective quantum response must modify, replace, or enlarge the exact bivariate response structure.

7.9 Infinitesimal Heisenberg relation

Suppose the two marginal families are written as

$$U(P) = \exp \left[-\frac{i}{\hbar} P_a \hat{X}^a \right], \quad (274)$$

$$V(\xi) = \exp \left[-\frac{i}{\hbar} \xi^\mu \hat{\Pi}_\mu \right]. \quad (275)$$

The mixed Weyl relation

$$U(P)V(\xi) = e^{-iP(e\xi)/\hbar} V(\xi)U(P)$$

then implies, formally on a suitable common invariant domain,

$$[\hat{X}^a, \hat{\Pi}_\mu] = i\hbar e^a{}_\mu I \quad (276)$$

up to the sign convention chosen in Eqs. (274)–(275).

The solder map therefore appears as the central bilinear coefficient of a Heisenberg-type algebra.

Equation (276) is not a result of the exact-response theory. It characterizes one possible enlarged projective quantization associated with a fixed classical soldering background.

7.10 Why the solder map cannot yet be promoted to the central operator

One might attempt to replace the classical coefficient in Eq. (276) by an operator,

$$[\hat{X}^a, \hat{\Pi}_\mu] \stackrel{?}{=} i\hbar \hat{e}^a{}_\mu. \quad (277)$$

This is a substantially stronger proposal.

If

$$\hat{e}^a{}_\mu$$

is central and strongly commuting, then the algebra decomposes spectrally into ordinary Heisenberg representations with different classical values of e . Such a construction still admits a superselection-type interpretation over classical soldering sectors.

If, on the other hand,

$$\hat{e}^a{}_\mu$$

fails to commute with

$$\hat{X}^a \quad \text{or} \quad \hat{\Pi}_\mu,$$

then the Jacobi identities introduce additional commutators and the resulting algebra lies beyond the exact response theory.

Therefore Eq. (277) cannot be adopted without specifying an enlarged algebra and verifying its consistency.

7.11 Central spectral soldering does not evade the no-go theorem

Suppose the response is spectrally decomposed as in Sec. 5, and consider a direct integral of Weyl systems labeled by

$$e \in \mathcal{E}_x : \\ \mathcal{H} = \int_{\mathcal{E}_x}^{\oplus} \mathcal{H}_e d\mu(e). \quad (278)$$

On each fiber,

$$U_e(P)V_e(\xi) = e^{-iP(e\xi)/\hbar} V_e(\xi)U_e(P). \quad (279)$$

The corresponding soldering operator acts centrally by multiplication:

$$(\hat{e}\Psi)(e) = e \Psi(e). \quad (280)$$

This theory contains genuine canonical noncommutativity inside each Weyl fiber, but the geometry label e itself remains a classical spectral variable.

Thus there are two distinct notions of quantumness:

1. noncommutativity of canonical matter or phase-space variables at fixed geometry;
2. noncommutativity of the geometric response variables themselves.

A direct integral of ordinary Weyl representations supplies the first but not necessarily the second.

This distinction prevents canonical quantization on a stochastic background from being misidentified with intrinsic quantum geometry.

7.12 Projective deformation of only one argument

Another possibility is to deform only one of the two exact composition laws.

For example, suppose generator composition remains exact,

$$W(P + Q, \xi) = W(P, \xi)W(Q, \xi), \quad (281)$$

while displacement composition becomes projective:

$$W(P, \xi + \zeta) = e^{i\beta(P; \xi, \zeta)} W(P, \xi)W(P, \zeta). \quad (282)$$

Associativity requires

$$\begin{aligned} \beta(P; \xi, \zeta) + \beta(P; \xi + \zeta, \chi) \\ = \beta(P; \zeta, \chi) + \beta(P; \xi, \zeta + \chi) \pmod{2\pi}. \end{aligned} \quad (283)$$

Likewise, one may retain exact displacement composition while introducing a generator cocycle

$$\alpha(P, Q; \xi).$$

However, such a one-sided deformation must also satisfy compatibility conditions between the two arguments. An arbitrary cocycle in one argument does not automatically produce a consistent two-variable response.

The classification of all such compatible deformations is a separate cohomological problem and is not attempted here.

7.13 The minimal algebraic change required for noncommutativity

The exact-response theorem allows a simple general conclusion.

If one demands

$$[W(P, \xi), W(Q, \zeta)] \neq 0 \quad (284)$$

for some labels, then at least one assumption entering the exact commutativity theorem must fail.

Therefore at least one of the following must occur:

1. generator composition is only projective or nonlinear;
2. displacement composition is only projective or nonlinear;
3. the response depends on path or transport data in addition to the endpoint displacement;
4. the local response algebra is enlarged by another noncommuting sector;
5. the label space itself acquires a non-Abelian or nonassociative composition law.

This conclusion is independent of which particular quantum-gravity model, if any, realizes the modification.

7.14 A quantum-response design principle

The previous discussion suggests a useful principle for future extensions of the MS reconstruction.

Principle 7.2 (Nontrivial quantum-response principle). *A genuinely noncommutative quantum extension of operational soldering must encode information that is absent from the exact local generator–displacement bicharacter. This additional information must enter through projective composition, noncommuting comparison, additional conjugate variables, or another algebraic obstruction to simultaneous exact additivity.*

The principle is not a unique quantization prescription. It specifies only what cannot remain unchanged if intrinsic noncommutative geometry is desired.

7.15 Relation between the classical symplectic form and a quantum cocycle

It is tempting to infer directly

$$\Omega_{\text{can}} \implies \text{Weyl cocycle.} \quad (285)$$

Such an implication is not automatic.

The classical response reconstruction determines

$$\Omega_{\text{can}}$$

as a two-form on cotangent phase space. A projective unitary representation requires, in addition, a choice of central extension of an appropriate transformation group.

For a linear symplectic vector space, the standard Weyl construction provides a canonical and highly natural such extension. In the full curved-spacetime setting, however, additional issues arise:

1. the solder map varies with spacetime position;
2. different tangent and cotangent fibers must be compared by transport;
3. curvature may obstruct a global identification with one fixed symplectic vector group;

4. torsion may enter the local expression

$$d(P_a e^a) = DP_a \wedge e^a + P_a T^a;$$

5. global phase consistency may require holonomy data.

Thus the existence of the classical symplectic form identifies the appropriate local two-form, but it does not by itself derive the global quantum representation.

7.16 Local projective quantization versus inter-event transport

At a single event x , a nondegenerate solder map supplies the local symplectic form

$$\omega_{e_x}.$$

One may therefore construct a local Weyl algebra

$$\mathfrak{B}_x = \mathfrak{B}(\mathcal{Z}_x, \omega_{e_x}).$$

At a second event y , one obtains another algebra

$$\mathfrak{B}_y = \mathfrak{B}(\mathcal{Z}_y, \omega_{e_y}).$$

A field-theoretic quantum geometry would require rules for comparing these local algebras. Schematically, for a path

$$\gamma : x \rightarrow y,$$

one would need a transport map

$$\mathcal{U}_\gamma : \mathfrak{B}_x \longrightarrow \mathfrak{B}_y. \quad (286)$$

Path dependence of

$$\mathcal{U}_\gamma$$

could contain geometric information not visible in any single local response algebra.

This provides a second logically distinct route to quantumness:

local projective structure and
nontrivial inter-event transport

need not encode the same physics.

The exact-response no-go theorem constrains the former only when the local response remains exact and commutative.

7.17 No derivation of quantum gravity

The constructions discussed in this section do not constitute a derivation of quantum gravity. In particular, the present analysis does not determine:

1. which projective extension, if any, is physically realized;
2. whether the geometric response variables themselves become noncommuting;
3. the Hilbert space of a gravitational field;
4. a Hamiltonian or path-integral dynamics for the projective response;

5. a quantum constraint algebra;
6. the value of Newton's constant;
7. a fundamental ultraviolet scale.

The purpose of the Weyl comparison is narrower. It identifies the specific algebraic ingredient that is absent from the exact response: a nontrivial central composition phase.

7.18 Exact and projective response structures

The distinction developed in this section may be summarized as

classical response : $\Theta(P, \xi) = P(e\xi)$, exact unitary response : $W(P, \xi)$ with double exact additivity, $\implies [W, W] = 0$, projective Weyl extension : $\mathcal{W}(z)\mathcal{W}(z') = e^{-i\omega_e(z, z')/(2\hbar)}\mathcal{W}(z + z')$, $\implies$ nontrivial central commutator.	(287)
---	-------

Thus the exact and projective theories occupy different algebraic classes.
The important structural observation is that the scalar operational pairing

$$P(e\xi)$$

can play two distinct roles:

$P(e\xi)$ = phase exponent of the exact response, $P(e\xi)$ = mixed central cocycle of an enlarged Weyl response.	(288)
---	-------

Only the second role produces Heisenberg-type noncommutativity.

7.19 Transition to the discussion of admissible quantum extensions

The results obtained so far establish a clear boundary.

The operational response

$$\Theta$$

is sufficient to reconstruct both the Lorentzian metric and the canonical cotangent symplectic form. Its exact unitary implementation, however, produces a commutative response algebra with a classical stochastic spectral representation. The associated Berry curvature vanishes, and the surviving information metric is positive semidefinite.

A noncommutative quantum extension must therefore contain additional structure. The Weyl construction demonstrates one mathematically consistent mechanism: replacing exact composition on a combined response space by a projective representation controlled by the soldered canonical symplectic form.

This mechanism is not uniquely selected by the MS reconstruction. Accordingly, the next section turns from the no-go theorem itself to the space of admissible extensions. We distinguish changes that genuinely evade the theorem from modifications that merely reparameterize the same commutative response sector, and we identify which open problems must be solved before any such extension can be regarded as a quantum theory of geometry.

8 Admissible Quantum Extensions Beyond Exact Response

The preceding sections establish a sharp boundary for the exact local operational response. If both generator composition and displacement composition are preserved as exact additive unitary laws, then the resulting response algebra is commutative, its regular spectrum consists of ordinary linear soldering maps, and every state restricted to that algebra admits a classical stochastic representation. The associated Berry curvature vanishes, while the surviving information metric is a positive semidefinite covariance form.

These results do not imply that an operational reconstruction of spacetime geometry is incompatible with quantum theory. They imply that a genuinely noncommutative quantum extension must contain information not present in the exact local bicharacter

$$(P, \xi) \mapsto W(P, \xi).$$

The purpose of this section is to classify, at a structural level, the ways in which an enlarged theory may evade the no-go theorem. We do not attempt to select a unique quantum-gravitational completion. Instead, we distinguish genuine escapes from modifications that leave the response sector effectively classical.

8.1 The assumptions that must be crossed

The exact-response quantum-geometry no-go theorem depends on a specific set of assumptions:

1. local response operators are unitary;
2. the response is strongly continuous;
3. generator composition is represented exactly and additively;
4. displacement composition is represented exactly and additively;
5. the relevant spectral characters are regular;
6. the geometric observable sector is restricted to the algebra generated by the exact local responses.

Intrinsic noncommutativity cannot appear unless at least one of these conditions is relaxed or the observable algebra is enlarged.

This yields the following structural statement.

Proposition 8.1 (Necessary enlargement of exact response). *Any quantum extension with genuinely noncommuting geometric observables must contain structure that is not representable within the exact local response algebra*

$$\mathcal{A}_W.$$

In particular, such a theory must modify at least one exact composition law, introduce additional noncommuting observables, or make inter-event comparison an independent quantum operation.

This proposition is deliberately weak in the constructive direction. It does not state which extension is physically correct. Its role is to prevent an intrinsically quantum geometry from being attributed to structures that remain entirely inside the commutative exact-response sector.

8.2 Projective composition

The most direct algebraic escape is projective composition.

Instead of

$$W(z)W(z') = W(z + z'),$$

one may allow

$$\mathcal{W}(z)\mathcal{W}(z') = e^{i\alpha(z,z')}\mathcal{W}(z + z'), \quad (289)$$

where

$$z, z' \in \mathcal{Z}_x$$

and

$$\alpha(z, z')$$

satisfies the cocycle condition required by associativity.

If the antisymmetric part of the cocycle is nontrivial,

$$\alpha(z, z') - \alpha(z', z) \neq 0 \pmod{2\pi}, \quad (290)$$

then the response operators need not commute.

The canonical Weyl extension discussed in Sec. 7 corresponds locally to

$$\alpha_e(z, z') = -\frac{1}{2\hbar}\omega_e(z, z'), \quad (291)$$

with

$$\omega_e((P, \xi), (Q, \zeta)) = P(e\zeta) - Q(e\xi).$$

This is the minimal familiar mechanism by which a classical symplectic form becomes quantum algebraic data.

However, introducing the cocycle is additional structure. The operational soldering theorem reconstructs the classical form

$$\omega_e,$$

but it does not by itself establish that the quantum response must realize its central extension.

8.3 Noncommuting inter-event transport

The local no-go theorem is fiberwise. It does not constrain the algebra of comparison operations between distinct events.

Let

$$\mathcal{H}_x$$

denote a local response Hilbert space at x , and let

$$U_\gamma : \mathcal{H}_x \longrightarrow \mathcal{H}_y$$

be a unitary transport operator associated with a path

$$\gamma : x \rightarrow y.$$

A covariant transport law for the response may take the form

$$U_\gamma W_x(P, \xi) U_\gamma^{-1} = W_y(L_\gamma^{-T} P, B_\gamma \xi), \quad (292)$$

where

$$L_\gamma$$

acts on internal generators and

$$B_\gamma$$

acts on tangent displacements.

Even if

$$\mathcal{A}_{W,x}$$

and

$$\mathcal{A}_{W,y}$$

are individually Abelian, the transport operator need not belong to either algebra. In particular,

$$[U_\gamma, W_x(P, \xi)] \neq 0 \quad (293)$$

is not excluded by the exact local response theorem.

This provides a conceptually different source of quantumness: noncommutativity may reside in the comparison of local geometries rather than in the local soldering response itself.

8.4 Quantum transport curvature

Suppose infinitesimal Hilbert-space transport is generated by operator-valued connection components

$$\hat{C}_\mu.$$

Locally,

$$U_{\delta x} = I - \frac{i}{\hbar} \hat{C}_\mu \delta x^\mu + O(\delta x^2). \quad (294)$$

If the spectral soldering operators are transported covariantly, their infinitesimal compatibility relation may take the schematic form

$$\partial_\mu \hat{e}^a{}_\nu - \Gamma_{\mu\nu}^\rho \hat{e}^a{}_\rho + \omega_{\mu b}^a \hat{e}^b{}_\nu + \frac{i}{\hbar} [\hat{C}_\mu, \hat{e}^a{}_\nu] = 0. \quad (295)$$

The final commutator is absent in a purely classical transport law. It represents the possibility that the Hilbert-space comparison structure acts nontrivially on the local geometric response sector.

The corresponding transport curvature is

$$\hat{\mathcal{F}}_{\mu\nu} = \partial_\mu \hat{C}_\nu - \partial_\nu \hat{C}_\mu - \frac{i}{\hbar} [\hat{C}_\mu, \hat{C}_\nu]. \quad (296)$$

This curvature should not be identified automatically with the spacetime Riemann curvature reconstructed from the soldered Lorentz connection. The two objects belong to different geometric structures.

The distinction is

R^{ab} = curvature of Lorentz-frame comparison, $\hat{\mathcal{F}}$ = curvature of Hilbert-space transport.
--

(297)

An enlarged quantum theory would have to specify whether and how these two curvatures are related.

8.5 Path dependence as additional response data

The exact local response depends only on an infinitesimal displacement

$$\xi.$$

A more general response may depend on the path by which the comparison is made:

$$W(P, \gamma) \tag{298}$$

rather than merely

$$W(P, \xi).$$

Then two paths

$$\gamma_1, \gamma_2 : x \rightarrow y$$

with the same endpoints need not define the same response:

$$W(P, \gamma_1) \neq W(P, \gamma_2). \tag{299}$$

For a closed loop

$$\gamma : x \rightarrow x,$$

the response may contain holonomy information

$$W(P, \gamma) \neq I. \tag{300}$$

Such path data do not contradict the local commutativity theorem because they are not determined by the local endpoint pair

$$(P, \xi)$$

alone.

This identifies transport and holonomy as natural candidates for the first genuinely new quantum data beyond exact soldering.

8.6 The classical closed-response phase and torsion

Already at the classical level, the Liouville form

$$\lambda = P_a e^a$$

contains information about closed displacement responses.

For a small closed loop

$$\gamma = \partial\Sigma,$$

one has

$$\oint_{\gamma} P_a e^a = \int_{\Sigma} d(P_a e^a). \tag{301}$$

If the generator is covariantly transported across the infinitesimal surface so that its covariant variation may be neglected to leading order, then

$$d(P_a e^a) \simeq P_a T^a. \tag{302}$$

The corresponding scalar phase is

$$\exp \left[-\frac{i}{\hbar} \oint_{\gamma} P_a e^a \right] \simeq \exp \left[-\frac{i}{\hbar} P_a \int_{\Sigma} T^a \right]. \quad (303)$$

Thus torsion may be operationally interpreted as a leading closed-loop translation-response defect.

This observation does not by itself supply quantum noncommutativity. The phase in Eq. (303) is still a scalar response unless the loop comparison is promoted to a noncommuting transport algebra.

8.7 Additional conjugate response sectors

A second broad escape from the no-go theorem is to enlarge the observable content.

The exact response probes the pairing

$$P(e\xi).$$

Its spectral variables

$$e^a_{\mu}$$

all commute.

A noncommutative theory may introduce another family of observables

$$\hat{\Pi}_a^{\mu}$$

canonically conjugate, in some appropriate sense, to the soldering data:

$$[\hat{e}^a_{\mu}, \hat{\Pi}_b^{\nu}] \neq 0. \quad (304)$$

The exact response algebra would then remain a commuting configuration subalgebra, analogous to the algebra of functions of position in ordinary quantum mechanics.

This provides an important interpretation of the no-go theorem: commutativity of the soldering-response sector need not be surprising if that sector represents only one polarization of a larger phase space.

8.8 Configuration subalgebras can be classical

In ordinary quantum mechanics, all bounded functions of the position operator commute:

$$[f(\hat{q}), g(\hat{q})] = 0.$$

Nevertheless the complete theory is quantum because the momentum operator does not commute with $\hat{q}$.

The exact response theorem may admit an analogous interpretation.

If

$$\hat{e}^a_{\mu}$$

represent a generalized configuration sector, then

$$\mathcal{A}_W$$

may be commutative while the complete geometric phase-space algebra is not.

Thus the no-go theorem rules out the stronger claim

$$\mathcal{A}_W = \text{complete algebra of quantum geometry} \quad (305)$$

whenever genuinely quantum geometric effects are required.

It does not rule out

$$\mathcal{A}_W \subsetneq \mathcal{A}_{\text{geom}}, \quad (306)$$

where

$$\mathcal{A}_{\text{geom}}$$

contains additional conjugate variables.

8.9 Finite calibration transformations

The exact local response describes infinitesimal generator–displacement pairing. Another possible source of additional structure is the finite comparison of generator calibrations.

Let

$$\mathcal{V}(g)$$

denote a unitary operator representing a finite change of calibration. Its action on the local response might take the form

$$\mathcal{V}(g)W(P, \xi)\mathcal{V}(g)^{-1} = W(f_g(P), B_g\xi). \quad (307)$$

If

$$f_g$$

and

$$B_g$$

are ordinary linear transformations preserving all exact additive structures, then the construction may remain inside a classical semidirect-product extension.

To generate genuinely new quantum structure, at least one of the following would be required:

1. noncommuting calibration operators;
2. projective calibration composition;
3. nonlinear action on the generator labels;
4. path-dependent comparison;
5. an enlarged operator algebra not reducible to automorphisms of $\mathcal{A}_W$.

Merely introducing a unitary implementation of an ordinary classical change of coordinates does not evade the no-go theorem in a physically substantive way.

8.10 A linearity obstruction for exact response automorphisms

The exact additive structure strongly constrains transformations of the generator labels.

Suppose an invertible unitary operator

$$\mathcal{V}$$

acts by

$$\mathcal{V}W(P, \xi)\mathcal{V}^{-1} = W(f(P), \xi) \quad (308)$$

and assume the response is generator-faithful.

Using exact generator composition,

$$\begin{aligned}
W(f(P + Q), \xi) &= \mathcal{V}W(P + Q, \xi)\mathcal{V}^{-1} \\
&= \mathcal{V}W(P, \xi)W(Q, \xi)\mathcal{V}^{-1} \\
&= W(f(P), \xi)W(f(Q), \xi) \\
&= W(f(P) + f(Q), \xi).
\end{aligned} \tag{309}$$

Generator faithfulness therefore gives

$$f(P + Q) = f(P) + f(Q). \tag{310}$$

Under continuity,

$$f$$

is real linear.

Thus a nonlinear generator-space geometry cannot arise merely by conjugating an exact faithful additive response while retaining its composition law unchanged.

Proposition 8.2 (Linearity of exact-response label automorphisms). *Every continuous faithful automorphism of the exact additive generator response induces a linear transformation of the generator vector space.*

This supplies another obstruction to obtaining curved momentum geometry from the exact response by a nonlinear relabeling alone.

8.11 Curved momentum space requires additional comparison structure

Suppose one wishes to replace the linear generator space V^* by a curved momentum manifold

$$\mathcal{P}.$$

If the exact additive labels $P \in V^*$ remain globally valid and all finite calibration changes act linearly, then the curvature of $\mathcal{P}$ is not encoded in the exact response algebra.

A genuinely curved comparison geometry would require additional structures such as

$$\nabla^{(p)}, \quad R^{(p)}, \quad \oplus, \tag{311}$$

where

$$\nabla^{(p)}$$

is a momentum-space connection,

$$R^{(p)}$$

its curvature, and

$$\oplus$$

a finite composition law that need not coincide with vector addition.

These objects describe comparison between distinct generator calibrations, not the local linear pairing between one generator and one displacement.

This distinction is parallel to the classical gravitational separation: local MS geometry may remain universal while curvature resides in the comparison between different local frames.

8.12 Associativity and nonlinear composition

A nonlinear generator composition

$$P \oplus Q \tag{312}$$

would evade the exact additive assumptions if

$$P \oplus Q \neq P + Q.$$

However, nonlinearity by itself does not guarantee nontrivial physics. If the composition is associative, smooth, possesses an identity and inverse locally, and is related to ordinary addition by a smooth change of coordinates,

$$P \oplus Q = F^{-1}(F(P) + F(Q)), \tag{313}$$

then it is only a nonlinear parametrization of an Abelian local group.

A physically nontrivial deformation therefore requires an invariant obstruction to such linearization.

Possible obstructions include:

1. a non-Abelian Lie bracket;
2. nonassociativity;
3. curvature of a comparison connection;
4. a nontrivial projective cocycle;
5. additional tensorial structures not removable by a coordinate change.

This observation prevents coordinate nonlinearity from being mistaken for quantum geometry.

8.13 Nonassociative response composition

A more radical possibility is nonassociative composition:

$$(P \oplus Q) \oplus R \neq P \oplus (Q \oplus R). \tag{314}$$

The associator

$$\mathfrak{A}(P, Q, R) = (P \oplus Q) \oplus R - P \oplus (Q \oplus R) \tag{315}$$

could encode curvature-like information in the finite comparison of generator states.

At the quantum level one may correspondingly encounter an associator operator

$$\Phi(P, Q, R), \tag{316}$$

so that operator products are associative only up to a specified coherent transformation.

Such structures lie entirely outside the exact-response theorem, whose composition laws are ordinary associative operator equalities.

No nonassociative extension is constructed in the present work.

8.14 Classical stochasticity is not sufficient

One possible response to the no-go theorem would be to allow a broad probability distribution

$$\mu(e)$$

over solder maps and interpret its fluctuations as quantum geometry.

The spectral theorem shows that this is insufficient.

Regardless of the width, multimodality, or non-Gaussianity of

$$\mu(e),$$

the response statistics remain classical if all observables belong to

$$\mathcal{A}_W.$$

In particular, a large variance

$$\text{Var}(e^a_\mu) \gg 0 \quad (317)$$

does not imply operator noncommutativity.

Similarly, a nontrivial correlation

$$\text{Cov}(e^a_\mu, e^b_\nu) \neq 0 \quad (318)$$

is compatible with an ordinary joint probability measure.

Thus

$$\text{geometric fluctuation} \not\Rightarrow \text{quantum geometry}. \quad (319)$$

8.15 Superposition language does not evade commutativity

The same caution applies to Hilbert-space superpositions.

A state may formally be written as a superposition over spectral response sectors,

$$|\Psi\rangle = \int_{\mathcal{E}_x}^{\oplus} \Psi(e) |e\rangle. \quad (320)$$

If all measurements are restricted to the commuting response algebra, only

$$|\Psi(e)|^2 \quad (321)$$

enters the corresponding spectral probabilities.

Relative phases between different spectral components cannot be detected by observables diagonal in the same response algebra.

Hence the phrase “superposition of geometries” does not, by itself, establish operational quantum geometry. To observe coherent phase information between different e -sectors, one needs observables that do not commute with the spectral projectors of

$$\mathcal{A}_W.$$

8.16 Necessary condition for coherent geometry interference

Let

$$E(\Delta)$$

denote a spectral projector for a measurable region

$$\Delta \subset \mathcal{E}_x.$$

An observable A capable of probing coherent interference between distinct response sectors must fail to commute with at least some of these projectors:

$$[A, E(\Delta)] \neq 0. \quad (322)$$

Every

$$A \in \mathcal{A}_W$$

commutes with all such projectors. Therefore no observable internal to the exact response algebra can detect coherent relative phases between different spectral soldering sectors.

This gives an operational criterion for a genuinely enlarged geometric observable algebra:

$$\mathcal{A}_{\text{geom}} \text{ must contain observables not commuting with } E. \quad (323)$$

8.17 Response classicality and superselection

If all physically realizable geometric observables commute with the spectral soldering algebra, then distinct spectral sectors behave as superselection sectors.

In that case, a density operator with off-diagonal components between different e -values is operationally equivalent, with respect to the geometric response observables, to its diagonalized counterpart.

Schematically,

$$\rho \longrightarrow \rho_{\text{diag}} = \int E(de) \rho E(de) \quad (324)$$

leaves all

$$A \in \mathcal{A}_W$$

expectation values unchanged:

$$\text{Tr}(\rho A) = \text{Tr}(\rho_{\text{diag}} A). \quad (325)$$

Thus overcoming response classicality requires observables that connect different spectral sectors.

8.18 A hierarchy of possible quantum extensions

The possible extensions may be organized by increasing structural departure from the exact theory. At the first level,

$$\text{classical stochastic response,} \quad (326)$$

one retains the exact algebra and allows arbitrary probability measures over solder maps. This does not evade the no-go theorem.

At the second level,

$$\text{canonical quantum variables on a spectral geometry,} \quad (327)$$

one introduces noncommuting matter or conjugate variables while the soldering map remains central. The theory is quantum, but the geometry itself remains response-classical.

At the third level,

$$\text{noncommuting comparison or conjugate geometric variables,} \quad (328)$$

the exact response algebra becomes a proper commuting subalgebra of a larger quantum-geometric algebra.

At the fourth level,

$$\text{projective, nonlinear, or nonassociative response composition,} \quad (329)$$

the composition law itself is changed.

The no-go theorem applies only to the first level when interpreted as a complete geometric algebra. The remaining levels evade at least one of its defining assumptions.

9 The role of $\hbar$ after the no-go theorem

The analysis also sharpens the interpretation of

$$\hbar.$$

In the exact response phase

$$e^{-iP(e\xi)/\hbar}, \quad (330)$$

$\hbar$ converts action into phase. This role is already meaningful before any noncommutative geometry is introduced.

In a Weyl extension, the same constant additionally controls the central commutator:

$$U(P)V(\xi) = e^{-iP(e\xi)/\hbar}V(\xi)U(P). \quad (331)$$

Thus

$$\hbar$$

can occur in two logically different ways:

$\hbar^{-1}\Theta = \text{action-to-phase conversion,}$ $\hbar = \text{scale of a nontrivial projective commutator.}$
--

(332)

Only the second role requires noncommutative quantum algebra.

9.1 No emergence of Newton's constant from exact response

The introduction of

$$\hbar$$

does not resolve the independent gravitational normalization identified in the classical reconstruction.

The exact response algebra contains no mechanism that fixes

$$\kappa = \frac{8\pi G}{c^4}.$$

Nor does the commutativity theorem generate a universal momentum scale from which G could be constructed.

Accordingly,

$$(\hbar, c, Q) \not\Rightarrow G \quad (333)$$

within the exact response framework.

Any relation between the gravitational constitutive scale and a quantum scale would require additional microscopic dynamics beyond the kinematical and algebraic results established here.

9.2 Three logically distinct structures

The combined classical and quantum analysis reveals three structures that should remain conceptually separate.

First,

$$g = e^* \eta \quad (334)$$

is the Lorentzian spacetime metric reconstructed from operational soldering.

Second,

$$\Omega_{\text{can}} = d(P_a e^a) \quad (335)$$

is the canonical cotangent symplectic form.

Third, a possible quantum projective curvature or cocycle,

$$\alpha(z, z'), \quad (336)$$

belongs to the unitary composition law.

The reconstruction of the first two does not automatically determine the third.

The logical relations are therefore

$$\Theta \longrightarrow (g, \Omega_{\text{can}}), \quad (337)$$

but not

$$\Theta \longrightarrow \text{unique quantum cocycle}. \quad (338)$$

9.3 What counts as a genuine evasion of the theorem

The preceding discussion suggests a practical criterion.

Definition 9.1 (Genuine quantum-response extension). *An extension of the exact operational response will be called genuinely quantum-geometric if it contains geometric observables or comparison operations whose joint statistics cannot be represented by a single ordinary probability measure over the spectral soldering space $\mathcal{E}_x$.*

Under this definition, the following do not by themselves constitute a genuine extension:

1. broadening the probability distribution over e ;
2. introducing non-Gaussian stochastic soldering fluctuations;
3. reparameterizing the generator labels nonlinearly;
4. calling a spectral decomposition a superposition of geometries;
5. constructing a positive Fisher metric on the same commutative orbit.

By contrast, the following may constitute genuine extensions:

1. observables connecting distinct spectral soldering sectors;
2. nontrivial projective response composition;
3. noncommuting quantum transport;
4. a conjugate geometric operator algebra;
5. non-Abelian or nonassociative finite comparison structures.

9.4 A minimal escape theorem

The preceding observations can be condensed into a final structural result.

Theorem 9.2 (Minimal escape from response classicality). *Let $\mathcal{A}_W$ be the commutative algebra generated by an exact local operational response. Suppose an enlarged geometric theory*

$$\mathcal{A}_{\text{geom}} \supset \mathcal{A}_W$$

exhibits operationally detectable quantum interference between distinct spectral soldering sectors.

Then

$$\mathcal{A}_{\text{geom}}$$

must contain at least one observable that does not commute with the spectral projections of $\mathcal{A}_W$.

Equivalently, a genuinely coherent quantum geometry cannot be obtained by enlarging only the probability distribution on the same commutative response spectrum.

Proof. If every observable in $\mathcal{A}_{\text{geom}}$ commuted with the spectral projectors of $\mathcal{A}_W$, then all geometric measurements would be block diagonal in the same spectral decomposition. Relative phases between distinct soldering sectors would be unobservable, and the state restricted to the geometric observables would be equivalent to an ordinary statistical mixture. Operational interference between distinct sectors therefore requires at least one observable that fails to commute with the spectral projectors. $\square$

9.5 Scope of the no-go theorem

The no-go theorem should therefore be read neither too weakly nor too strongly.

It is stronger than the statement that one particular tetrad quantization happens to commute. The commutativity follows before any tetrad operator is assumed, directly from exact response composition.

It is weaker than a theorem excluding quantum gravity. It leaves open all extensions in which additional comparison, conjugate, projective, or nonassociative structures are introduced.

Its precise content is

exact local unitary generator–displacement response + double exact additivity $\implies$ commutative local response algebra $\implies$ classical stochastic spectral representation.	(339)
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Any theory claiming intrinsically quantum geometry while retaining all these assumptions must therefore identify additional observables outside the exact response algebra.

9.6 Implications for the MS reconstruction programme

Within the MS reconstruction programme, the result suggests a natural division of labor.

The classical compact-state and operational-response structures are well suited to reconstructing:

1. relativistic chronometry;
2. massive generator geometry;
3. Lorentz observer transformations;
4. soldering;
5. the spacetime metric;
6. the canonical cotangent pairing;
7. the classical symplectic form.

They do not, under exact unitary promotion alone, determine:

1. a noncommutative geometric algebra;
2. a unique Weyl cocycle;
3. noncommuting Hilbert-space transport;
4. a quantum gravitational Hamiltonian;
5. a microscopic origin of G ;
6. a universal ultraviolet scale.

The classical reconstruction and the quantum completion problem are therefore separated by a genuine algebraic boundary rather than by a mere absence of calculation.

9.7 A possible research hierarchy

The analysis suggests that future developments should be approached in a specific order.

The first question is whether the exact response algebra is the complete geometric observable algebra. The present theorem gives a negative answer if intrinsic quantum geometry is required.

The second question is which additional observables are operationally motivated. Natural candidates include:

inter-event comparison operators, closed-loop response operators, conjugate geometric variables, finite calibration transformations.	(340)
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Only after such an enlarged algebra is defined should one ask whether it possesses a distinguished state, dynamics, semiclassical limit, or relation to a gravitational coupling.

This order avoids introducing ultraviolet scales or deformed dispersion relations before the underlying quantum algebra has been identified.

9.8 Transition to the final discussion

We have now completed the structural analysis required for the main claim of the paper.

The classical operational response reconstructs both Lorentzian and canonical cotangent geometry. Exact unitary promotion of the same response, with both additive composition laws preserved, forces a commutative observable algebra. Its spectral degrees of freedom are ordinary linear solder maps, its states admit classical probability representations, and its immediate quantum information geometry does not restore the missing noncommutativity.

Projective composition provides one explicit mechanism for crossing this boundary, while non-commuting transport, conjugate geometric variables, and more general comparison structures provide other possibilities. None is uniquely selected by the exact MS response itself.

The final section summarizes these results, clarifies their relation to the preceding classical reconstruction, and states the principal limitations and open problems without promoting any particular extension to a completed theory of quantum gravity.

10 Discussion and Conclusions

The purpose of this paper has been to determine how much quantum geometric structure follows from the operational generator–displacement response when its classical composition laws are preserved exactly in a Hilbert-space representation.

The answer is restrictive but structurally informative.

The classical response

$$\Theta_x(P, \xi) = P(e_x \xi) \quad (341)$$

contains enough information to reconstruct a nondegenerate solder map and hence the Lorentzian spacetime metric. The same pairing also determines the canonical cotangent Liouville form and symplectic structure,

$$\lambda = P_a e^a, \quad \Omega_{\text{can}} = d\lambda. \quad (342)$$

Thus the operational response has two distinct classical geometric realizations:

$\begin{aligned} \Theta &\longrightarrow e \longrightarrow g = e^* \eta, \\ \Theta &\longrightarrow \lambda = P_a e^a \longrightarrow \Omega_{\text{can}} = d\lambda. \end{aligned}$	(343)
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The central question was whether exponentiating this response while retaining its exact additive structure can, without further input, produce intrinsically noncommutative quantum geometry. The results of the preceding sections show that it cannot.

10.1 Summary of the exact-response results

The quantum response was represented by a strongly continuous unitary family

$$W(P, \xi)$$

satisfying exact composition in both arguments,

$$W(P + Q, \xi) = W(P, \xi)W(Q, \xi), \quad (344)$$

$$W(P, \xi + \zeta) = W(P, \xi)W(P, \zeta). \quad (345)$$

The key algebraic observation is that the same operator

$$W(P + Q, \xi + \zeta)$$

may be decomposed in the two possible orders. Consistency of those decompositions forces

$$[W(P, \xi), W(Q, \zeta)] = 0 \quad (346)$$

for all admissible labels.

Consequently, the local response algebra

$$\mathcal{A}_W = C^* (\{W(P, \xi)\}) \quad (347)$$

is commutative.

This result is stronger than the statement that one particular set of geometric operators may be chosen to commute. No operator-valued coframe was assumed in the derivation. The commutativity follows directly from simultaneous exact additivity in the two operational arguments.

Joint spectral theory then implies that the exact response can be decomposed into scalar continuous bicharacters. Under the regularity conditions stated in the paper, those bicharacters have the form

$$\chi_e(P, \xi) = \exp \left[-\frac{i}{\hbar} P(e\xi) \right], \quad (348)$$

where

$$e : T_x M \longrightarrow V_x$$

is an ordinary real linear response map.

Accordingly,

$$W(P, \xi) = \int_{\mathcal{E}_x} \exp \left[-\frac{i}{\hbar} P(e\xi) \right] d\mathbf{E}(e). \quad (349)$$

When the spectral coordinate functions are defined on the relevant domains, one may introduce strongly commuting spectral soldering operators

$$\hat{e}^a_\mu = \int_{\mathcal{E}_x} e^a_\mu d\mathbf{E}(e), \quad (350)$$

and write

$$W(P, \xi) = \exp \left[-\frac{i}{\hbar} P_a \xi^a \hat{e}^a_\mu \right]. \quad (351)$$

The important logical point is that

$$\hat{e}^a_\mu$$

is obtained from the spectral representation of W ; it is not a fundamental operator introduced before the response algebra is determined.

10.2 Classical stochasticity versus intrinsic quantumness

Because $\mathcal{A}_W$ is commutative, every state restricted to this algebra admits an ordinary probability representation on its spectrum. For the regular response sector this gives

$$\omega(W(P, \xi)) = \int_{\mathcal{O}_x} \exp \left[-\frac{i}{\hbar} P(e\xi) \right] d\mu_\omega(e). \quad (352)$$

Thus, with respect to the exact response observables alone, the statistics of the theory are reproduced by a classical random variable

$$e \in \mathcal{O}_x.$$

If the probability measure is supported on nondegenerate maps,

$$\det e \neq 0,$$

then every spectral realization defines an ordinary Lorentzian metric

$$g_e = e^* \eta. \quad (353)$$

The theory then describes, as far as the exact response algebra is concerned, a probability distribution over classical soldered geometries.

This distinction is important because nonzero fluctuations do not by themselves establish intrinsic quantumness. One may have

$$\text{Cov}(e^a_\mu, e^b_\nu) \neq 0 \quad (354)$$

while the complete response statistics still possess an ordinary joint probability representation.

Likewise,

$$\langle g_{\mu\nu} \rangle - g_{\mu\nu}(\langle e \rangle) = \eta_{ab} \text{Cov}(e^a_\mu, e^b_\nu) \quad (355)$$

can be nonzero without implying a noncommutative geometry.

The appropriate conclusion is therefore not that the response cannot fluctuate, but that exact-response fluctuations alone are response-classical.

10.3 The role of the quantum geometric tensor

The Hilbert-space geometry of the response orbit provides an independent test of the same issue. For a fixed displacement ξ , let

$$|\psi(P; \xi)\rangle = W(P, \xi)|\psi_0\rangle. \quad (356)$$

The infinitesimal generators

$$A_a(\xi) = \xi^\mu \hat{e}_{a\mu}$$

strongly commute. The quantum geometric tensor therefore reduces to

$$Q_{ab}^{(\xi)} = \frac{1}{\hbar^2} \text{Cov}(A_a(\xi), A_b(\xi)). \quad (357)$$

Its imaginary part vanishes,

$$\text{Im } Q_{ab}^{(\xi)} = 0, \quad (358)$$

and hence the corresponding Berry curvature vanishes.

The real part is a positive semidefinite information metric,

$$g_{ab}^{\text{FS}} = \frac{1}{\hbar^2} \text{Cov}(A_a, A_b) \geq 0. \quad (359)$$

Two consequences follow.

First, the nonzero classical cotangent symplectic form

$$\Omega_{\text{can}}$$

is not reproduced automatically as a Berry curvature of the exact response orbit:

$$\Omega_{\text{can}} \neq \mathcal{F}^{\text{B}} \quad (360)$$

in general.

Second, the positive semidefinite information metric cannot be identified with the ambient Lorentzian generator metric

$$\eta_{ab}.$$

The two objects have different signatures and different operational meanings.

The quantum information metric measures statistical distinguishability between nearby response states. The Lorentzian metric determines causal and massive generator geometry. Their coexistence does not imply their identity.

10.4 Why the appearance of $\hbar$ is not sufficient

The exact classical response naturally produces the phase

$$\exp\left[-\frac{i}{\hbar}\Theta(P, \xi)\right]. \quad (361)$$

It is tempting to interpret the appearance of $\hbar$ as evidence that the response has already become quantum mechanical. The no-go theorem shows that this is insufficient.

In the exact response algebra,

$$\hbar$$

sets the conversion between an action-valued response and a dimensionless phase. It does not, by itself, generate a nontrivial operator commutator.

Thus one must distinguish

$$\text{phase structure} \quad \text{from} \quad \text{noncommutative quantum structure}. \quad (362)$$

The former is already present in an ordinary scalar character. The latter requires an additional algebraic obstruction to simultaneous classical composition.

10.5 Exact response and projective quantization

The comparison with Weyl quantization makes the missing ingredient explicit. For the combined response label space

$$\mathcal{Z}_x = V_x^* \oplus T_x M,$$

a nondegenerate classical solder map defines the antisymmetric pairing

$$\omega_e((P, \xi), (Q, \zeta)) = P(e\xi) - Q(e\xi). \quad (363)$$

A Weyl system associated with this symplectic vector space obeys

$$\mathcal{W}_e(z)\mathcal{W}_e(z') = \exp\left[-\frac{i}{2\hbar}\omega_e(z, z')\right]\mathcal{W}_e(z+z'). \quad (364)$$

The corresponding group commutator is nontrivial:

$$\mathcal{W}_e(z)\mathcal{W}_e(z')\mathcal{W}_e(z)^{-1}\mathcal{W}_e(z')^{-1} = \exp\left[-\frac{i}{\hbar}\omega_e(z, z')\right]I. \quad (365)$$

This is precisely what is absent from the exact response.

The difference may be summarized as

exact response : ordinary composition $\longrightarrow$ commutative algebra, Weyl response : projective composition $\longrightarrow$ central noncommutativity.	(366)
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The canonical symplectic form therefore supplies a natural candidate for a projective cocycle, but the existence of that classical form does not force the projective representation. Choosing such a representation is an additional quantization step.

10.6 A change in the role of the operational phase

The comparison also suggests a useful conceptual distinction.

In the exact theory,

$$\exp\left[-\frac{i}{\hbar}P(e\xi)\right] \quad (367)$$

is the response itself.

In a Weyl-type extension, the same scalar can instead appear as the mixed group commutator of two distinct translation sectors. If

$$U(P) = \mathcal{W}(P, 0), \quad V(\xi) = \mathcal{W}(0, \xi),$$

then

$$U(P)V(\xi) = \exp\left[-\frac{i}{\hbar}P(e\xi)\right]V(\xi)U(P). \quad (368)$$

Hence the same operational pairing may play two mathematically distinct roles:

$P(e\xi)$: phase exponent of a classical or exact response, $P(e\xi)$: central cocycle of an enlarged quantum algebra.	(369)
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The second role contains genuinely new algebraic information. It is not obtained by exponentiating the classical response more carefully; it requires an enlarged response structure.

10.7 What the no-go theorem excludes

The results of this paper exclude a specific class of quantum extensions.

They exclude the possibility that all of the following may hold simultaneously:

1. the quantum geometric observables are generated solely by the local response family $W(P, \xi)$;
2. generator composition remains exactly additive;
3. displacement composition remains exactly additive;
4. the response remains unitary and strongly continuous;
5. the resulting local response geometry is intrinsically noncommutative.

The first four conditions imply the negation of the fifth.

Equivalently, if intrinsic noncommutativity is required, then at least one structural element of the exact response theory must be modified or enlarged.

10.8 What the no-go theorem does not exclude

The theorem is local and algebraic. Its scope should not be extended beyond what has been proved.

It does not exclude:

1. projective unitary representations;
2. noncommuting inter-event transport;
3. quantum holonomy or connection variables outside $\mathcal{A}_W$;
4. additional canonical response sectors;
5. nonlinear finite generator calibration;
6. non-Abelian composition of enlarged labels;
7. nonassociative comparison structures;
8. entanglement with degrees of freedom outside the exact response algebra;
9. quantum matter on a response-classical geometry;
10. field-theoretic nonlocal correlations between distinct spacetime points.

In particular, the local result

$$[W_x(P, \xi), W_x(Q, \zeta)] = 0 \quad (370)$$

does not determine the algebra of comparison operators between different events.

This leaves inter-event transport as one of the most natural places at which additional quantum structure may enter.

10.9 The distinction between local response and transport

The classical MS reconstruction already separates two geometric roles.

The local response determines how a generator couples to an infinitesimal displacement at one event:

$$\Theta_x(P, \xi) = P(e_x \xi). \quad (371)$$

A connection, by contrast, determines how internal states and generators are compared between distinct events.

The same distinction remains meaningful after quantization.

It is therefore possible in principle for every local algebra

$$\mathcal{A}_{W,x}$$

to be commutative while the transport between those algebras contains nontrivial operator structure.

For a path

$$\gamma : x \rightarrow y,$$

one may schematically introduce a comparison operator

$$\mathcal{U}_\gamma.$$

A quantum transport law would act by

$$\mathcal{U}_\gamma W_x(P, \xi) \mathcal{U}_\gamma^{-1} = W_y(P_\gamma, \xi_\gamma), \quad (372)$$

possibly with additional projective or operator-valued structure.

The present paper does not specify such a theory. The point is only that the local commutativity theorem does not constrain all possible noncommutativity of inter-event comparison.

10.10 Implications for quantum extensions of MS geometry

The results impose a useful ordering on future extensions of the motion–staticity reconstruction. The classical programme establishes a chain of the form

compact state structure $\longrightarrow$ Lorentzian massive geometry	(373)
$\longrightarrow$ operational soldering	
$\longrightarrow g, \lambda, \Omega_{\text{can}}.$	

The present work shows that the next arrow cannot simply be

$$\Theta \longrightarrow W_{\text{exact}} \longrightarrow \text{noncommutative quantum geometry}. \quad (374)$$

Instead,

$$\Theta \longrightarrow W_{\text{exact}} \longrightarrow \text{commutative spectral geometry}. \quad (375)$$

Therefore a genuinely quantum continuation requires a new structural step between operational soldering and noncommutative geometry.

This is a useful constraint on the programme. It prevents a complex response phase or a fluctuating spectral coframe from being interpreted prematurely as a complete quantum geometry.

10.11 Possible locations of the missing quantum input

The no-go theorem does not select a unique escape mechanism, but it narrows the possibilities. Several logically distinct modifications remain available.

Projective composition. The exact composition laws may be replaced by a central extension controlled by a two-cocycle. The soldered canonical symplectic form provides a natural local candidate for such a cocycle.

Additional conjugate variables. The response may represent only one half of a larger canonical algebra. Introducing a second conjugate sector can convert the scalar pairing into a nontrivial commutator.

Quantum transport. Local response algebras may remain commutative while inter-event comparison is implemented by noncommuting transport operators.

Finite nonlinear calibration. The infinitesimal response may remain linear even when finite comparison of generator calibrations is nonlinear or non-Abelian.

Enlarged composition geometry. The generator or displacement labels may cease to form ordinary Abelian groups at the relevant microscopic level.

Each possibility changes a different assumption of the exact-response theorem. Determining which, if any, is physically justified requires additional principles not supplied by the present construction.

10.12 No derivation of a fundamental ultraviolet scale

Nothing in the exact-response analysis produces a universal ultraviolet momentum or length scale. The parameter

$$\hbar$$

sets the action-to-phase conversion, but it does not determine a momentum-space curvature radius.

Likewise, the spectral covariance

$$\text{Cov}(e, e)$$

may introduce state-dependent fluctuation scales, but these are not automatically universal constants.

No relation of the form

$$\hbar \longrightarrow M_* \tag{376}$$

or

$$(\hbar, c) \longrightarrow G \tag{377}$$

is established here.

The numerical gravitational coupling remains a separate constitutive normalization in the classical gravitational completion of the MS programme.

Accordingly, the present no-go theorem should not be interpreted as a derivation of the Planck scale.

10.13 Relation to canonical quantization

The analysis also clarifies the relation between operational response and ordinary canonical quantization.

Classically, the same soldering response that determines the spacetime metric gives

$$\lambda = P_a e^a$$

and hence

$$\Omega_{\text{can}} = d\lambda.$$

Canonical quantization can therefore be applied after the classical phase-space structure has been reconstructed.

However, such canonical quantization is an additional operation. It is not equivalent to replacing the scalar response phase by an operator-valued response while keeping both classical additive laws exact.

The distinction may be summarized as

$$\begin{array}{l} \text{operational reconstruction : } \Theta \longrightarrow \Omega_{\text{can}}, \\ \text{quantization : } \Omega_{\text{can}} \longrightarrow \text{projective or noncommutative algebra.} \end{array} \quad (378)$$

The first arrow has been derived within the classical response framework. The second is not fixed by that derivation.

10.14 Novelty and relation to standard mathematics

Several mathematical ingredients used in this paper are standard.

The tautological one-form and canonical symplectic form on a cotangent bundle are standard structures of Hamiltonian geometry. The spectral theory of commuting normal operators and the representation of Abelian unitary groups by projection-valued measures are standard functional analysis. The Gelfand representation of commutative C^* -algebras and the probability representation of their states are standard. The quantum geometric tensor, Berry curvature, Fubini–Study metric, and Weyl relations are likewise established structures.

The claim of the present work is therefore not the discovery of any of these mathematical objects individually.

The structural contribution is the chain connecting them to the operational soldering response:

$$\begin{array}{l} \Theta \longrightarrow e, g, \lambda, \Omega_{\text{can}}, \\ \Theta \xrightarrow{\text{exact unitary realization}} W, \\ \text{double exact additivity} \longrightarrow \mathcal{A}_W \text{ commutative,} \\ \mathcal{A}_W \text{ commutative} \longrightarrow \text{classical stochastic spectral representation.} \end{array} \quad (379)$$

The resulting no-go statement identifies a constraint on the quantization of an operationally reconstructed spacetime geometry.

10.15 Limitations

The analysis has several explicit limitations.

First, it is local and fiberwise. A global field theory requires consistent algebras and states over extended spacetime regions.

Second, the spectral classification assumes the regularity conditions required to identify the relevant characters with continuous bilinear responses.

Third, the paper does not construct a dynamical law for the spectral measure

$$\mu_\omega(e).$$

Fourth, it does not derive field equations for expectation values or correlation functions of the spectral geometry.

Fifth, it does not determine whether degenerate solder maps must be excluded dynamically. Spectral nondegeneracy is logically stronger than collective response faithfulness.

Sixth, the analysis concerns the algebra generated by local exact responses. It does not classify all possible enlarged quantum observable algebras.

Seventh, no claim is made that a projective Weyl extension is the unique or physically correct quantum continuation.

Eighth, neither the Born rule nor the Hilbert-space structure of a complete gravitational theory has been derived from the MS state geometry.

These limitations are part of the logical boundary of the result.

10.16 A hierarchy of reconstructed and additional structures

The results obtained across the classical and present quantum analyses suggest a useful hierarchy:

$$\begin{aligned}
& \text{compact MS state} \longrightarrow \text{relativistic massive kinematics,} \\
& \text{operational response } \Theta \longrightarrow \text{soldering } e, \\
& \quad e + \eta \longrightarrow \text{Lorentzian metric } g, \\
& \quad \Theta \longrightarrow \text{Liouville form } \lambda, \\
& \quad \lambda \longrightarrow \text{canonical symplectic form } \Omega_{\text{can}}, \\
& \quad \Theta + \hbar \longrightarrow \text{exact response phase,} \\
& \text{exact unitary composition} \longrightarrow \text{commutative response algebra,} \\
& \text{intrinsic quantum noncommutativity} \longrightarrow \text{additional structure required.}
\end{aligned} \tag{380}$$

This hierarchy separates what is reconstructed from what must still be supplied.

In particular,

$$g$$

and

$$\Omega_{\text{can}}$$

are classical consequences of the operational response, whereas a nontrivial quantum algebra is not.

10.17 Final conclusion

The main result of this paper can be stated in a compact form.

The operational generator–displacement response is sufficiently rich to reconstruct both the Lorentzian spacetime metric and the canonical symplectic structure of cotangent phase space. Nevertheless, if this same response is promoted to a unitary operator family while its additivity in both arguments is retained exactly, the resulting local response algebra is necessarily commutative.

Its spectral components are ordinary bilinear response maps,

$$e : T_x M \rightarrow V_x,$$

and every state restricted to the response algebra is represented by an ordinary probability measure over those maps. The associated response orbit has vanishing Berry curvature and a positive semidefinite information metric. Neither structure supplies an intrinsically noncommutative quantum geometry.

The conclusion is therefore a boundary statement:

$$\boxed{\begin{array}{c} \text{exact quantization of classical operational response} \\ \text{preserving both additive composition laws} \\ \implies \text{response-classical geometry.} \end{array}} \quad (381)$$

Consequently, if the MS reconstruction is to admit an intrinsically quantum geometric extension, the missing quantum structure cannot be obtained by exact exponentiation of the classical soldering response alone. It must enter through an additional algebraic ingredient, such as projective composition, a conjugate response sector, noncommuting transport, or another controlled failure of simultaneous exact additivity.

In this sense the classical reconstruction and the quantum no-go theorem fit together rather than compete. The same operational structure that is strong enough to determine classical Lorentzian and symplectic geometry is also strong enough to identify the point at which a genuine quantum extension must depart from classical composition.

A Positive Generator Cone and Additive Group Completion

The exact operational response is physically introduced on positive massive generators, whereas several results in the main text are most naturally formulated on the full additive vector space V^* . This appendix records the relation between these two domains.

Let

$$(V^*, \eta^*)$$

be a four-dimensional Lorentzian covector space of signature

$$(-, +, +, +).$$

Choose a future orientation and define the open future timelike cone

$$C^+ = \{P \in V^* : \eta^*(P, P) < 0, P \text{ future}\} \cup \{0\}. \quad (382)$$

For every fixed $Q > 0$, the positive massive shell

$$\Sigma_Q^+ = \{P \in V^* : \eta^*(P, P) = -Q^2, P \text{ future}\}$$

generates the same positive cone under positive rescaling:

$$\text{cone}(\Sigma_Q^+) = C^+. \quad (383)$$

A.1 Additive closure of the future cone

If

$$P, Q \in C^+,$$

then

$$P + Q \in C^+.$$

Indeed, for nonzero future timelike covectors the reverse Lorentzian Cauchy–Schwarz inequality gives

$$-\eta^*(P, Q) \geq \sqrt{-\eta^*(P, P)} \sqrt{-\eta^*(Q, Q)}. \quad (384)$$

Consequently,

$$\begin{aligned} \eta^*(P + Q, P + Q) &= \eta^*(P, P) + 2\eta^*(P, Q) + \eta^*(Q, Q) \\ &< 0, \end{aligned} \quad (385)$$

unless one of the terms is zero in the trivial manner. Future orientation is preserved as well.

A.2 The cone generates the full vector space

The positive cone is not a group because additive inverses are absent. Nevertheless, its Grothendieck group completion is canonically isomorphic to the full real vector space V^* .

Proposition A.1. *Every*

$$R \in V^*$$

can be written as

$$R = P - Q \quad (386)$$

with

$$P, Q \in C^+.$$

Proof. Choose any future unit timelike covector

$$U \in V^*, \quad \eta^*(U, U) = -1.$$

For sufficiently large $\lambda > 0$,

$$Q = \lambda U$$

is future timelike, and

$$P = R + \lambda U$$

is also future timelike. Indeed,

$$\eta^*(R + \lambda U, R + \lambda U) = -\lambda^2 + 2\lambda\eta^*(R, U) + \eta^*(R, R), \quad (387)$$

which is negative for sufficiently large λ , while its time orientation agrees with U . Therefore

$$R = P - Q.$$

□

A.3 Uniqueness of the unitary group completion

Suppose

$$W(P, \xi), \quad P \in C^+,$$

is unitary and satisfies

$$W(P + Q, \xi) = W(P, \xi)W(Q, \xi).$$

For

$$R = P - Q,$$

define

$$W(R, \xi) = W(P, \xi)W(Q, \xi)^{-1}. \quad (388)$$

If

$$P_1 - Q_1 = P_2 - Q_2,$$

then

$$P_1 + Q_2 = P_2 + Q_1.$$

Hence

$$W(P_1, \xi)W(Q_2, \xi) = W(P_2, \xi)W(Q_1, \xi).$$

Since all generator responses commute for fixed ξ , one obtains

$$W(P_1, \xi)W(Q_1, \xi)^{-1} = W(P_2, \xi)W(Q_2, \xi)^{-1}.$$

Thus the extension is well defined and gives a unitary representation of the additive group

$$(V^*, +).$$

B An Interchange-Law Proof of Response Commutativity

The proof of the exact-response commutativity theorem may be viewed as a continuous operator-valued analogue of an interchange-law argument.

Consider four labels

$$P, Q \in V^*, \quad \xi, \zeta \in T_x M.$$

Define the four elementary responses

$$A = W(P, \xi), \quad B = W(P, \zeta), \quad C = W(Q, \xi), \quad D = W(Q, \zeta).$$

Exact composition in the generator direction followed by exact composition in the displacement direction gives

$$W(P + Q, \xi + \zeta) = ABCD. \quad (389)$$

Reversing the order of the two decompositions gives

$$W(P + Q, \xi + \zeta) = ACBD. \quad (390)$$

Therefore

$$ABCD = ACBD.$$

Multiplying from the left by A^{-1} and from the right by D^{-1} gives

$$BC = CB. \quad (391)$$

Since the four labels are arbitrary,

$$[W(P, \xi), W(Q, \zeta)] = 0.$$

The argument is structurally similar to the Eckmann–Hilton mechanism: two compatible exact composition structures sharing a common identity force commutativity. In the present case the two compositions act on different label variables, so the conclusion appears through the rectangular interchange relation rather than through two binary operations on one set.

B.1 Why unitarity is used

The cancellation step above requires invertibility. Unitarity supplies this automatically:

$$W(P, \xi)^{-1} = W(P, \xi)^\dagger.$$

More generally, the same algebraic proof holds for any response valued in a group. The Hilbert-space structure is not needed for the finite commutativity theorem itself. It becomes essential only when spectral and quantum-geometric consequences are considered.

B.2 Semigroup-valued responses

If the response were merely semigroup-valued and noninvertible, the interchange identity

$$ABCD = ACBD$$

would not in general imply

$$BC = CB.$$

Thus the exact-response no-go theorem relies on reversible response composition at the finite operator level.

This identifies another possible route outside the theorem: a genuinely irreversible response theory need not satisfy the same cancellation argument. Such a theory would, however, lie outside the unitary framework considered in the present work.

C Classification of Continuous Bicharacters on Real Vector Spaces

This appendix provides a direct classification of the scalar bicharacters used in the spectral response theorem.

Let A and B be finite-dimensional real vector spaces. Consider a continuous map

$$\chi : A \times B \longrightarrow U(1)$$

such that

$$\chi(a + a', b) = \chi(a, b)\chi(a', b), \quad (392)$$

$$\chi(a, b + b') = \chi(a, b)\chi(a, b'). \quad (393)$$

C.1 Characters of a real vector group

For fixed $a \in A$,

$$b \mapsto \chi(a, b)$$

is a continuous character of the additive group B .

Every continuous character of $B \simeq \mathbb{R}^n$ has the form

$$\chi(a, b) = e^{-i\ell_a(b)/\hbar} \quad (394)$$

for a unique

$$\ell_a \in B^*.$$

Uniqueness may be seen by restricting to every line

$$b = tv.$$

The function

$$t \mapsto \chi(a, tv)$$

is a continuous character of $(\mathbb{R}, +)$, hence

$$\chi(a, tv) = e^{-itc(a,v)/\hbar}$$

for a unique real number $c(a, v)$.

C.2 Linearity in the first argument

Equation (392) implies

$$\ell_{a+a'} = \ell_a + \ell_{a'}.$$

Continuity gives

$$\ell_{ta} = t\ell_a, \quad t \in \mathbb{R}.$$

Therefore

$$a \mapsto \ell_a$$

is a real linear map

$$L : A \longrightarrow B^*.$$

Define

$$b(a, b) = (La)(b). \quad (395)$$

Then b is a real bilinear form and

$$\chi(a, b) = \exp \left[-\frac{i}{\hbar} b(a, b) \right]. \quad (396)$$

C.3 Application to operational soldering

Set

$$A = V_x^*, \quad B = T_x M.$$

Finite-dimensional duality gives

$$\text{Bil}(V_x^*, T_x M; \mathbb{R}) \simeq \text{Hom}(T_x M, V_x).$$

Thus there exists a unique

$$e : T_x M \longrightarrow V_x$$

such that

$$b(P, \xi) = P(e\xi). \quad (397)$$

Consequently every continuous scalar bicharacter is of the form

$$\chi_e(P, \xi) = \exp \left[-\frac{i}{\hbar} P(e\xi) \right]. \quad (398)$$

No Lorentz metric is needed for this classification. The Lorentzian structure enters only when one interprets e as a solder map between the internal Lorentz space and tangent space.

D Joint Spectral Construction of the Soldering Operators

The spectral representation in the main text may alternatively be constructed directly from the commuting one-parameter subgroups.

Choose bases

$$\{E^a\} \subset V_x^*, \quad \{X_\mu\} \subset T_x M.$$

For every pair (a, μ) , define

$$F_{a\mu}(s, t) = W(sE^a, tX_\mu). \quad (399)$$

Exact composition gives

$$F_{a\mu}(s + s', t) = F_{a\mu}(s, t)F_{a\mu}(s', t), \quad (400)$$

$$F_{a\mu}(s, t + t') = F_{a\mu}(s, t)F_{a\mu}(s, t'). \quad (401)$$

The finite commutativity theorem implies that all members of all these families commute.

D.1 Bilinear Stone generators

For fixed t , Stone's theorem gives a self-adjoint generator

$$A_{a\mu}(t)$$

such that

$$F_{a\mu}(s, t) = \exp \left[-\frac{i}{\hbar} s A_{a\mu}(t) \right]. \quad (402)$$

The second composition law implies, in the joint spectral representation,

$$A_{a\mu}(t + t') = A_{a\mu}(t) + A_{a\mu}(t'). \quad (403)$$

Strong continuity then gives

$$A_{a\mu}(t) = tA_{a\mu}(1). \quad (404)$$

Define

$$\hat{e}^a_{\mu} = A_{a\mu}(1). \quad (405)$$

Then

$$W(sE^a, tX_{\mu}) = \exp\left[-\frac{i}{\hbar}st\hat{e}^a_{\mu}\right]. \quad (406)$$

D.2 Reconstruction for arbitrary labels

Write

$$P = P_a E^a, \quad \xi = \xi^{\mu} X_{\mu}.$$

Repeated use of exact composition gives

$$W(P, \xi) = \prod_{a, \mu} W(P_a E^a, \xi^{\mu} X_{\mu}). \quad (407)$$

All factors strongly commute, hence

$$W(P, \xi) = \exp\left[-\frac{i}{\hbar}P_a \xi^{\mu} \hat{e}^a_{\mu}\right]. \quad (408)$$

The operators

$$\hat{e}^a_{\mu}$$

are therefore not independently postulated quantum tetrad variables. They are the joint Stone generators of the exact unitary bicharacter.

D.3 Joint projection-valued measure

Because the family

$$\{\hat{e}^a_{\mu}\}$$

strongly commutes, there exists a joint projection-valued measure

$$\mathbf{E}$$

on a measurable subset

$$\mathcal{E}_x \subset \mathbb{R}^{16} \simeq \text{Hom}(T_x M, V_x)$$

such that

$$\hat{e}^a_{\mu} = \int_{\mathcal{E}_x} e^a_{\mu} d\mathbf{E}(e). \quad (409)$$

Functional calculus then yields

$$W(P, \xi) = \int_{\mathcal{E}_x} \exp\left[-\frac{i}{\hbar}P_a e^a_{\mu} \xi^{\mu}\right] d\mathbf{E}(e). \quad (410)$$

This construction avoids treating the Gelfand spectrum as a priori norm-continuous in the response labels. Strong continuity is handled at the level of the commuting one-parameter unitary groups and their joint spectral measure.

E Strong Commutativity and Domains of Unbounded Generators

The infinitesimal soldering operators need not be bounded. Statements about their commutators must therefore be distinguished from statements about the finite unitary groups.

E.1 Strong commutativity

Two self-adjoint operators A and B strongly commute if their spectral projection measures commute:

$$E_A(\Delta)E_B(\Gamma) = E_B(\Gamma)E_A(\Delta) \quad (411)$$

for all Borel sets

$$\Delta, \Gamma \subset \mathbb{R}.$$

Equivalently,

$$e^{isA}e^{itB} = e^{itB}e^{isA} \quad (412)$$

for every

$$s, t \in \mathbb{R}.$$

The exact-response commutativity theorem therefore implies strong commutativity directly at the finite-unitary level.

E.2 Ordinary commutators

The formal relation

$$[A, B] = 0$$

is weaker and requires a common domain on which both

$$AB \quad \text{and} \quad BA$$

are defined.

Accordingly, throughout the main text an ordinary commutator relation between unbounded generators is to be understood only on a specified common invariant core. The representation-independent statement is strong commutativity.

E.3 Joint functional calculus

For a strongly commuting family

$$\hat{e}^a_\mu,$$

every Borel function

$$f(e)$$

defines, on its natural domain,

$$f(\hat{e}) = \int f(e)d\mathbf{E}(e). \quad (413)$$

In particular,

$$P_a \xi^\mu \hat{e}_\mu^a$$

is self-adjoint on its natural spectral domain, and the response exponential is unambiguously defined by

$$W(P, \xi) = \exp \left[-\frac{i}{\hbar} P_a \xi^\mu \hat{e}_\mu^a \right]. \quad (414)$$

F Faithfulness Does Not Imply Spectral Nondegeneracy

The distinction between operational faithfulness and pointwise spectral nondegeneracy is important enough to warrant an explicit example.

F.1 A finite-dimensional counterexample

Let

$$T = V = \mathbb{R}^4.$$

For $A = 0, 1, 2, 3$, define four rank-one linear maps

$$e_A : T \longrightarrow V$$

by

$$e_A(\xi) = \xi^A E_A, \quad (415)$$

with no sum over A .

Each map is degenerate:

$$\text{rank}(e_A) = 1.$$

Nevertheless,

$$\bigcap_{A=0}^3 \ker e_A = \{0\}. \quad (416)$$

Let the spectral measure assign nonzero probability to all four maps. Suppose

$$W(P, \xi) = I$$

for every generator P on every spectral component. Then

$$P(e_A \xi) = 0$$

for every P and every A , hence

$$e_A \xi = 0$$

for all A . Equation (416) gives

$$\xi = 0.$$

Thus the total response may be displacement-faithful even though every individual spectral geometry is degenerate.

F.2 Spectral nondegeneracy as an additional condition

If the spectral theory is intended to represent an ensemble of ordinary Lorentzian geometries, one must require

$$\mathbb{E}(\{e : \det e = 0\}) = 0. \quad (417)$$

This condition does not follow merely from collective faithfulness of the response family. The logical distinction is therefore

response faithfulness : no common invisible displacement, spectral nondegeneracy : almost every individual e is invertible.
--

(418)

G Probability Measures, Moments, and Response Tomography

Let a normal state on the response von Neumann algebra be represented by a probability measure

$$\mu(e)$$

on the spectral soldering space. Define the characteristic response functional

$$\mathcal{A}(P, \xi) = \int_{\mathcal{E}_x} \exp \left[-\frac{i}{\hbar} P_a e^a_{\mu} \xi^{\mu} \right] d\mu(e). \quad (419)$$

G.1 First moments

If the first moments exist,

$$i\hbar \left. \frac{\partial \mathcal{A}}{\partial \xi^{\mu}} \right|_{\xi=0} = P_a \langle e^a_{\mu} \rangle. \quad (420)$$

Differentiating also with respect to P_a ,

$$i\hbar \left. \frac{\partial^2 \mathcal{A}}{\partial P_a \partial \xi^{\mu}} \right|_{P=0, \xi=0} = \langle e^a_{\mu} \rangle. \quad (421)$$

Thus the mean solder map can be reconstructed from infinitesimal response tomography.

G.2 Second moments

Similarly,

$$-\hbar^2 \left. \frac{\partial^4 \mathcal{A}}{\partial P_a \partial P_b \partial \xi^{\mu} \partial \xi^{\nu}} \right|_{P=0, \xi=0} = \langle e^a_{\mu} e^b_{\nu} \rangle. \quad (422)$$

The covariance is

$$C^{ab}_{\mu\nu} = \langle e^a_{\mu} e^b_{\nu} \rangle - \langle e^a_{\mu} \rangle \langle e^b_{\nu} \rangle. \quad (423)$$

G.3 Cumulant functional

When

$$\mathcal{A}(P, \xi) \neq 0$$

near the identity, define

$$\mathcal{K}(P, \xi) = \log \mathcal{A}(P, \xi). \quad (424)$$

Derivatives of $\mathcal{K}$ generate connected response cumulants. For example,

$$-\hbar^2 \frac{\partial^4 \mathcal{K}}{\partial P_a \partial P_b \partial \xi^\mu \partial \xi^\nu} \Big|_0 = C_{\mu\nu}^{ab}. \quad (425)$$

Higher derivatives encode non-Gaussian classical stochastic geometry. Their nonvanishing does not imply noncommutativity.

G.4 Mean metric and fluctuation correction

For each nondegenerate spectral map,

$$g_{\mu\nu}(e) = \eta_{ab} e^a_\mu e^b_\nu.$$

Therefore

$$\begin{aligned} \langle g_{\mu\nu} \rangle &= \eta_{ab} \langle e^a_\mu e^b_\nu \rangle \\ &= \eta_{ab} \bar{e}^a_\mu \bar{e}^b_\nu + \eta_{ab} C_{\mu\nu}^{ab}. \end{aligned} \quad (426)$$

Hence

$$\langle g_{\mu\nu} \rangle - g_{\mu\nu}(\bar{e}) = \eta_{ab} C_{\mu\nu}^{ab}. \quad (427)$$

This is a purely stochastic correction and need not be of quantum origin.

H Detailed Derivation of the Quantum Geometric Tensor

This appendix records the detailed calculation underlying Sec. 6.

Let

$$|\psi(\lambda)\rangle = U(\lambda)|\psi_0\rangle \quad (428)$$

with

$$U(\lambda) = \exp \left[-\frac{i}{\hbar} \lambda^A T_A \right],$$

where the self-adjoint generators T_A strongly commute.

H.1 Parameter derivatives

Strong commutativity gives

$$\partial_A |\psi\rangle = -\frac{i}{\hbar} T_A |\psi\rangle. \quad (429)$$

The quantum geometric tensor is

$$\begin{aligned} Q_{AB} &= \langle \partial_A \psi | \partial_B \psi \rangle - \langle \partial_A \psi | \psi \rangle \langle \psi | \partial_B \psi \rangle \\ &= \frac{1}{\hbar^2} (\langle T_A T_B \rangle - \langle T_A \rangle \langle T_B \rangle). \end{aligned} \quad (430)$$

H.2 Symmetric and antisymmetric parts

In the general noncommuting case,

$$\text{Re } Q_{AB} = \frac{1}{2\hbar^2} \langle \{\Delta T_A, \Delta T_B\} \rangle, \quad (431)$$

$$\text{Im } Q_{AB} = \frac{1}{2i\hbar^2} \langle [T_A, T_B] \rangle. \quad (432)$$

For the exact response,

$$[T_A, T_B] = 0$$

on a common invariant core, or more precisely the operators strongly commute. Hence

$$\text{Im } Q_{AB} = 0. \quad (433)$$

H.3 Positivity

For any real coefficients v^A ,

$$\begin{aligned} v^A v^B \text{Re } Q_{AB} &= \frac{1}{\hbar^2} \left\langle \left(v^A \Delta T_A \right)^2 \right\rangle \\ &\geq 0. \end{aligned} \quad (434)$$

Thus the Fubini–Study metric cannot possess Lorentzian signature on a nondegenerate four-dimensional response parameter space.

H.4 Application to the full (P, ξ) orbit

For

$$|\psi(P, \xi)\rangle = W(P, \xi)|\psi_0\rangle,$$

define

$$T_a^{(P)} = \xi^\mu \hat{e}_{a\mu}, \quad (435)$$

$$T_\mu^{(\xi)} = P_a \hat{e}^a{}_\mu. \quad (436)$$

Then

$$Q_{ab} = \frac{1}{\hbar^2} \xi^\mu \xi^\nu C_{ab\mu\nu}, \quad (437)$$

$$Q_{\mu\nu} = \frac{1}{\hbar^2} P_a P_b C^{ab}{}_{\mu\nu}, \quad (438)$$

$$Q_{a\mu} = \frac{1}{\hbar^2} \xi^\nu P_b C_a{}^b{}_{\nu\mu}, \quad (439)$$

with all blocks real.

Therefore the Berry two-form of the complete local exact-response orbit vanishes.

I Weyl Cocycles, Coboundaries, and Central Extensions

The comparison between the exact response and a Weyl system depends only on the cohomology class of the projective phase, not on a particular phase convention.

I.1 Projective composition

Let

$$\mathcal{W}(z), \quad z \in \mathcal{Z},$$

satisfy

$$\mathcal{W}(z)\mathcal{W}(z') = c(z, z')\mathcal{W}(z + z'), \quad (440)$$

where

$$c(z, z') \in U(1).$$

Associativity requires

$$c(z, z')c(z + z', z'') = c(z', z'')c(z, z' + z''). \quad (441)$$

I.2 Change of phase convention

Redefine

$$\mathcal{W}'(z) = b(z)\mathcal{W}(z), \quad b(z) \in U(1). \quad (442)$$

Then

$$c'(z, z') = \frac{b(z)b(z')}{b(z + z')}c(z, z'). \quad (443)$$

Thus cocycles differing by such a factor are projectively equivalent.

I.3 Symplectic Weyl cocycle

For a real antisymmetric bilinear form ω ,

$$c_\omega(z, z') = \exp \left[-\frac{i}{2\hbar} \omega(z, z') \right] \quad (444)$$

satisfies the cocycle condition.

The corresponding group commutator is independent of the symmetric coboundary ambiguity:

$$\mathcal{W}(z)\mathcal{W}(z')\mathcal{W}(z)^{-1}\mathcal{W}(z')^{-1} = \exp \left[-\frac{i}{\hbar} \omega(z, z') \right] I. \quad (445)$$

Hence the antisymmetric symplectic form is the projectively invariant content responsible for noncommutativity.

I.4 Soldered cocycle

For

$$z = (P, \xi), \quad z' = (Q, \zeta),$$

the soldered symplectic form is

$$\omega_e(z, z') = P(e\zeta) - Q(e\xi). \quad (446)$$

The associated Weyl cocycle is

$$c_e(z, z') = \exp \left[-\frac{i}{2\hbar} (P(e\zeta) - Q(e\xi)) \right]. \quad (447)$$

This cocycle vanishes on each marginal sector:

$$\omega_e((P, 0), (Q, 0)) = 0, \quad (448)$$

$$\omega_e((0, \xi), (0, \zeta)) = 0, \quad (449)$$

but not in general on mixed generator–displacement pairs.

J Lorentz Covariance of the Soldered Symplectic Pairing

Let

$$\Lambda \in SO^+(1, 3)$$

act on the internal vector space V . The solder map and generator transform as

$$e \mapsto e' = \Lambda e, \quad (450)$$

$$P \mapsto P' = \Lambda^{-T} P. \quad (451)$$

Then

$$\begin{aligned} P'(e'\xi) &= (\Lambda^{-T} P)(\Lambda e\xi) \\ &= P(e\xi). \end{aligned} \quad (452)$$

Consequently,

$$\begin{aligned} \omega_{e'}((P', \xi), (Q', \zeta)) &= P'(e'\zeta) - Q'(e'\xi) \\ &= P(e\zeta) - Q(e\xi) \\ &= \omega_e((P, \xi), (Q, \zeta)). \end{aligned} \quad (453)$$

Thus the soldered local Weyl cocycle, if introduced, is invariant under internal local Lorentz changes of frame.

This covariance does not imply that the projective Weyl extension is selected dynamically. It shows only that such an extension is compatible with the internal Lorentz gauge symmetry of the classical soldered geometry.

K Covariant Decomposition of the Liouville Two-Form

Let

$$\lambda = P_a e^a.$$

Introduce a Lorentz connection

$$\omega^a_b$$

and define

$$DP_a = dP_a - \omega^b_a P_b, \quad (454)$$

$$T^a = de^a + \omega^a_b \wedge e^b. \quad (455)$$

Then

$$\begin{aligned} d\lambda &= dP_a \wedge e^a + P_a de^a \\ &= DP_a \wedge e^a + P_a T^a. \end{aligned} \quad (456)$$

K.1 Cancellation of connection terms

Explicitly,

$$\begin{aligned} DP_a \wedge e^a + P_a T^a &= dP_a \wedge e^a - \omega^b_a P_b \wedge e^a \\ &\quad + P_a de^a + P_a \omega^a_b \wedge e^b. \end{aligned} \quad (457)$$

Relabeling internal indices in the last term shows cancellation of the two connection contributions.

Thus $d\lambda$ is independent of the auxiliary Lorentz frame used to write the covariant decomposition.

K.2 Torsion-free branch

If

$$T^a = 0,$$

then

$$d\lambda = DP_a \wedge e^a. \quad (458)$$

If torsion is nonzero,

$$P_a T^a$$

is part of the same canonical two-form expressed in Lorentz-covariant variables. It should not be interpreted as an additional independent symplectic structure.

K.3 Zero section

Let

$$s_0 : M \longrightarrow T^*M$$

be the zero section. Since

$$p = 0$$

on $s_0(M)$,

$$s_0^* \lambda = 0. \quad (459)$$

Consequently,

$$s_0^* d\lambda = 0. \quad (460)$$

Thus the zero section is a Lagrangian submanifold of the canonical cotangent symplectic geometry.

L Counterexamples When Exact Double Additivity Is Relaxed

The no-go theorem requires both exact composition laws. This appendix gives simple examples showing that either law may be relaxed while allowing noncommutativity.

L.1 Exact generator composition alone

Let

$$\mathcal{H} = \mathbb{C}^2,$$

let $P \in \mathbb{R}$, and let $\xi \in \mathbb{R}$. Define

$$A(\xi) = \cos \xi \sigma_z + \sin \xi \sigma_x, \quad (461)$$

and

$$W(P, \xi) = \exp \left[-\frac{i}{\hbar} P A(\xi) \right]. \quad (462)$$

For every fixed ξ ,

$$W(P + Q, \xi) = W(P, \xi) W(Q, \xi). \quad (463)$$

Thus generator composition is exact.

However,

$$[A(\xi), A(\zeta)] = 2i \sin(\zeta - \xi) \sigma_y,$$

so responses at distinct displacement labels generally fail to commute:

$$[W(P, \xi), W(Q, \zeta)] \neq 0. \quad (464)$$

The displacement composition law is not exact because

$$A(\xi + \zeta) \neq A(\xi) + A(\zeta).$$

L.2 Exact displacement composition alone

A dual example is obtained by defining

$$B(P) = \cos P \sigma_z + \sin P \sigma_x$$

and

$$W(P, \xi) = \exp \left[-\frac{i}{\hbar} \xi B(P) \right]. \quad (465)$$

For fixed P ,

$$W(P, \xi + \zeta) = W(P, \xi) W(P, \zeta),$$

but responses associated with distinct generator labels need not commute.

Thus neither exact composition law separately is sufficient for the no-go theorem.

L.3 Projective composition

A Weyl system satisfies neither the combined exact-response commutativity conclusion nor a trivial group commutator. Instead,

$$U(P)V(\xi) = e^{-iP(e\xi)/\hbar} V(\xi)U(P). \quad (466)$$

Hence projective composition explicitly evades the exact-response theorem.

L.4 Noncommuting inter-event transport

Even if every local algebra

$$\mathcal{A}_{W,x}$$

is Abelian, introduce a unitary transport

$$U_\gamma : \mathcal{H}_x \rightarrow \mathcal{H}_y$$

satisfying

$$U_\gamma W_x(P, \xi) U_\gamma^{-1} = W_y(P_\gamma, \xi_\gamma). \quad (467)$$

For two different paths

$$\gamma_1, \gamma_2 : x \rightarrow y,$$

one may have

$$U_{\gamma_2}^{-1} U_{\gamma_1} \neq I. \quad (468)$$

Thus local response commutativity does not imply trivial quantum transport or trivial holonomy.

M Local Algebras and Inter-Event Comparison

The main no-go theorem is fiberwise. A field-theoretic extension would require a family

$$x \mapsto \mathcal{A}_{W,x}$$

of local commutative response algebras.

M.1 A bundle of response algebras

At every event,

$$\mathcal{A}_{W,x} = C^*(W_x(P, \xi))$$

is Abelian.

One may regard the collection

$$\{\mathcal{A}_{W,x}\}_{x \in M}$$

as a bundle or net of local observable algebras, depending on the global formulation.

M.2 Transport automorphisms

For a path

$$\gamma : x \rightarrow y,$$

suppose there exists a unitary comparison map

$$U_\gamma.$$

It induces

$$\alpha_\gamma(A) = U_\gamma A U_\gamma^{-1}, \quad A \in \mathcal{A}_{W,x}. \quad (469)$$

Although

$$\mathcal{A}_{W,x}$$

and

$$\mathcal{A}_{W,y}$$

are individually Abelian, the family of transport operators

$$\{U_\gamma\}$$

need not commute.

M.3 Closed transport

For a loop

$$\gamma : x \rightarrow x,$$

the induced automorphism

$$\alpha_\gamma$$

may be nontrivial:

$$\alpha_\gamma \neq \text{id}. \quad (470)$$

Thus one possible location for genuinely quantum geometric information is not the local response algebra but the comparison functor between local response algebras.

The present paper does not construct such a theory; the observation only clarifies the scope of the local no-go result.

N A Hierarchy of Classicality and Quantumness

Several distinct notions of “quantum geometry” can be separated within the operational framework.

N.1 Level I: sharp classical soldering

A single nondegenerate map

$$e : T_x M \rightarrow V_x$$

gives

$$W(P, \xi) = e^{-iP(e\xi)/\hbar} I.$$

There are no response fluctuations.

N.2 Level II: classical stochastic soldering

A probability measure

$$\mu(e)$$

gives

$$\mathcal{A}(P, \xi) = \int e^{-iP(e\xi)/\hbar} d\mu(e).$$

The response fluctuates, but the observable algebra remains classical.

N.3 Level III: quantum matter on a spectral geometry

One may take a direct integral

$$\mathcal{H} = \int^{\oplus} \mathcal{H}_e d\mu(e)$$

in which each fiber carries a noncommutative matter algebra while e remains a central spectral label.

This theory is quantum in its matter sector but not necessarily in the geometry label itself.

N.4 Level IV: projective local response

A nontrivial Weyl cocycle gives

$$\mathcal{W}(z)\mathcal{W}(z') = c(z, z')\mathcal{W}(z + z').$$

The local response-related algebra is now intrinsically noncommutative.

N.5 Level V: noncommutative geometric response

If geometric response operators themselves obey noncentral commutation relations, the spectral classical-random representation fails.

Such a theory requires an enlarged algebra beyond the exact response axioms considered here.

N.6 Level VI: quantum inter-event comparison

Even with Abelian local response algebras, noncommuting or path-dependent transport operators may carry genuinely quantum comparison data.

These levels should not be conflated. In particular,

$$\text{geometric fluctuation} \not\Rightarrow \text{noncommutative geometry}, \quad (471)$$

and

$$\text{quantum matter} \not\Rightarrow \text{quantized soldering}. \quad (472)$$

O Assumption Map for the Exact-Response No-Go Theorem

For reference, the logical dependence of the principal results is collected here.

O.1 Classical reconstruction assumptions

The classical soldering result uses:

1. additivity or homogeneous extension in the generator variable;
2. additivity and continuity in infinitesimal displacement;
3. smooth dependence on the spacetime point;
4. operational faithfulness;
5. equality of tangent and internal dimensions.

These give

$$\Theta(P, \xi) = P(e\xi)$$

and, when e is nondegenerate,

$$g = e^* \eta.$$

O.2 Cotangent reconstruction assumptions

No new dynamical assumption is needed to obtain

$$p = e^* P, \quad \lambda = P_a e^a, \quad \Omega = d\lambda.$$

These follow from the canonical definition of the cotangent Liouville form once the physical covector has been identified.

O.3 Quantum commutativity assumptions

The finite commutativity theorem uses:

1. unitarity, or more generally invertibility;
2. exact generator composition;
3. exact displacement composition.

Strong continuity is not required for the finite commutativity proof.

O.4 Spectral representation assumptions

To pass from commutativity to self-adjoint infinitesimal generators and the joint spectral representation one additionally uses:

1. strong continuity;
2. standard regularity assumptions for the one-parameter unitary groups;
3. joint spectral theory for strongly commuting self-adjoint operators.

O.5 Classical stochastic representation assumptions

The probability representation uses:

1. commutativity of the response algebra;
2. a state restricted to that algebra;
3. the Gelfand–Riesz representation, or equivalently a spectral measure in a normal representation.

O.6 Quantum geometric tensor assumptions

The covariance formulas additionally require existence of the relevant first and second moments of the self-adjoint response generators.

The vanishing of Berry curvature follows from strong commutativity, while positivity of the Fubini–Study metric follows from its covariance form.

P Summary of the Appendix Results

The technical results collected above refine the main argument in the following way:

$$\begin{aligned}
& C^+ \longrightarrow V^* \quad \text{by additive group completion,} \\
& \text{double exact composition} \longrightarrow [W(P, \xi), W(Q, \zeta)] = 0, \\
& \quad \text{strong continuity} \longrightarrow \text{strongly commuting Stone generators,} \\
& \quad \text{joint spectral theory} \longrightarrow \hat{e}^a_\mu \text{ with a common projection-valued measure,} \\
& \quad W(P, \xi) = \int e^{-iP(e\xi)/\hbar} dE(e), \\
& \quad \text{states on } \mathcal{A}_W \longrightarrow \text{ordinary probability measures on response space,} \\
& \quad \text{response fluctuations} \longrightarrow \text{classical stochastic moments and cumulants,} \\
& \quad \text{exact response QGT} \longrightarrow \mathcal{F}^B = 0, \\
& \quad \text{Re } Q \longrightarrow \text{positive semidefinite covariance geometry,} \\
& \quad \text{Weyl projective extension} \longrightarrow \text{nontrivial central commutator,} \\
& \quad \text{local response commutativity} \not\Rightarrow \text{trivial inter-event transport.}
\end{aligned} \tag{473}$$

The principal boundary established by the paper is therefore not a statement that quantum geometry is impossible. It is the more specific statement that exact preservation of the classical local generator–displacement response composition leaves the corresponding geometric observable sector commutative and classically representable.

Intrinsic quantum geometry must consequently enter through additional structure not contained in that exact local response algebra.